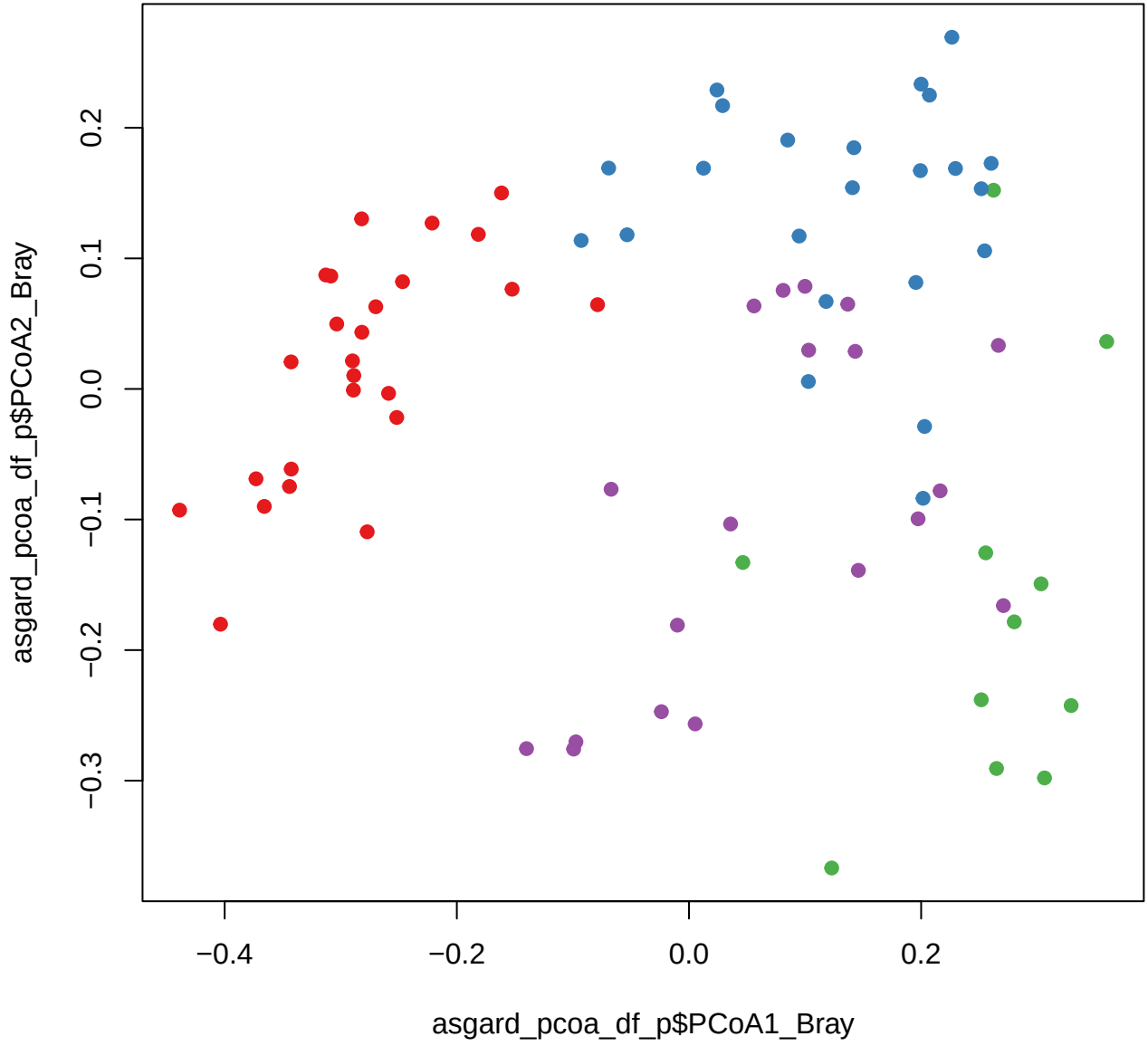
asgard_pcoa_df_p$PCoA2_Bray
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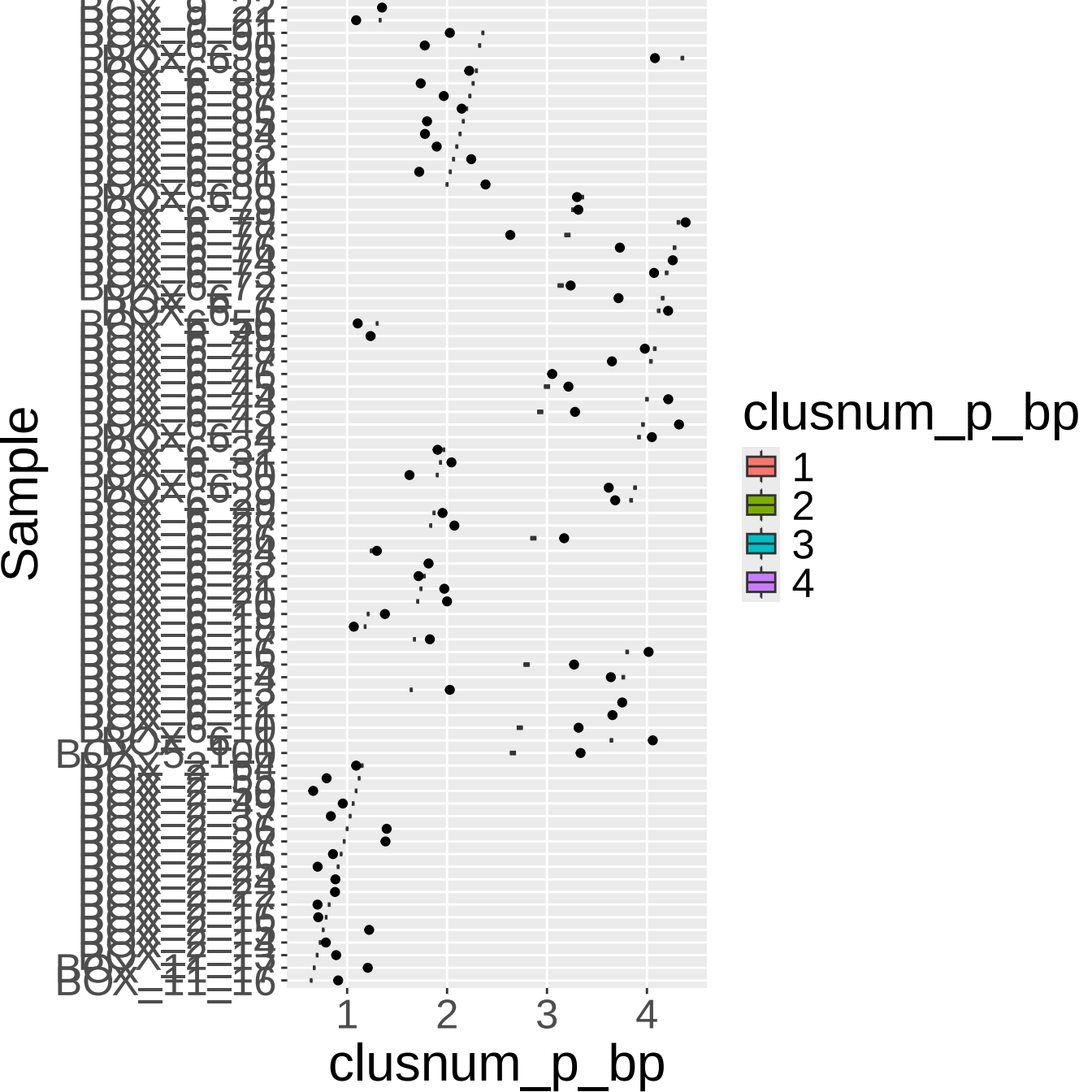
asgard_pcoa_df_p$PCoA1_Bray


Sample
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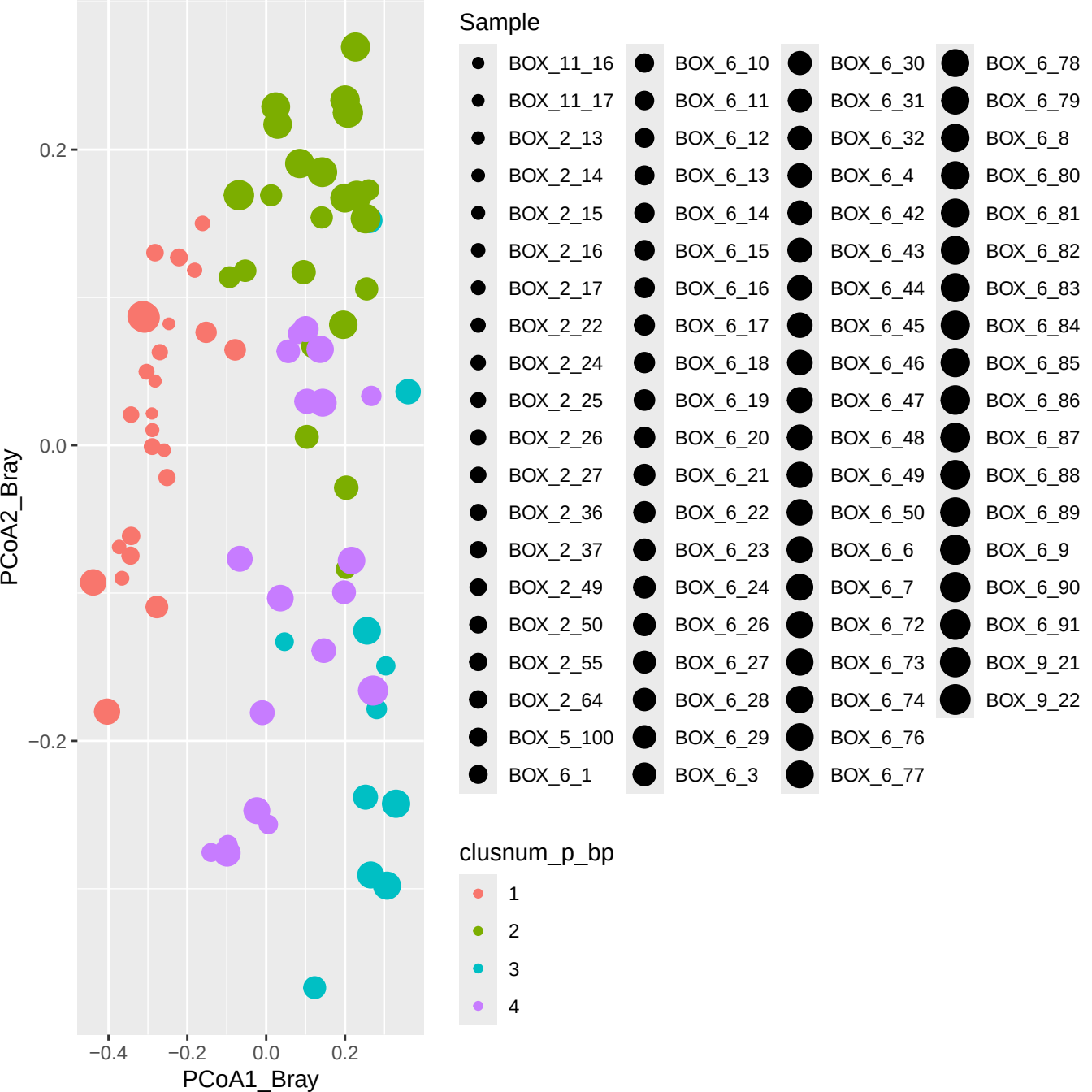
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BOX_2_26
BOX_2_27
BOX_2_36
BOX_2_37
BOX_2_49
BOX_2_50
BOX_2_55
BOX_2_64
BOX_5_100
BOX_6_1
BOX_6_10
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BOX_9_22
clusnum_p_bp
1
2
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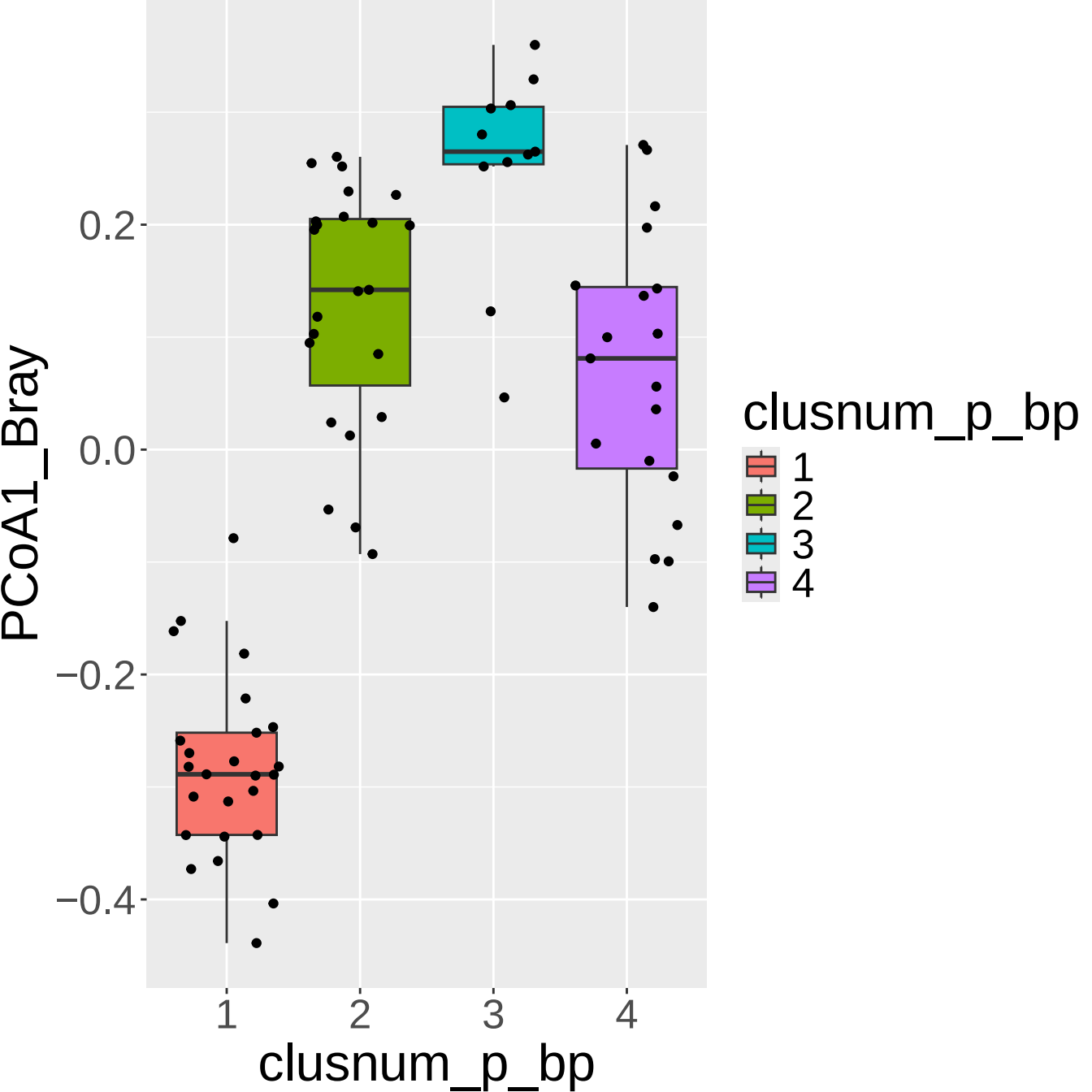
PCoA1_Bray


PCoA1_Bray
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clusnum_p_bp
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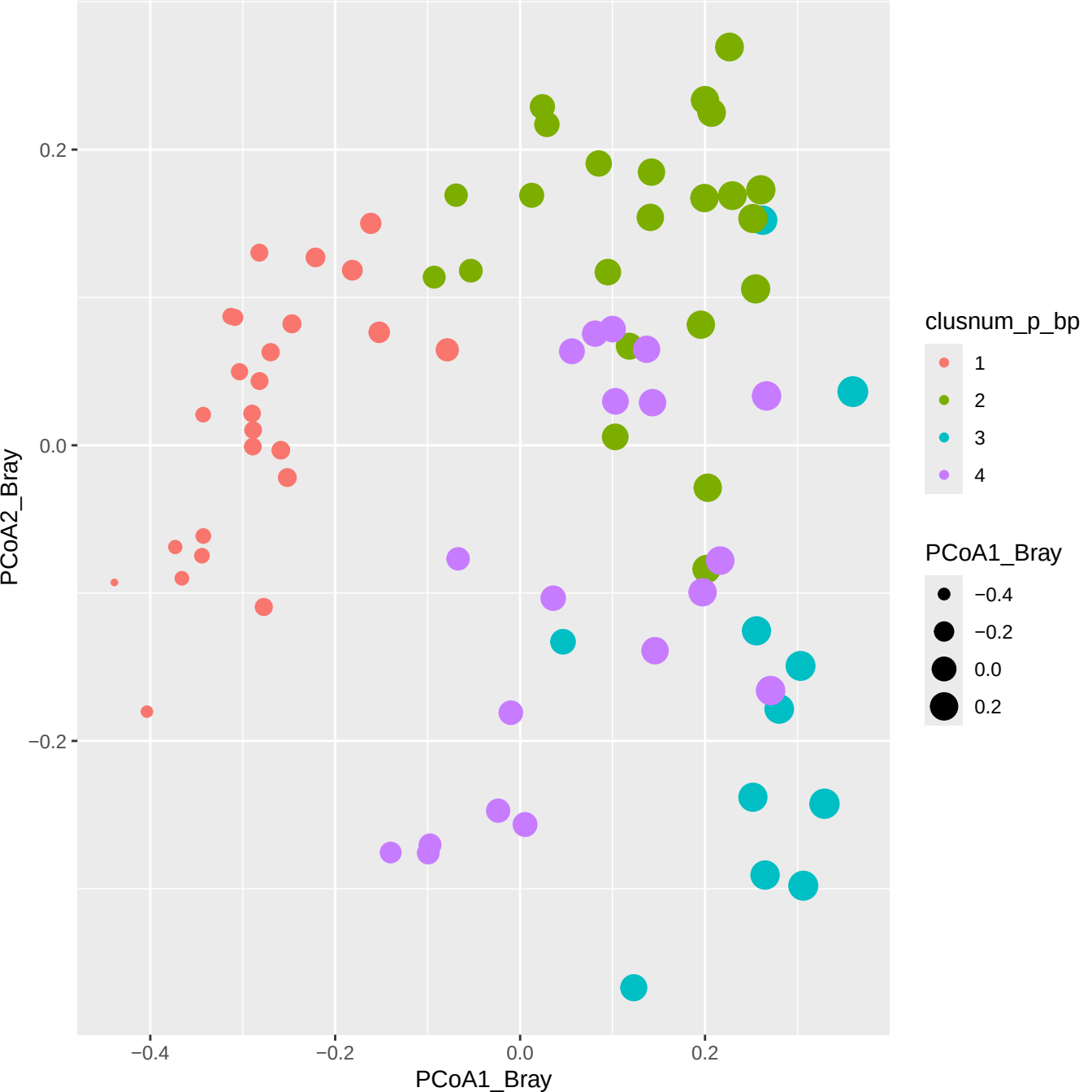
3
4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−0.2
0.0
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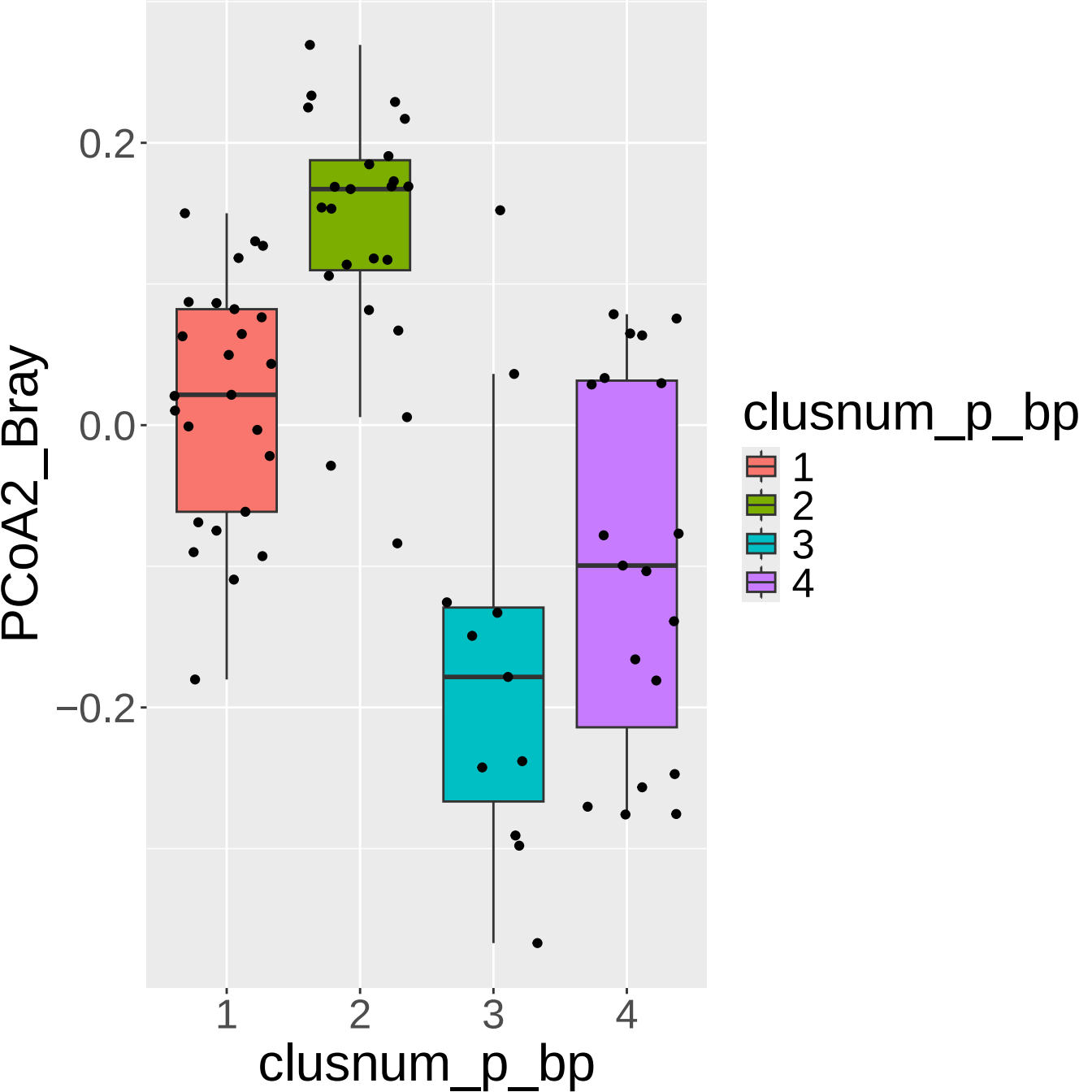
PCoA2_Bray
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clusnum_p_bp
clusnum_p_bp
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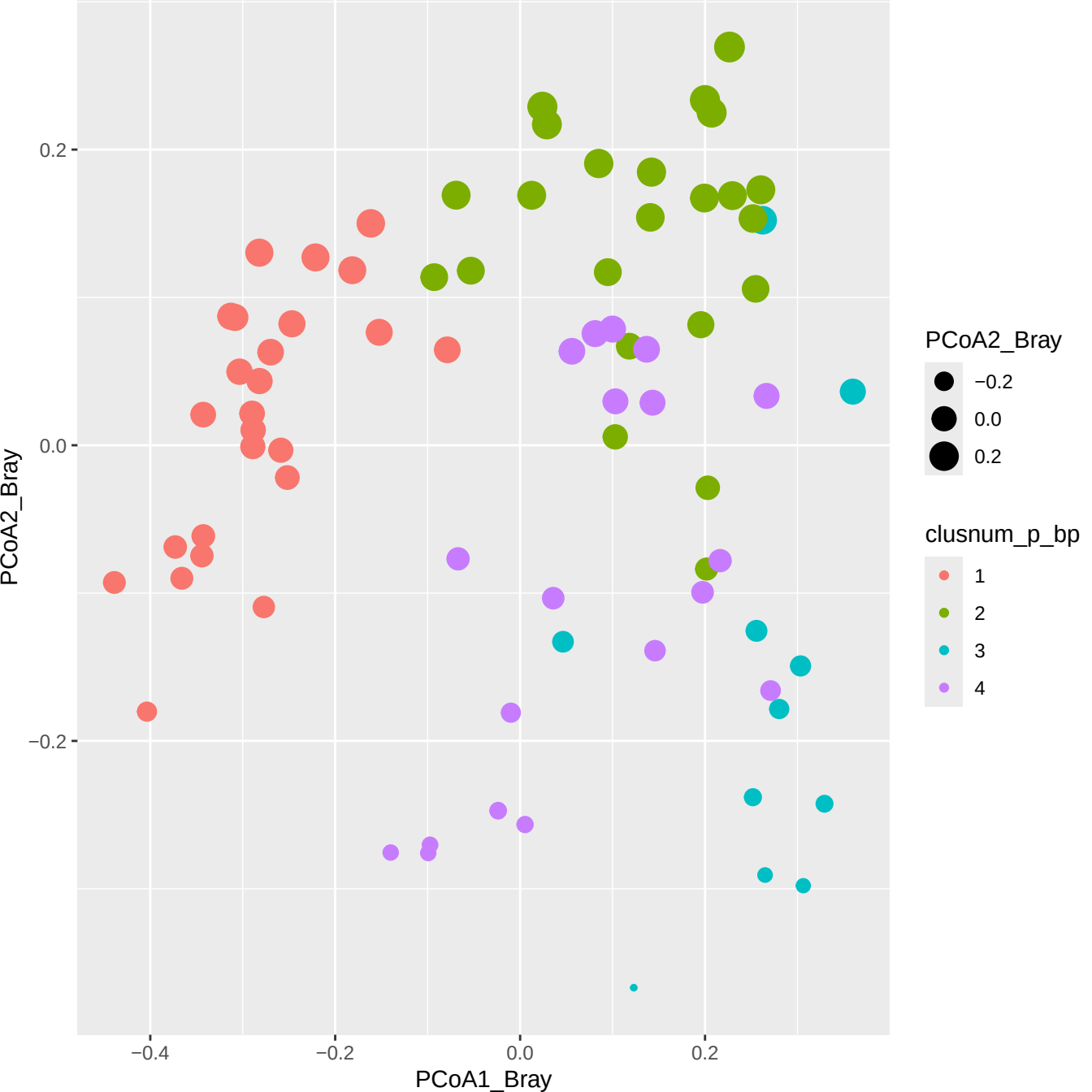
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0.0
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PCoA2_Bray
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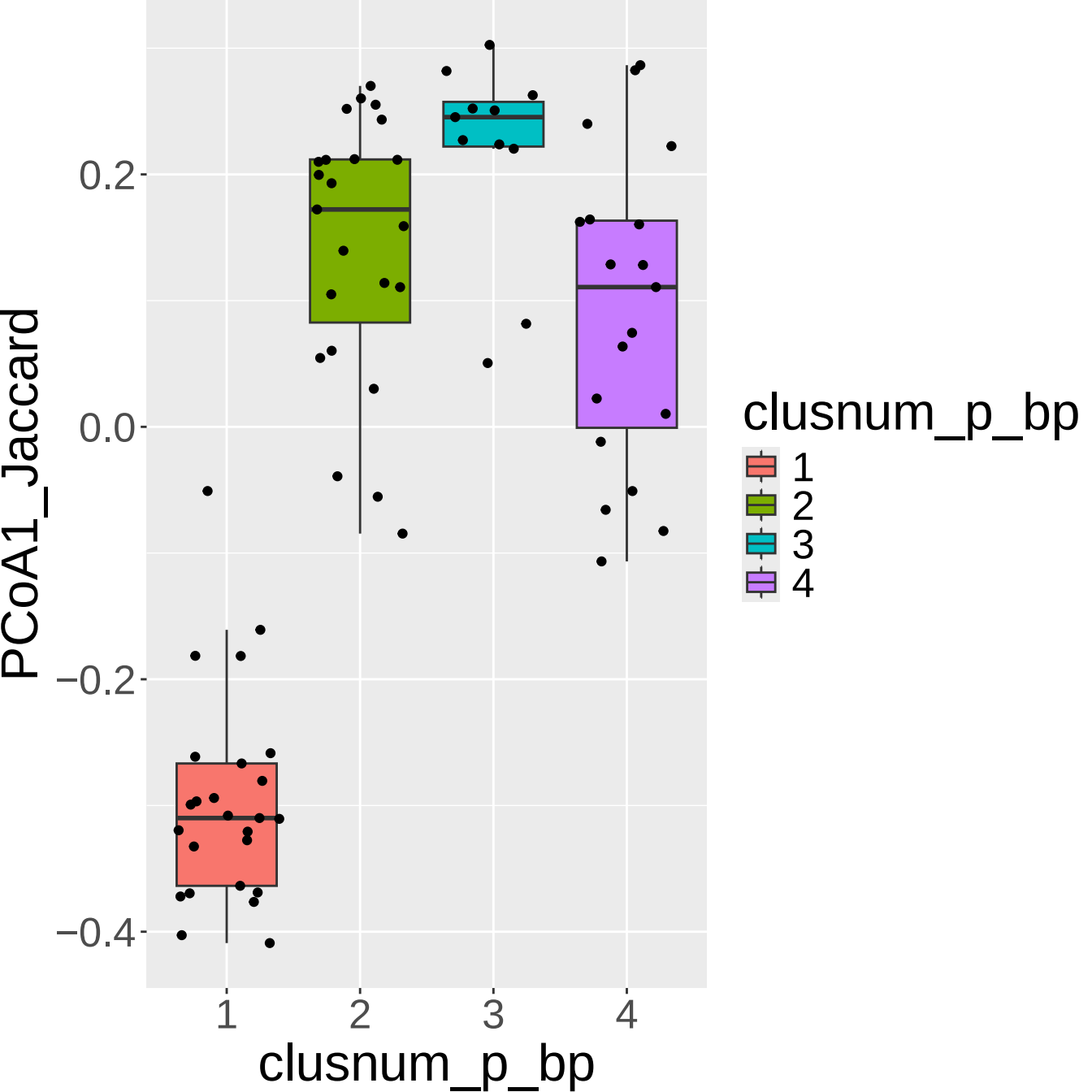
PCoA1_Jaccard
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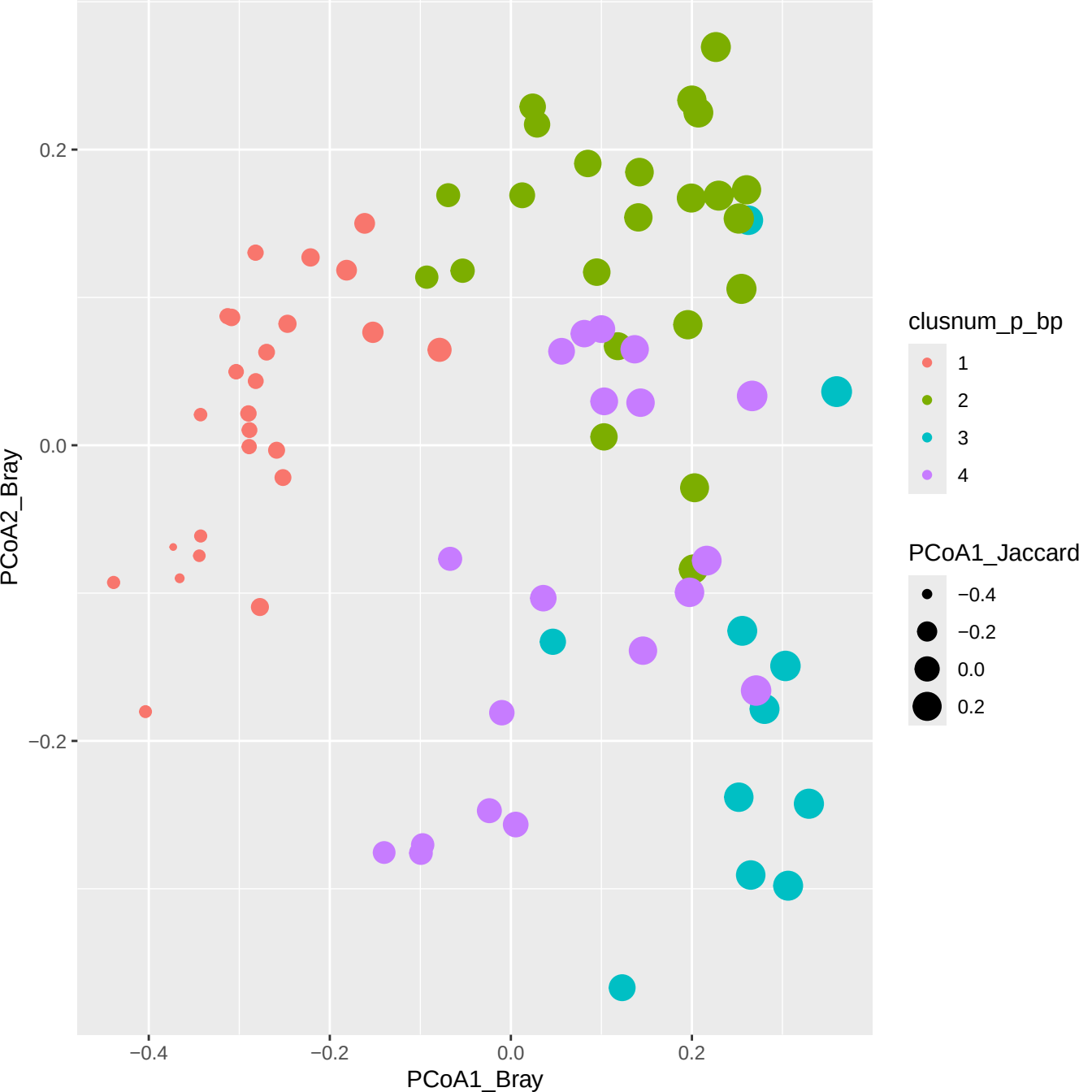
−0.2
−0.4
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4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PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
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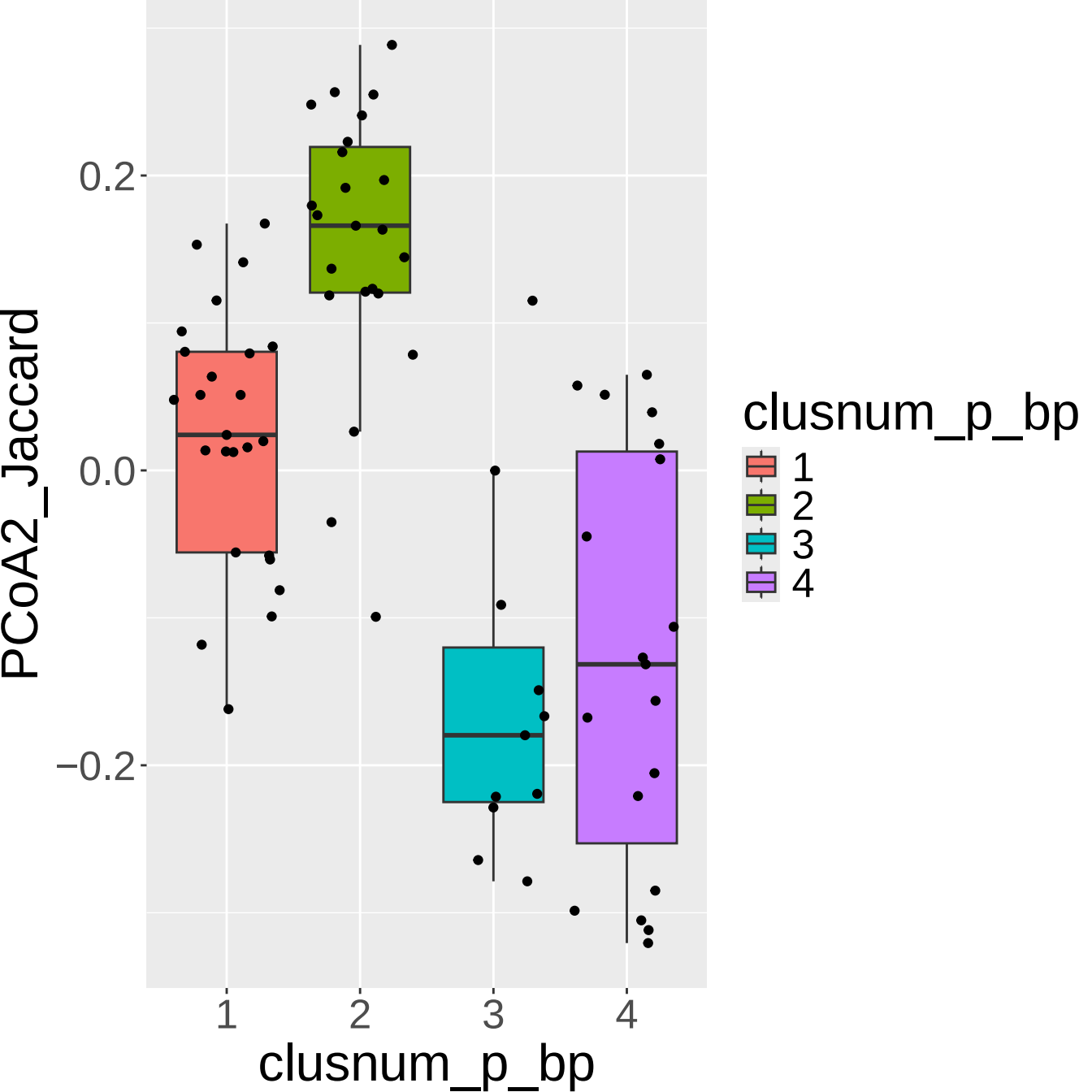
3
4
PCoA1_Jaccard
−0.4
−0.2
0.0
0.2


PCoA2_Jaccard
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1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
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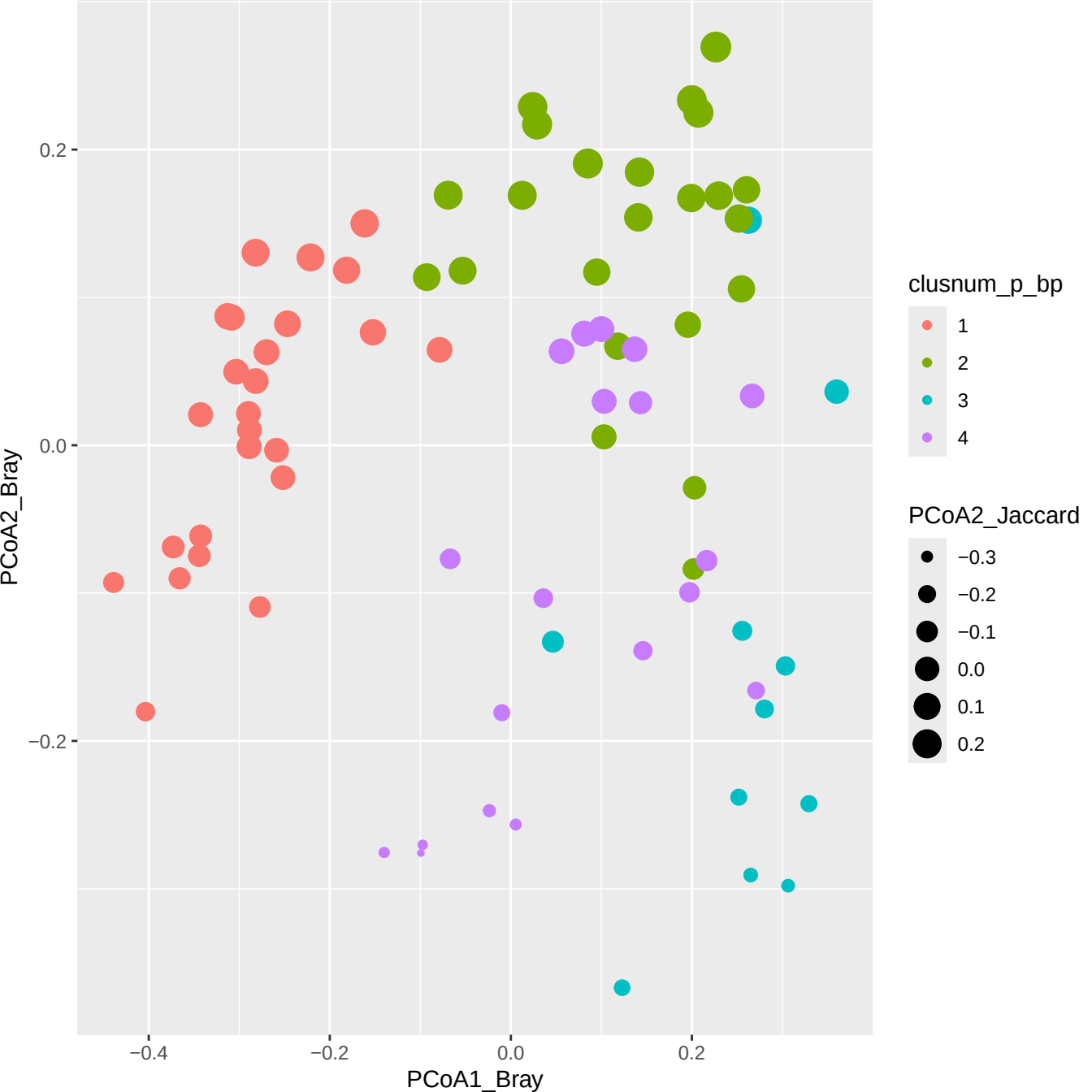
3
4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−0.2
PCoA2_Bray
−0.4
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0.0
0.2
PCoA1_Bray
clusnum_p_bp
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3
4
PCoA2_Jaccard
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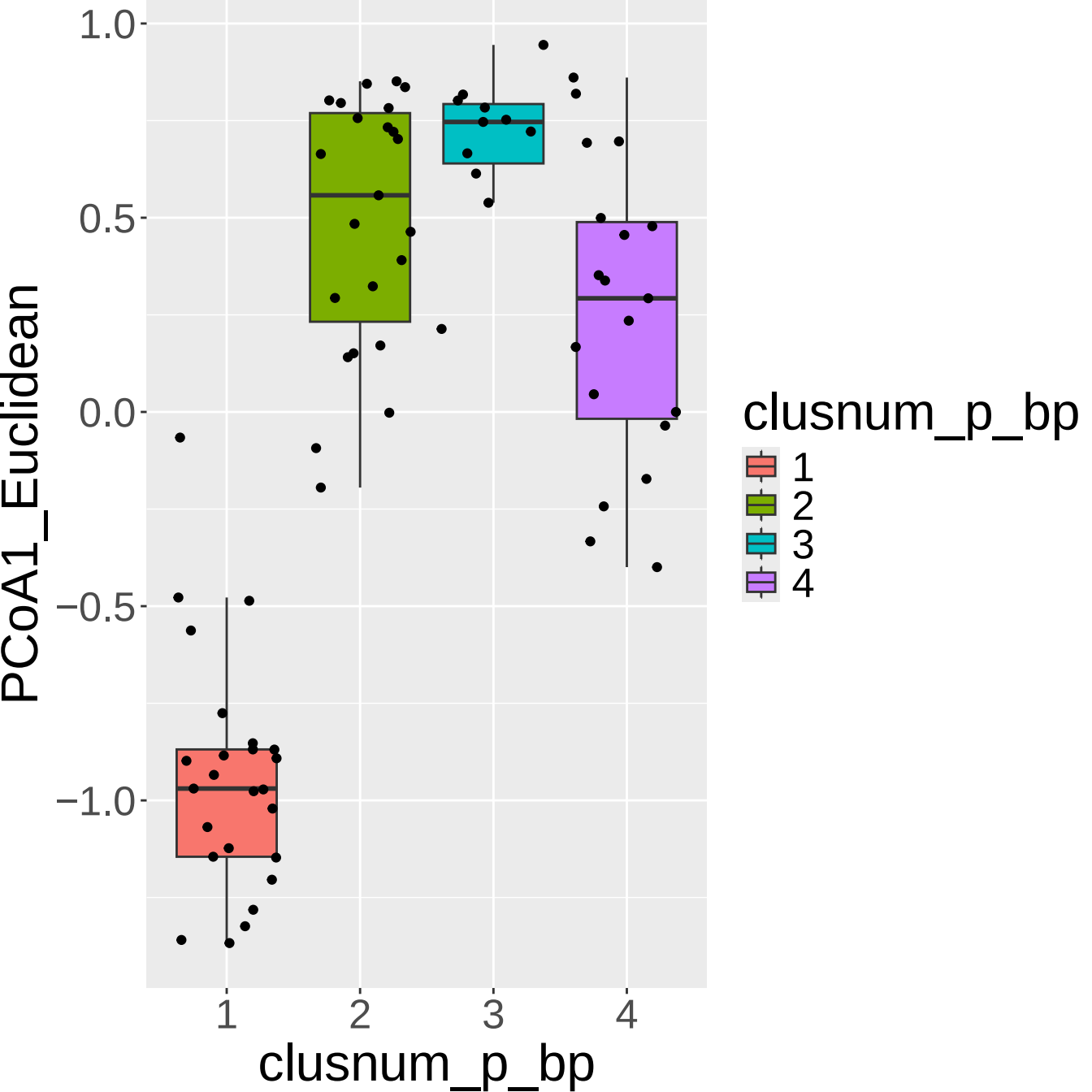
−0.2
−0.1
0.0
0.1
0.2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0.0
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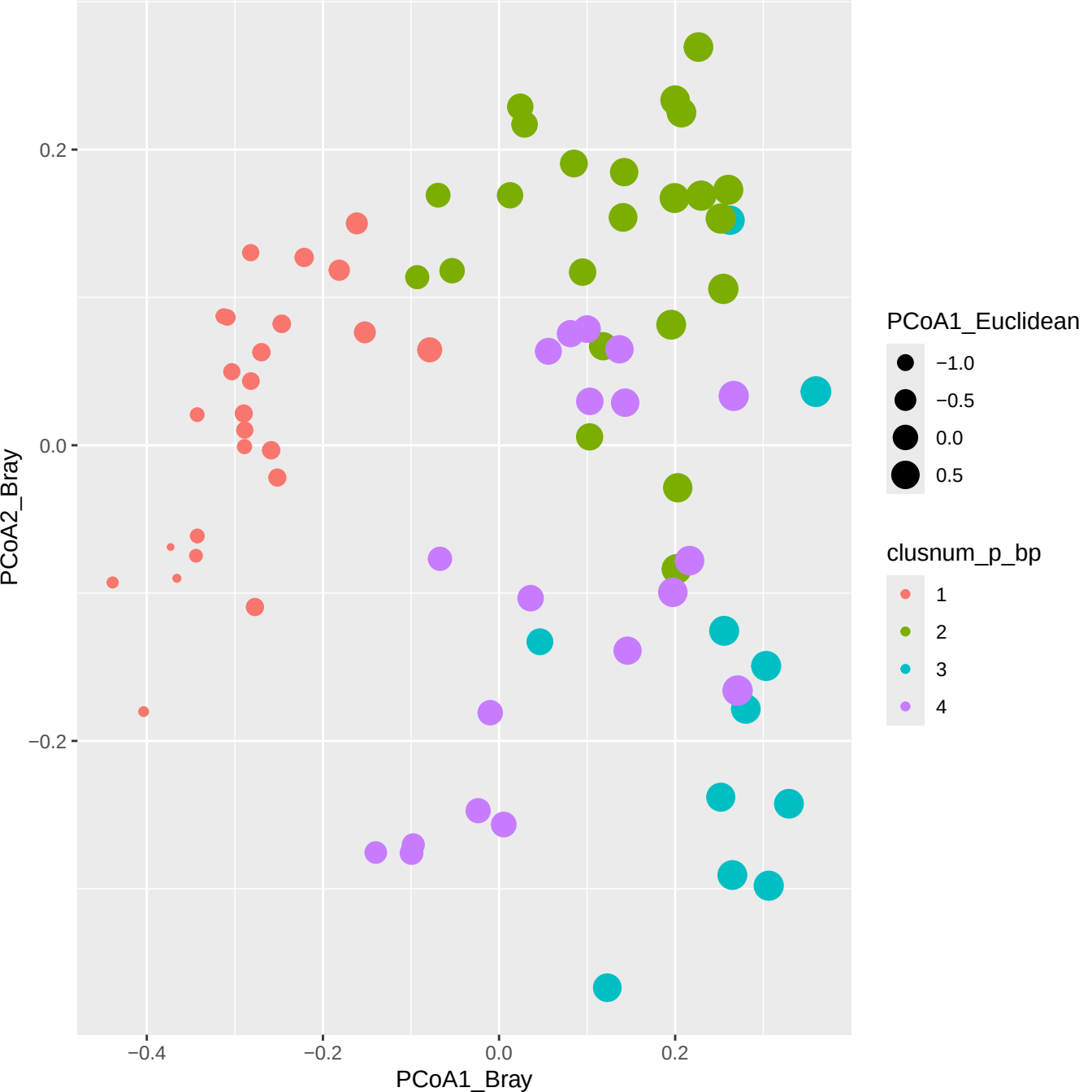
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4


0.2
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−0.4
−0.2
0.0
0.2
PCoA1_Bray
PCoA1_Euclidean
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2
3
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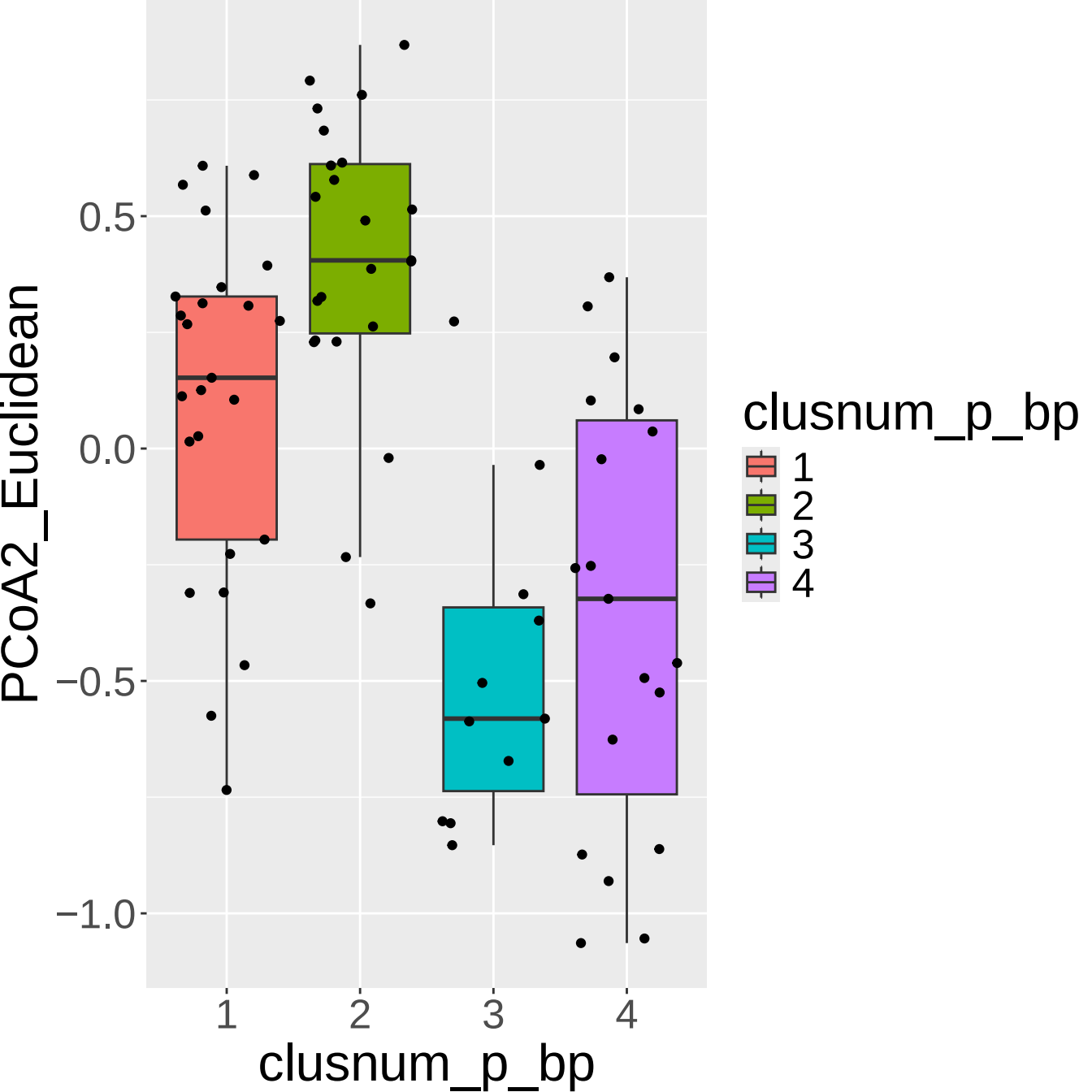
PCoA2_Euclidean
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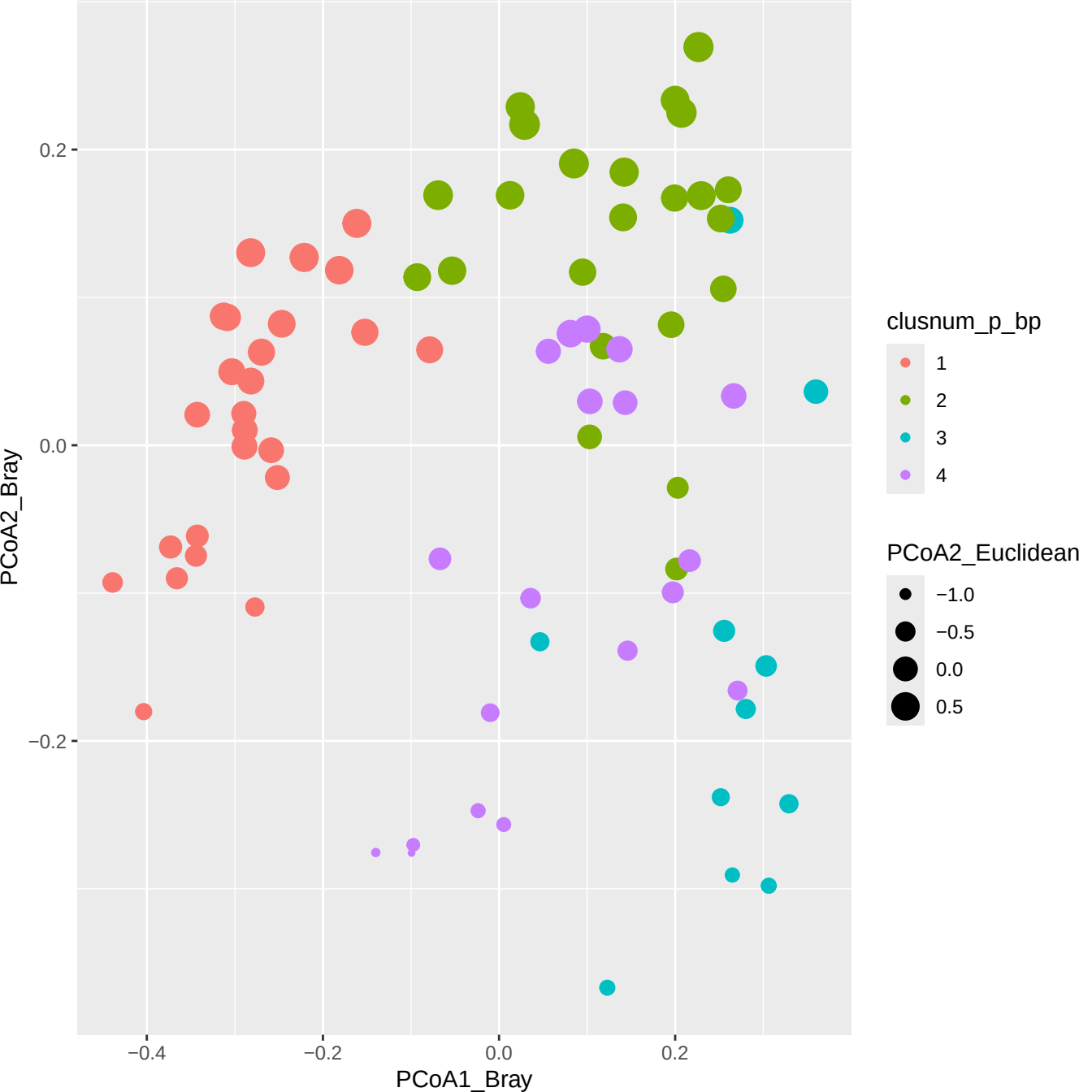
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4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0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
PCoA2_Euclidean
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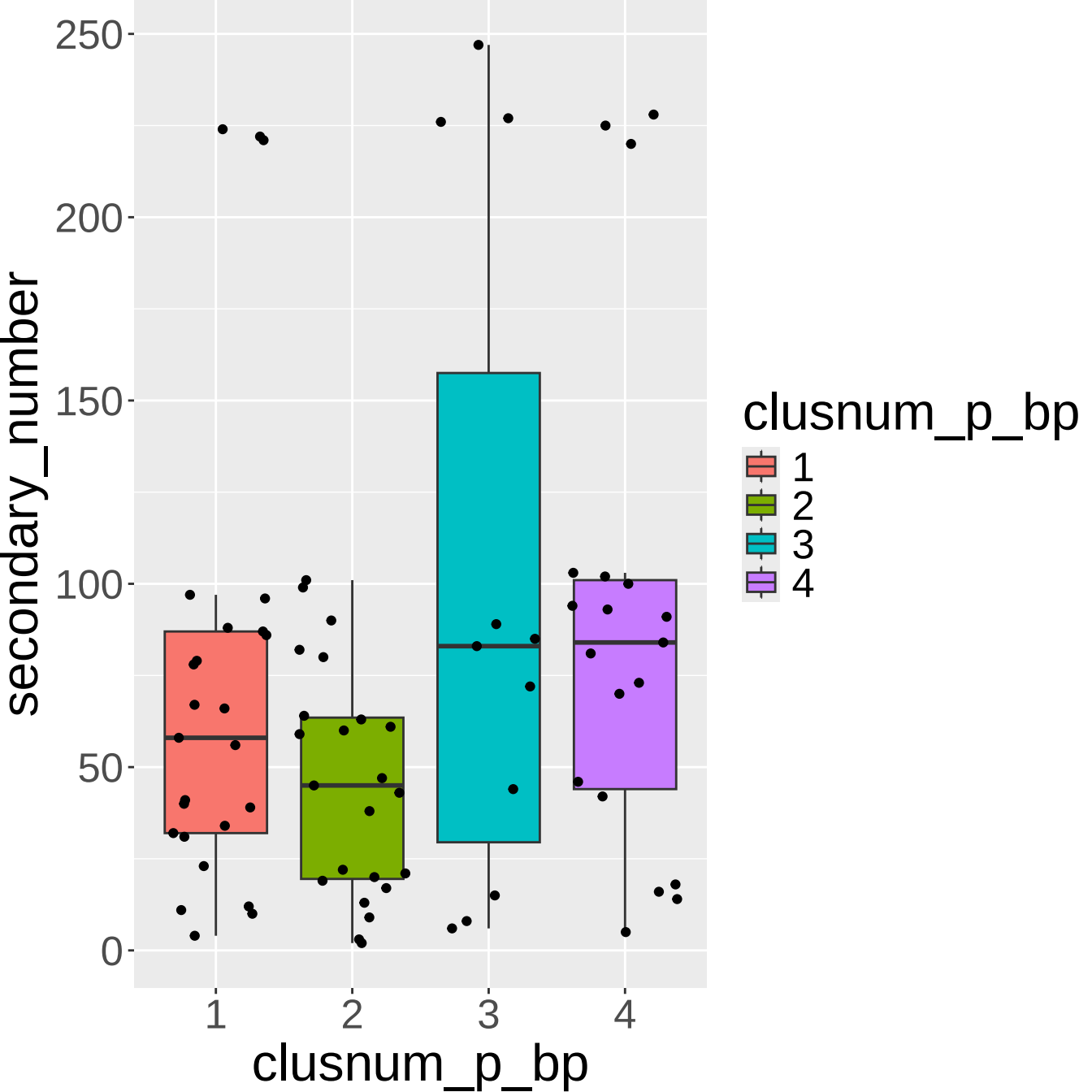
−0.5
0.0
0.5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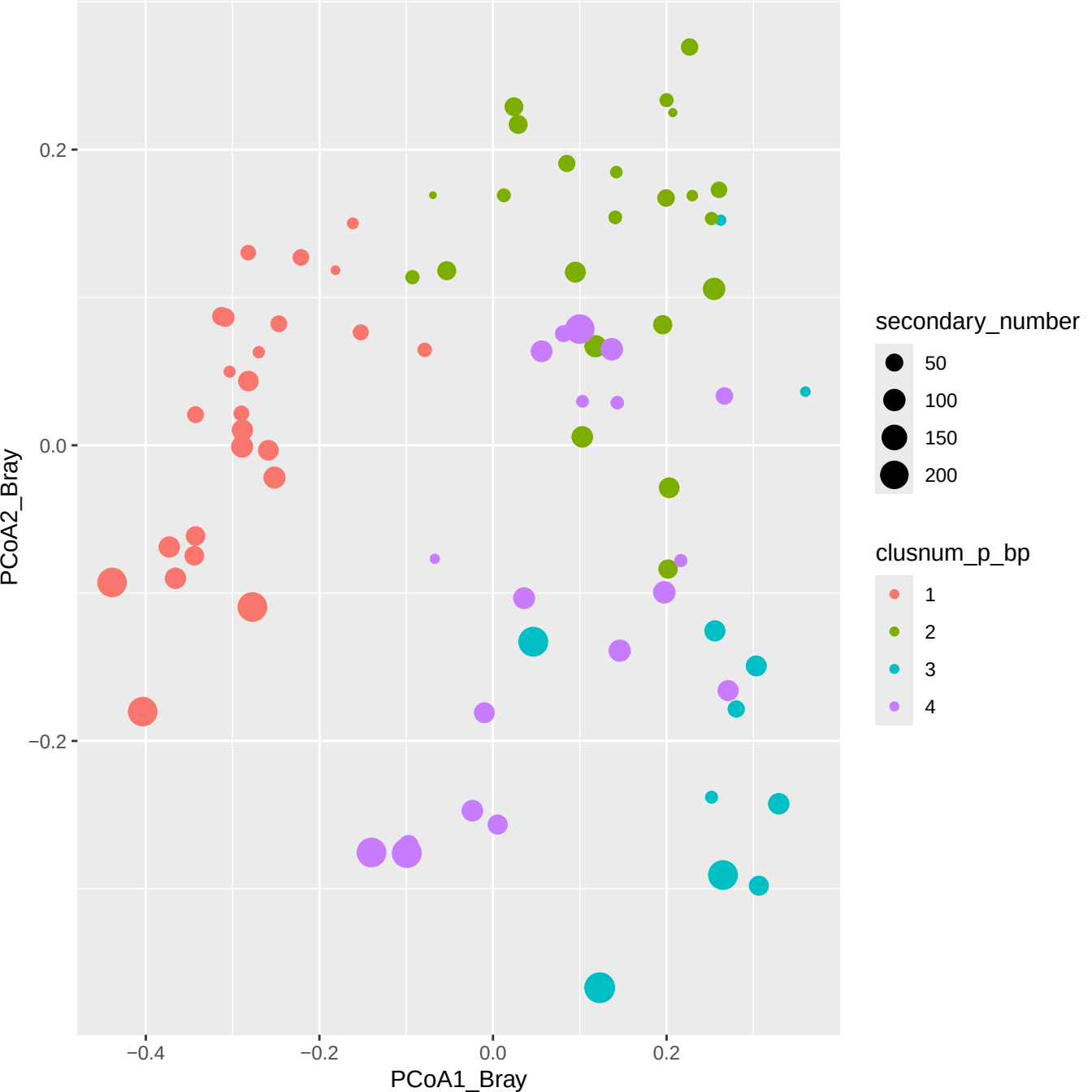
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4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−0.2
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PCoA1_Bray
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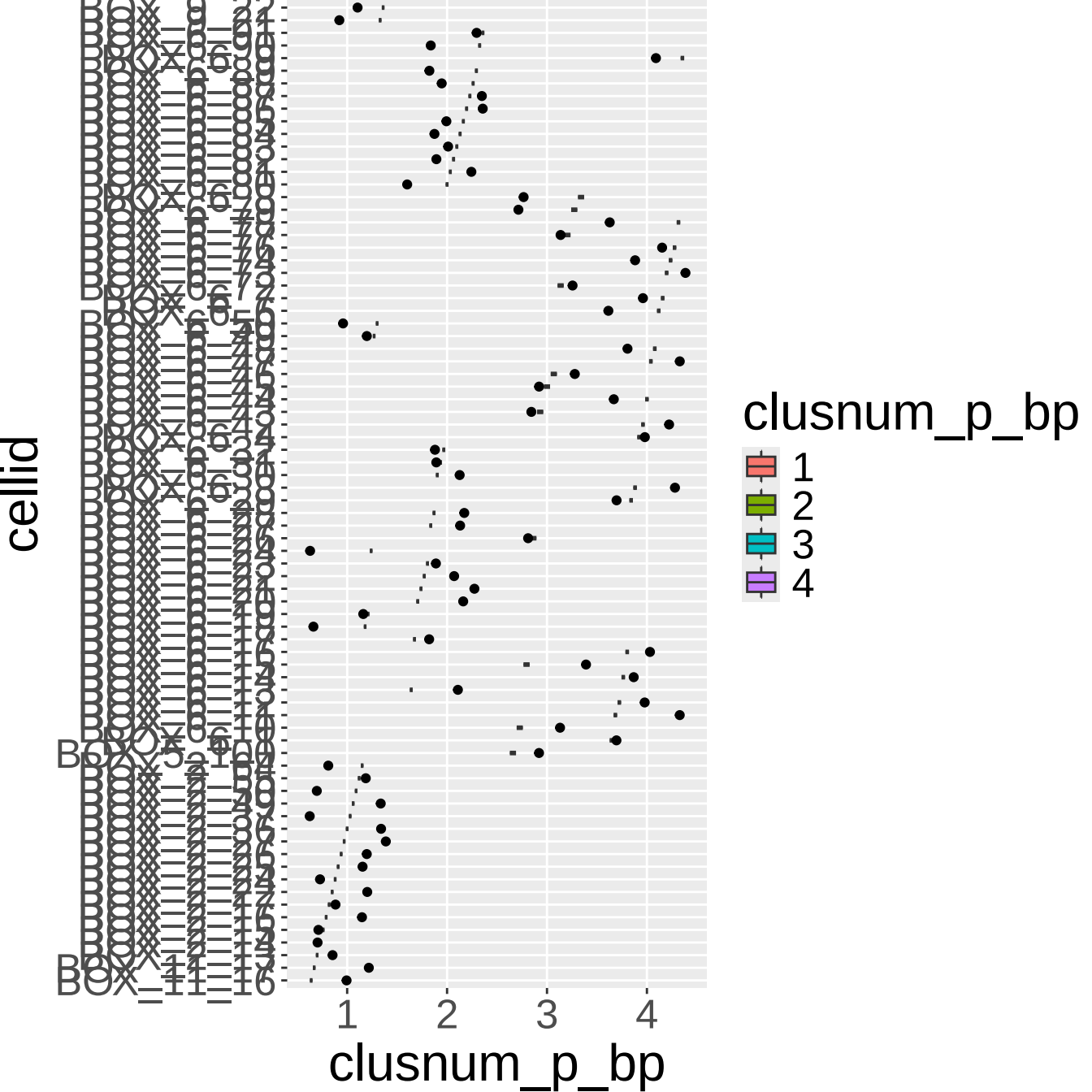
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3
4
clusnum_p_bp
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3
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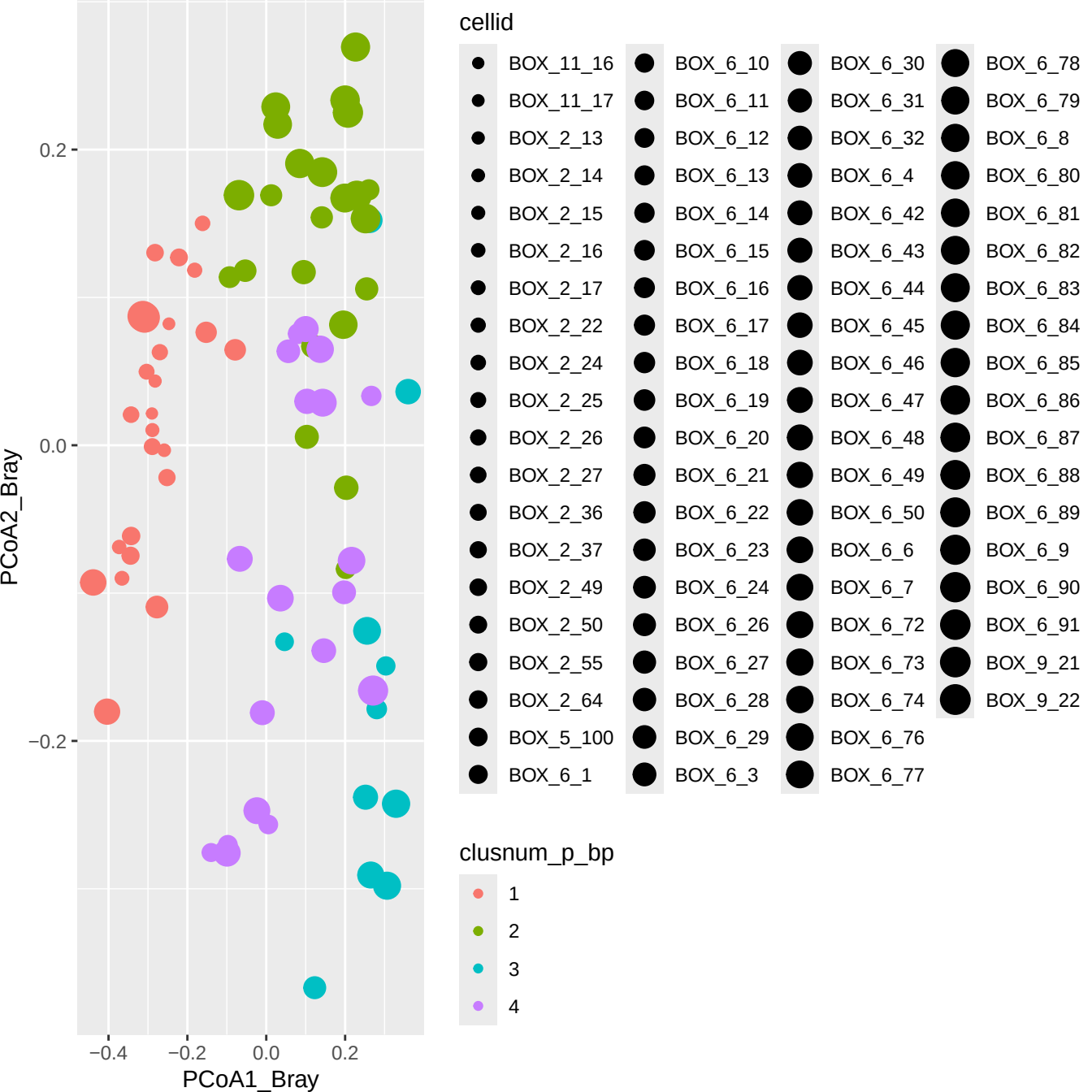
0.2
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−0.2
PCoA2_Bray
−0.4
−0.2
0.0
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PCoA1_Bray
cellid
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BOX_11_17
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BOX_2_14
BOX_2_15
BOX_2_16
BOX_2_17
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3
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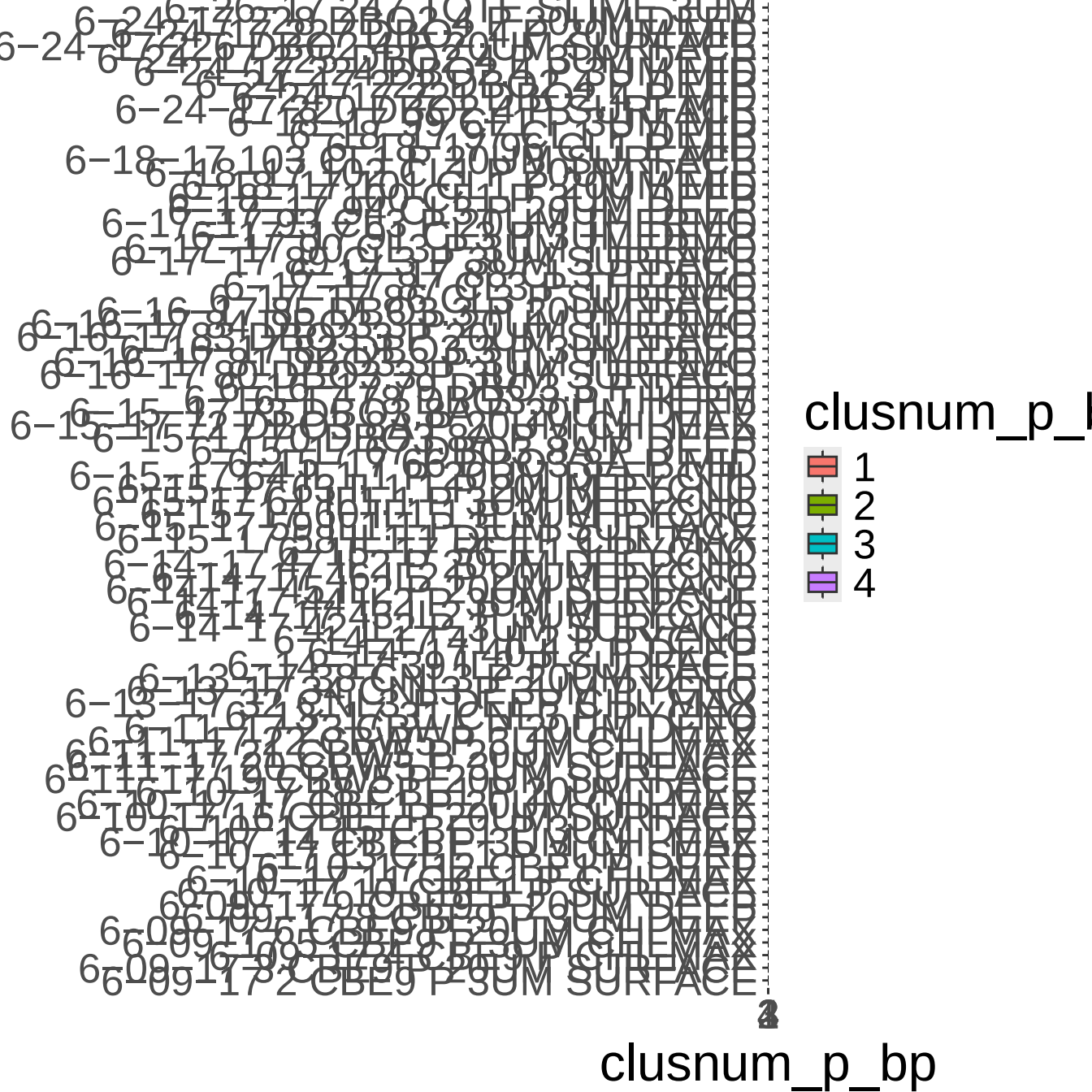
clusnum_p_bp
clusnum_p_
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3
4

P 3UM SURFACE
P 20UM SURFACE
P CHL MAX
P 3UM CHLMAX
P 20UM CHLMAX
P 3UM DEEP
P 20UM DEEP
P SURFACE
P CHL MAX
P DEEP
P 3UM SURF
P 3UM CHLMAX
P 3UM DEEP
P 20UM SURFACE
P 20UM CHLMAX
P 20UM DEEP
5 P 20UM SURFACE
5 P 3UM SURFACE
5 P 20UM CHLMAX
5 P 3UM CHLMAX

6−11−17 23 CBW5 P 20UM DEEP
6−13−17 31 CNL3 P PYCNO
6−13−17 32 CNL3 P DEEP CHL MAX
6−13−17 34 CNL3 P 3UM PYCNO
6−13−17 38 CNL3 P 20UM DEEP
6−14−17 39 IL2 P SURFACE
6−14−17 40 IL2 P DEEP
6−14−17 41 IL2 P PYCNO
6−14−17 42 IL2 P 3UM SURFACE
6−14−17 43 IL2 P 3UM PYCNO
6−14−17 44 IL2 P 3UM DEEPCHL
6−14−17 45 IL2 P 20UM SURFACE
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6−15−17 56 IL1.1 PYCNO
6−15−17 58 IL1.1 DEEP CHL MAX
6−15−17 59 IL1.1 P 3UM SURFACE
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6−15−17 67 DBO3.8A P DEEP
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6−16−17 78 DBO3.3 P THERM
6−16−17 79 DBO3.3 P DEEP
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6−16−17 81 DBO3.3 P 3UM THERMO
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6−17−17 88 CL3 P DEEP
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6−17−17 90 CL3 P 3UM THERMO
6−17−17 91 CL3 P 3UM DEEP

6−17-
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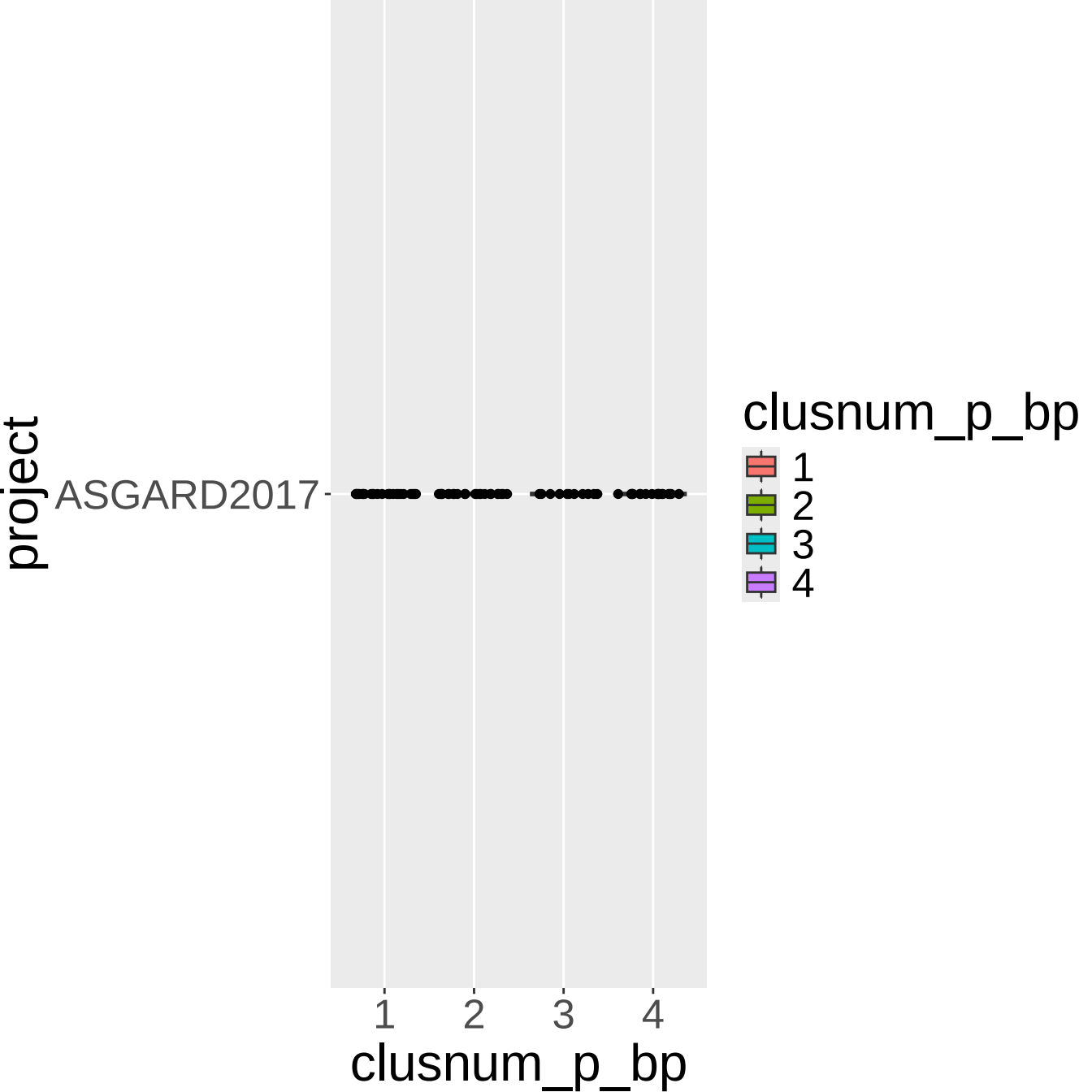
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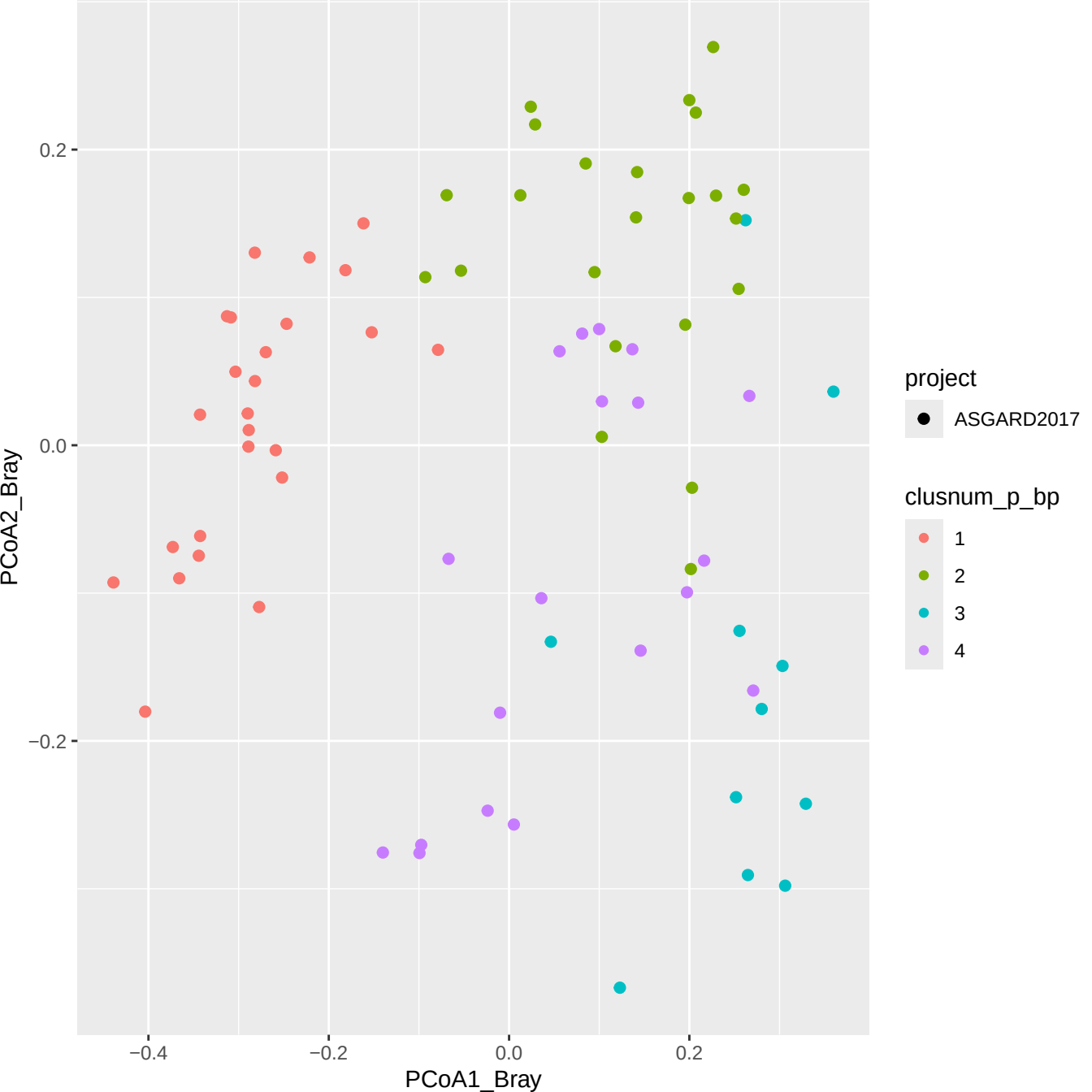
2
3
4


0.2
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PCoA1_Bray
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ASGARD2017
clusnum_p_bp
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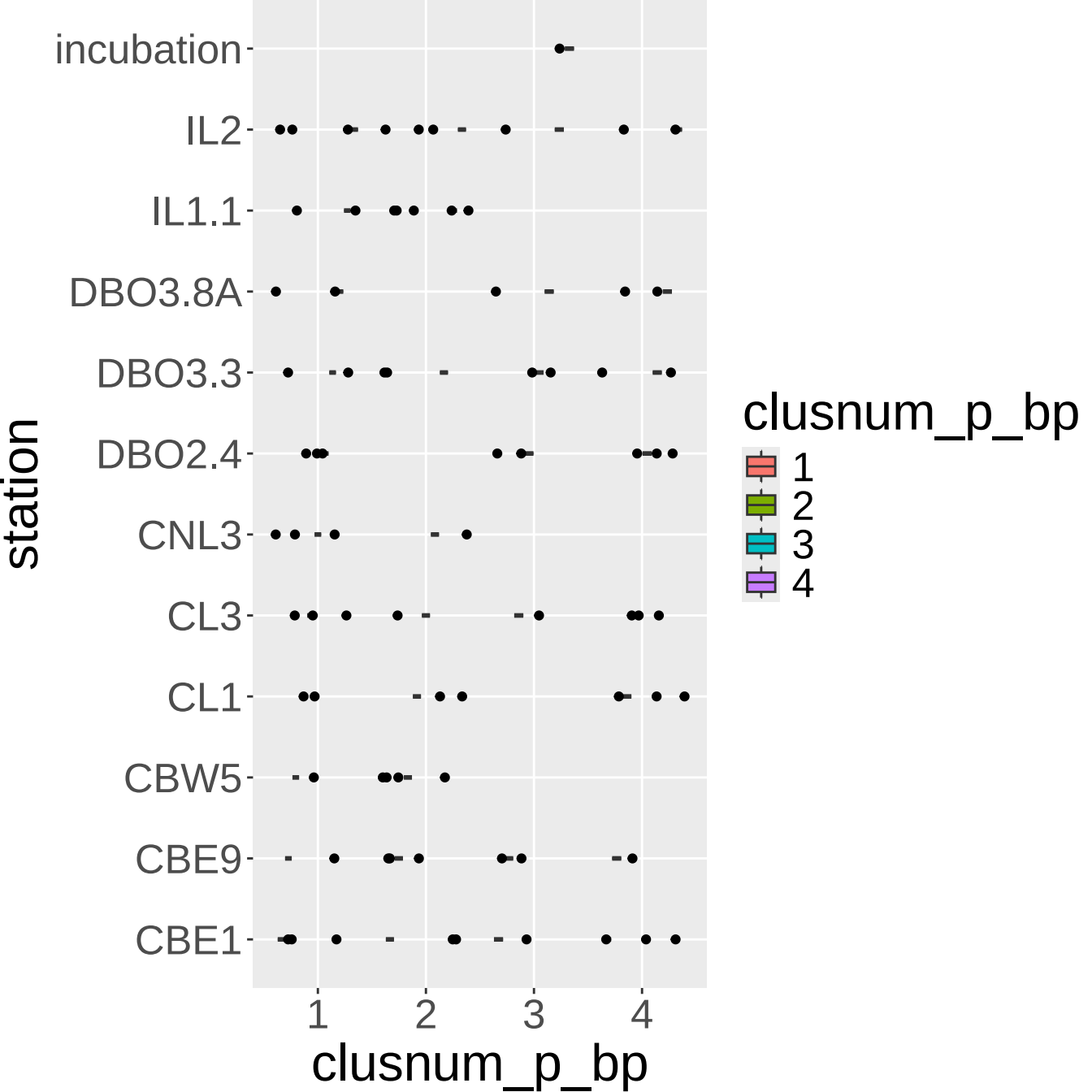
3
4


incubation
IL2
IL1.1
DBO3.8A
DBO3.3
DBO2.4
CNL3
CL3
CL1
CBW5
CBE9
CBE1
station
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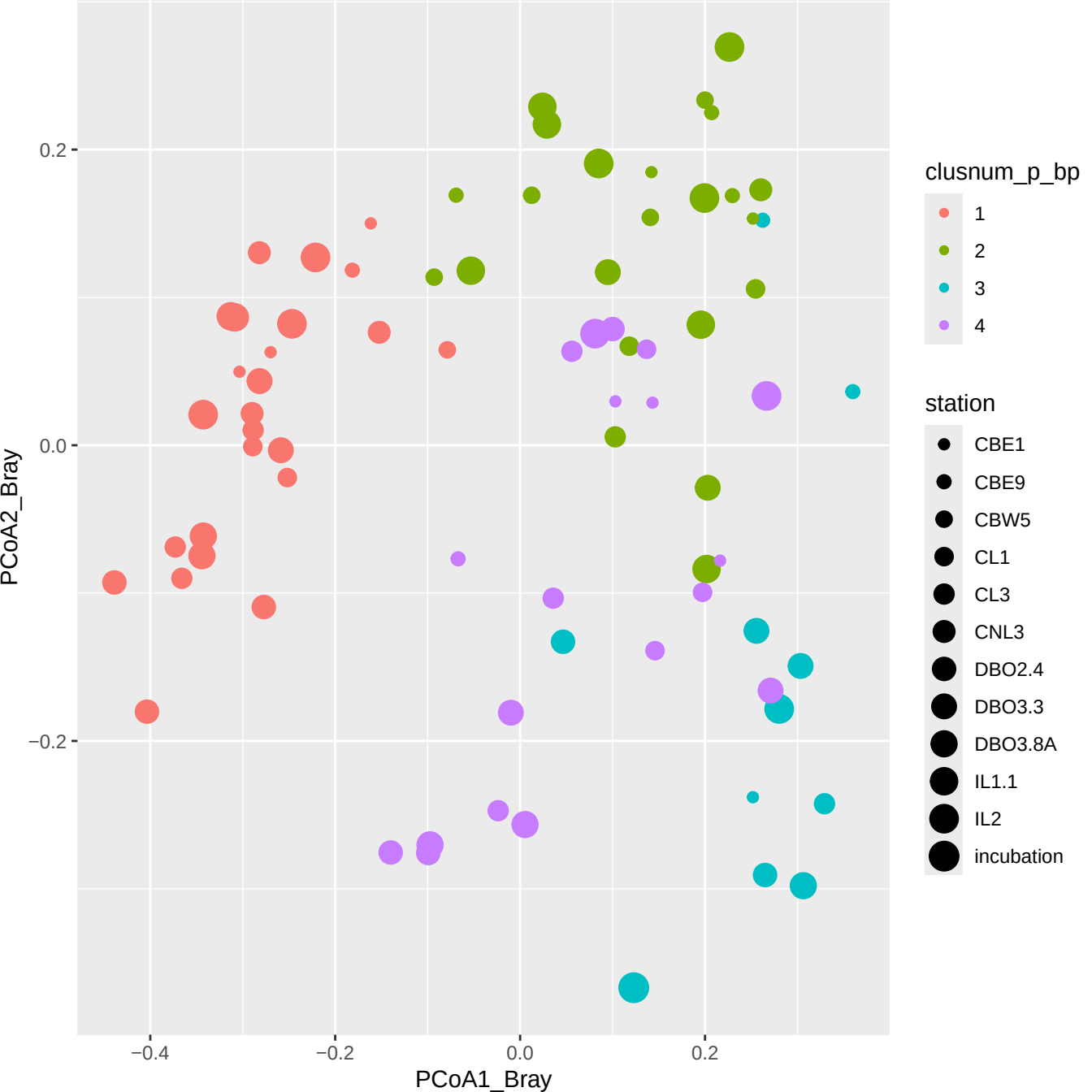
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4


0.2
0.0
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PCoA2_Bray
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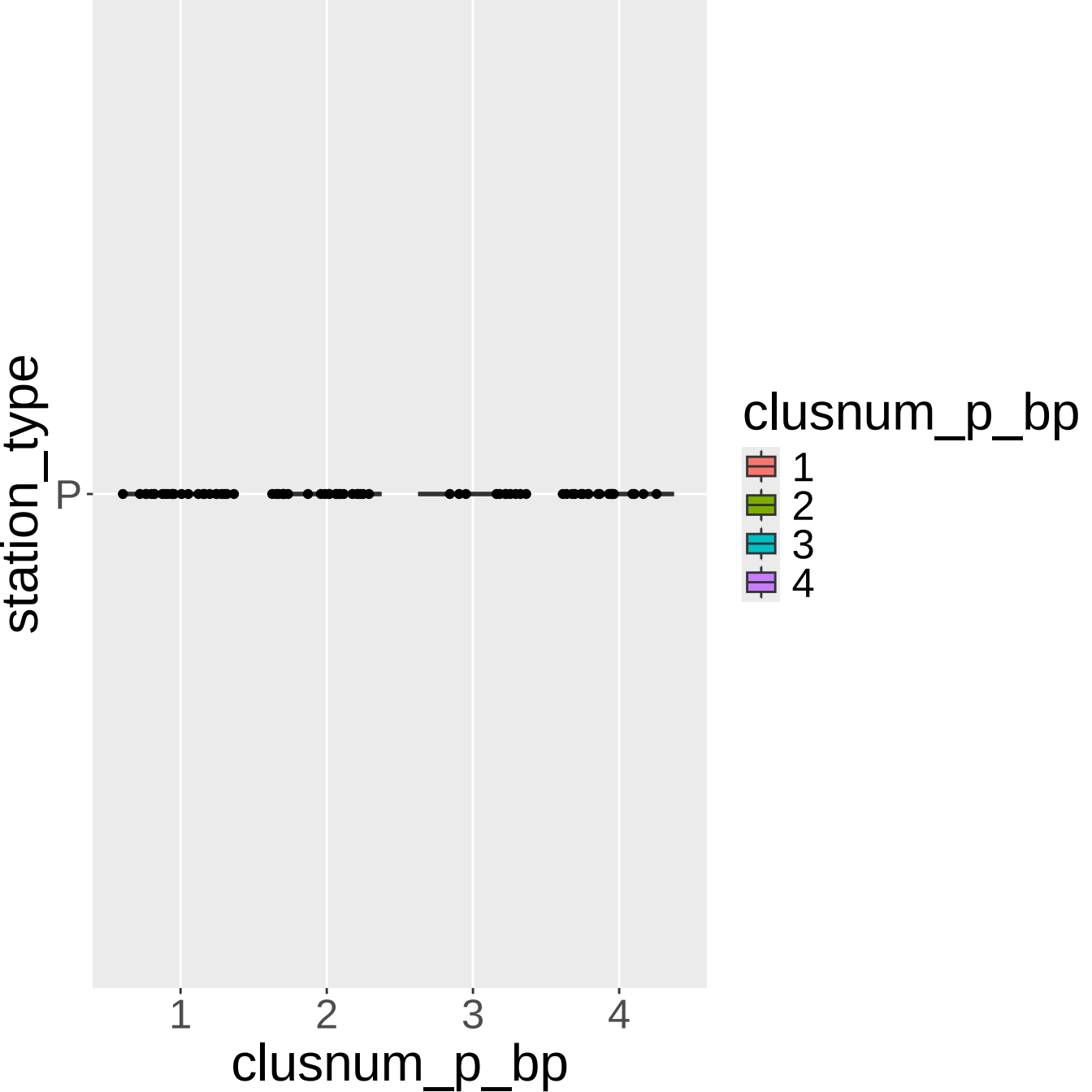
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
station
CBE1
CBE9
CBW5
CL1
CL3
CNL3
DBO2.4
DBO3.3
DBO3.8A
IL1.1
IL2
incubation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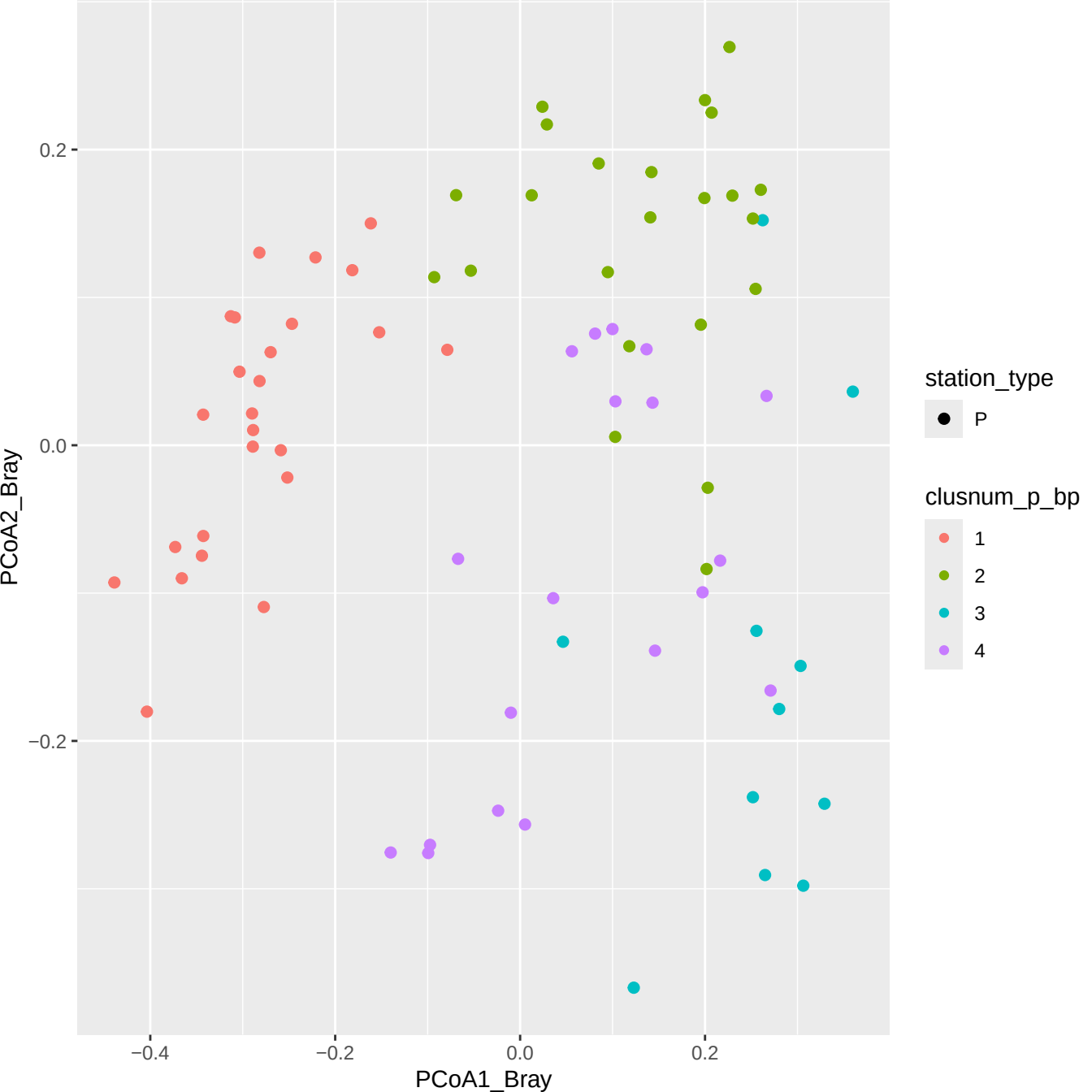
0.2
0.0
−0.2
PCoA2_Bray
−0.4
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0.0
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PCoA1_Bray
station_type
P
clusnum_p_bp
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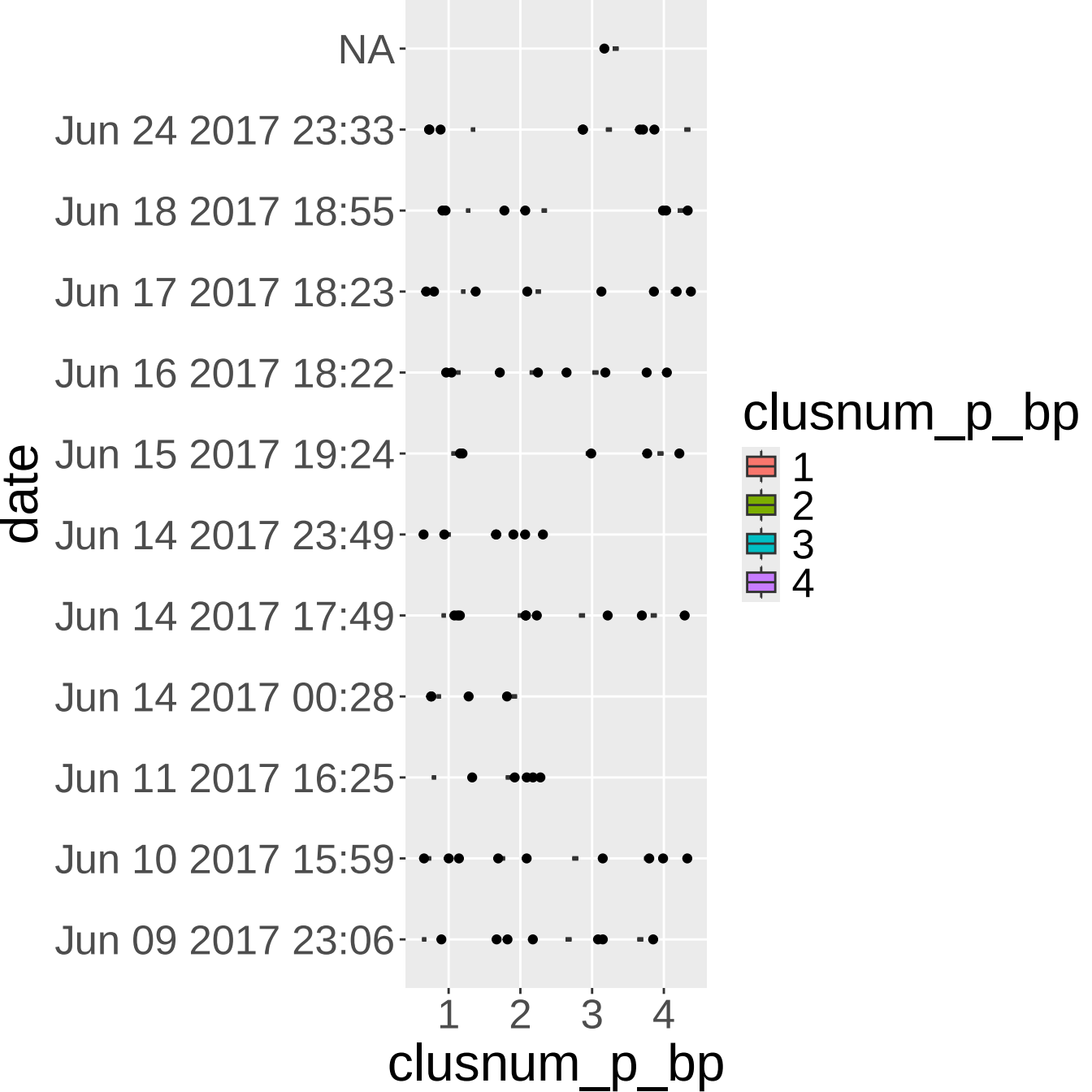
3
4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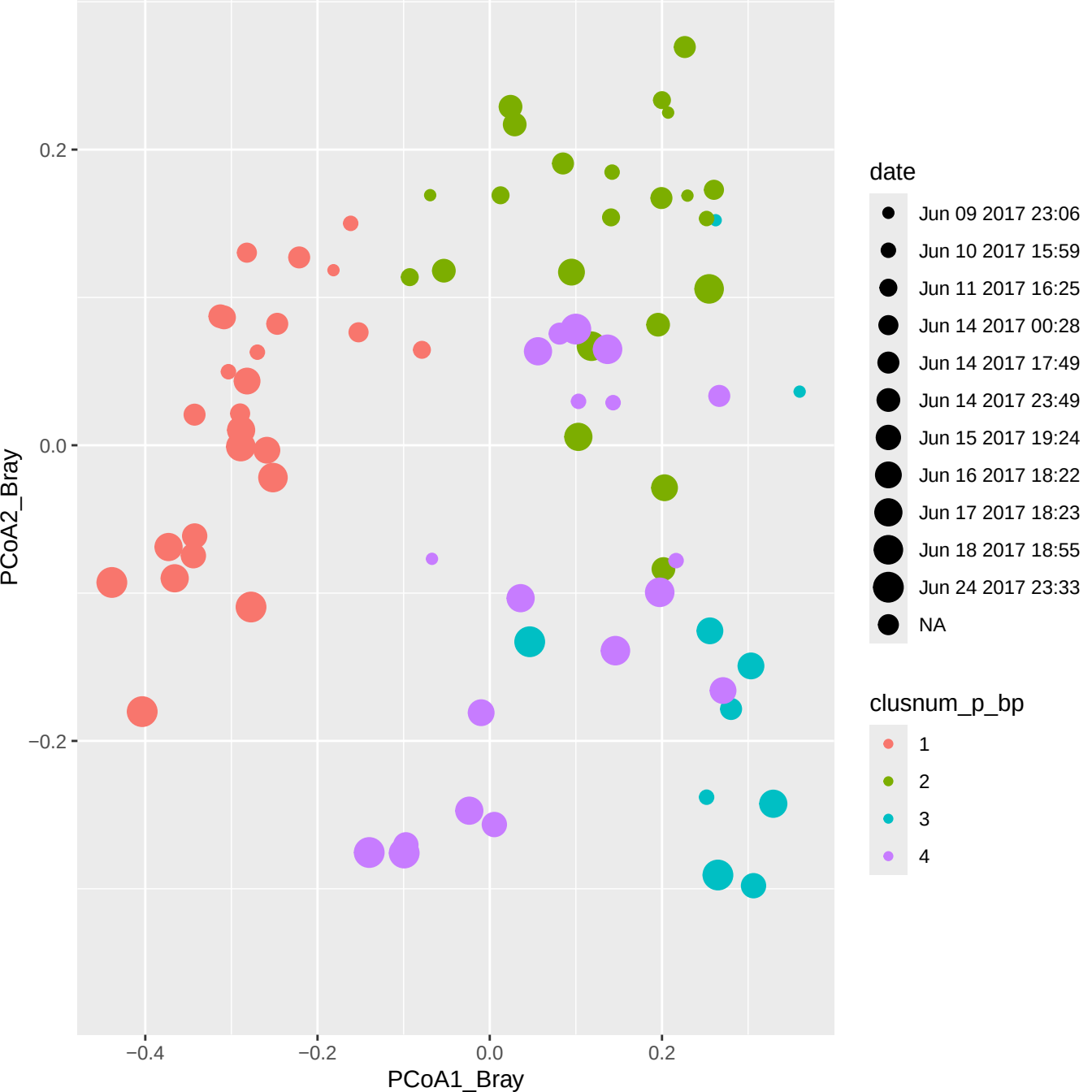
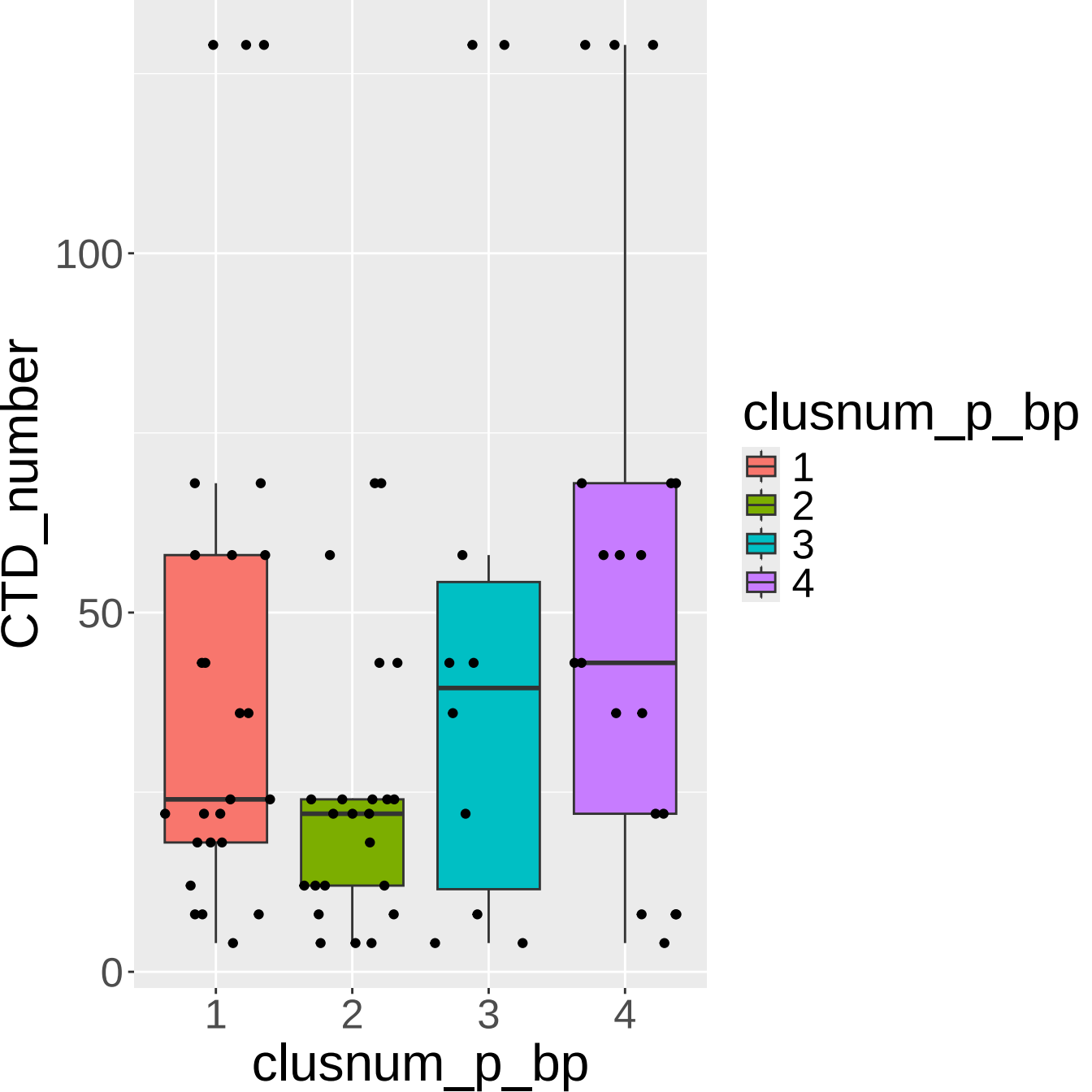
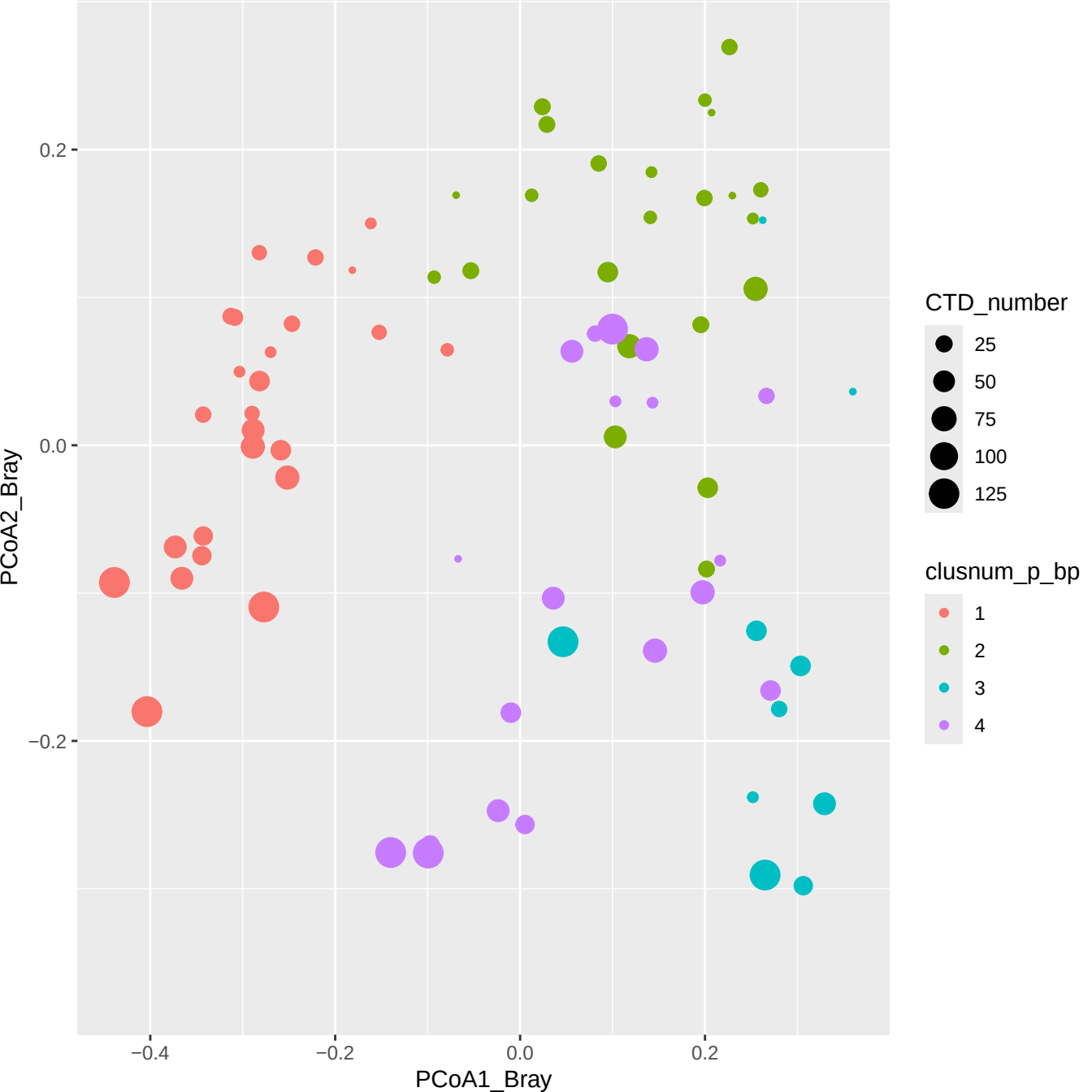
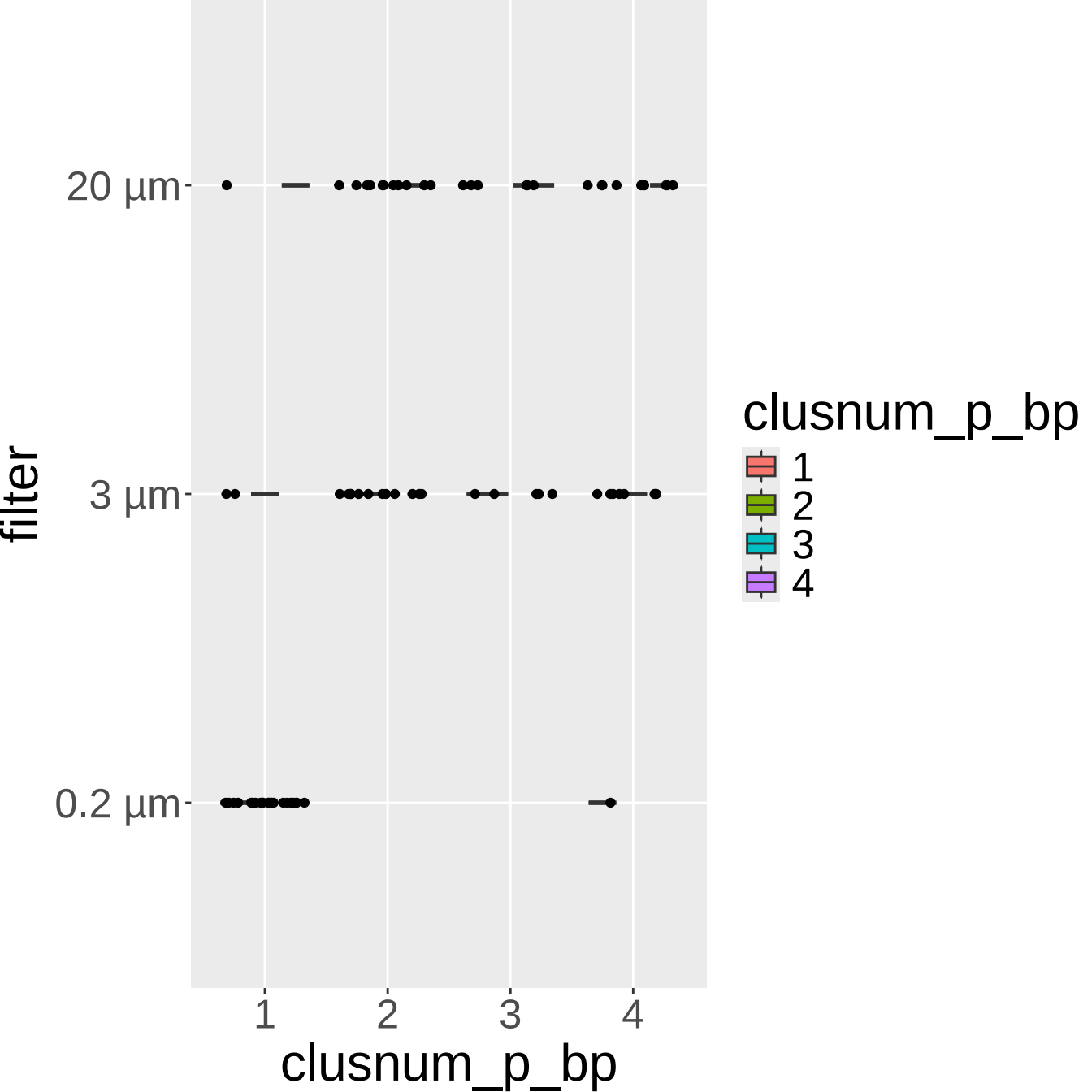
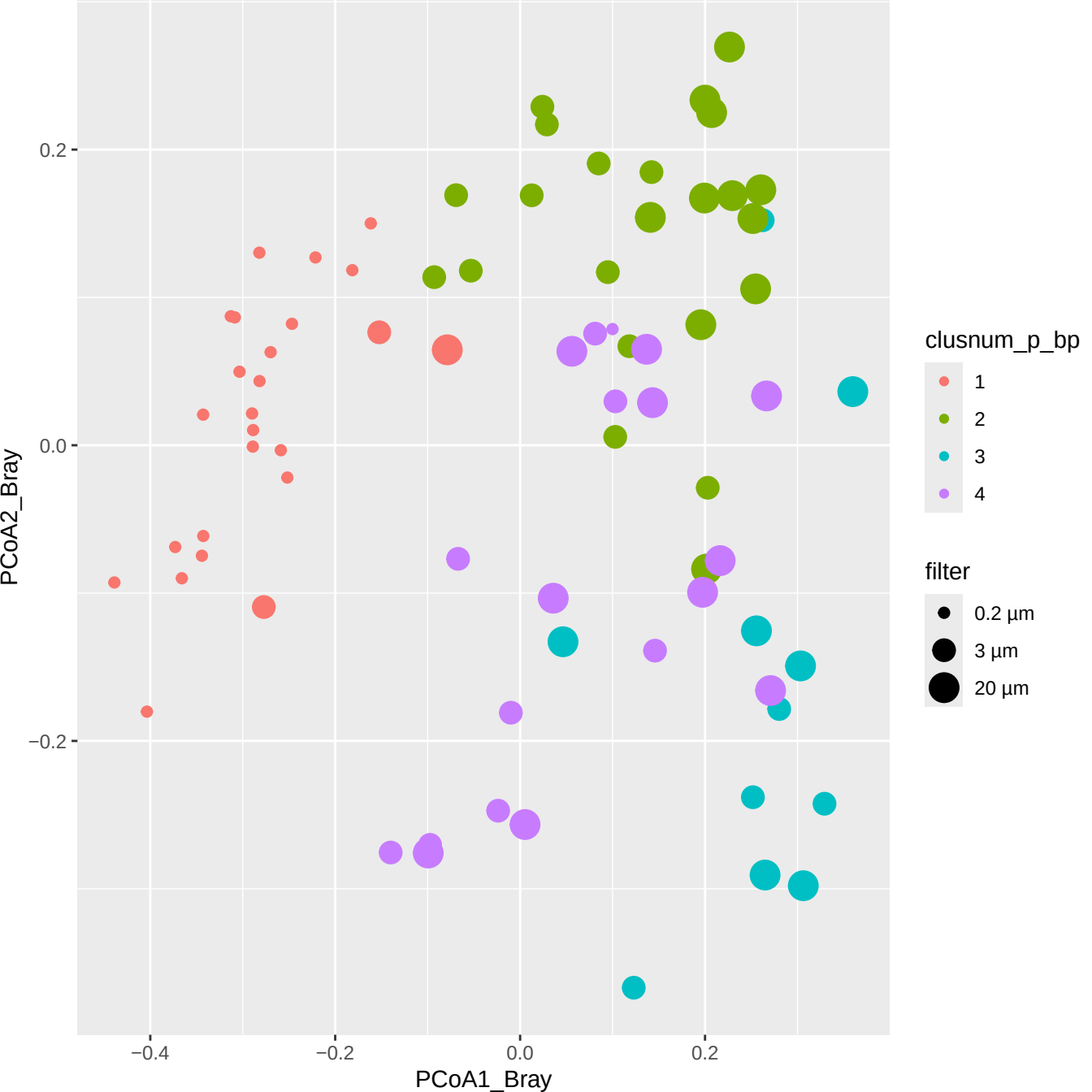
Jun 14 2017 23:49
Jun 14 2017 17:49
Jun 14 2017 00:28
Jun 11 2017 16:25
Jun 10 2017 15:59
Jun 09 2017 23:06
date
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

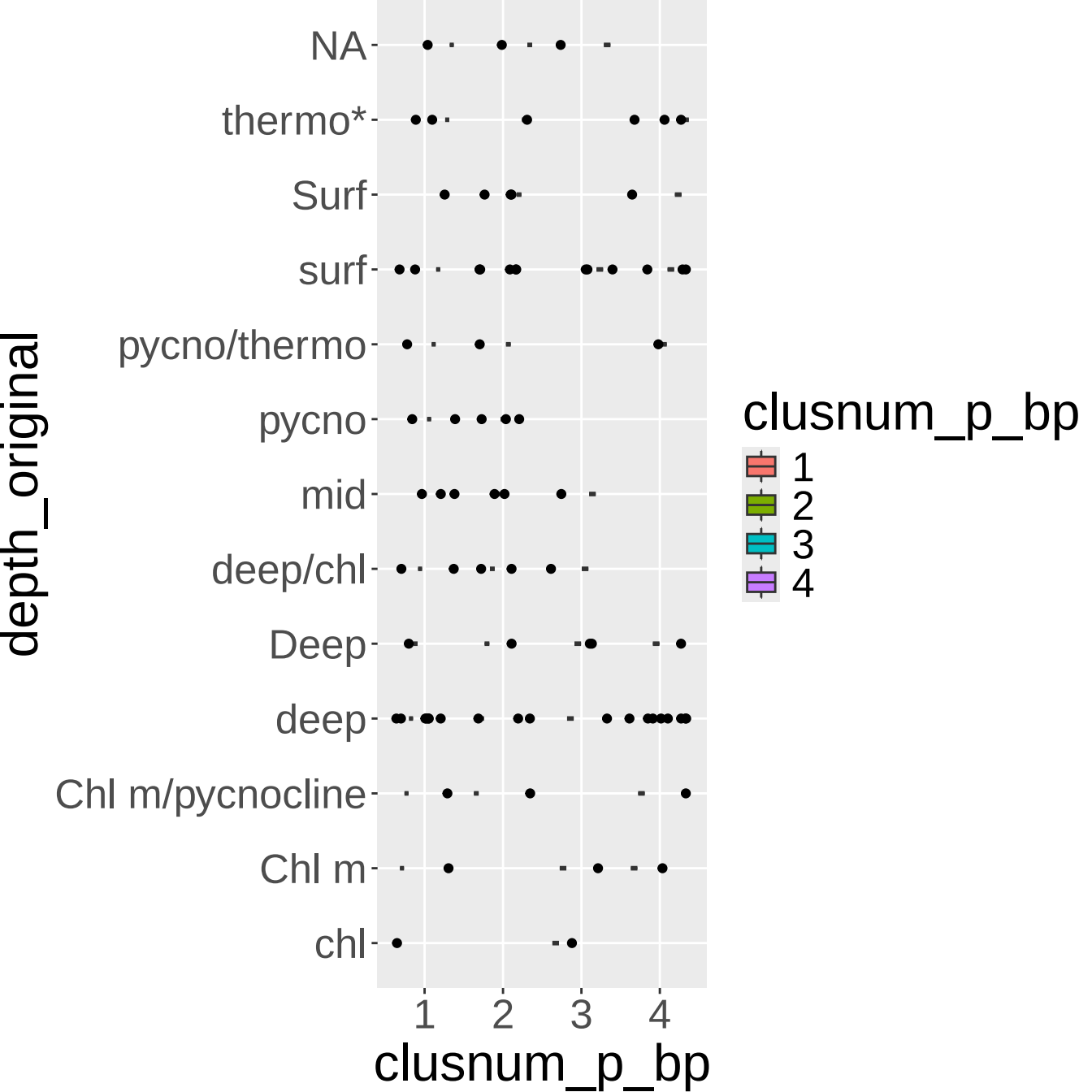

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
date
Jun 09 2017 23:06
Jun 10 2017 15:59
Jun 11 2017 16:25
Jun 14 2017 00:28
Jun 14 2017 17:49
Jun 14 2017 23:49
Jun 15 2017 19:24
Jun 16 2017 18:22
Jun 17 2017 18:23
Jun 18 2017 18:55
Jun 24 2017 23:33
NA
clusnum_p_bp
1
2
3
4

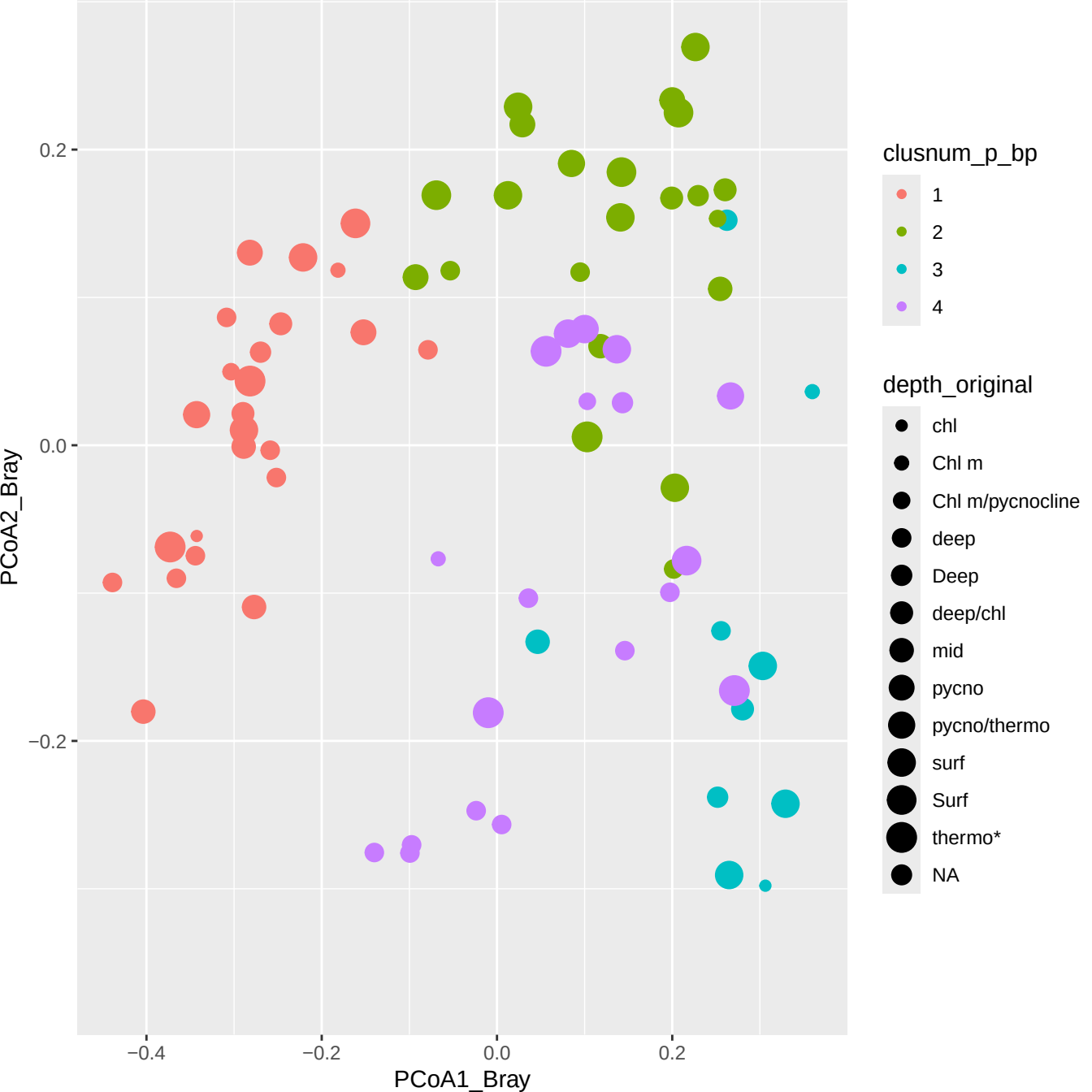

CTD_number
100
50
0
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

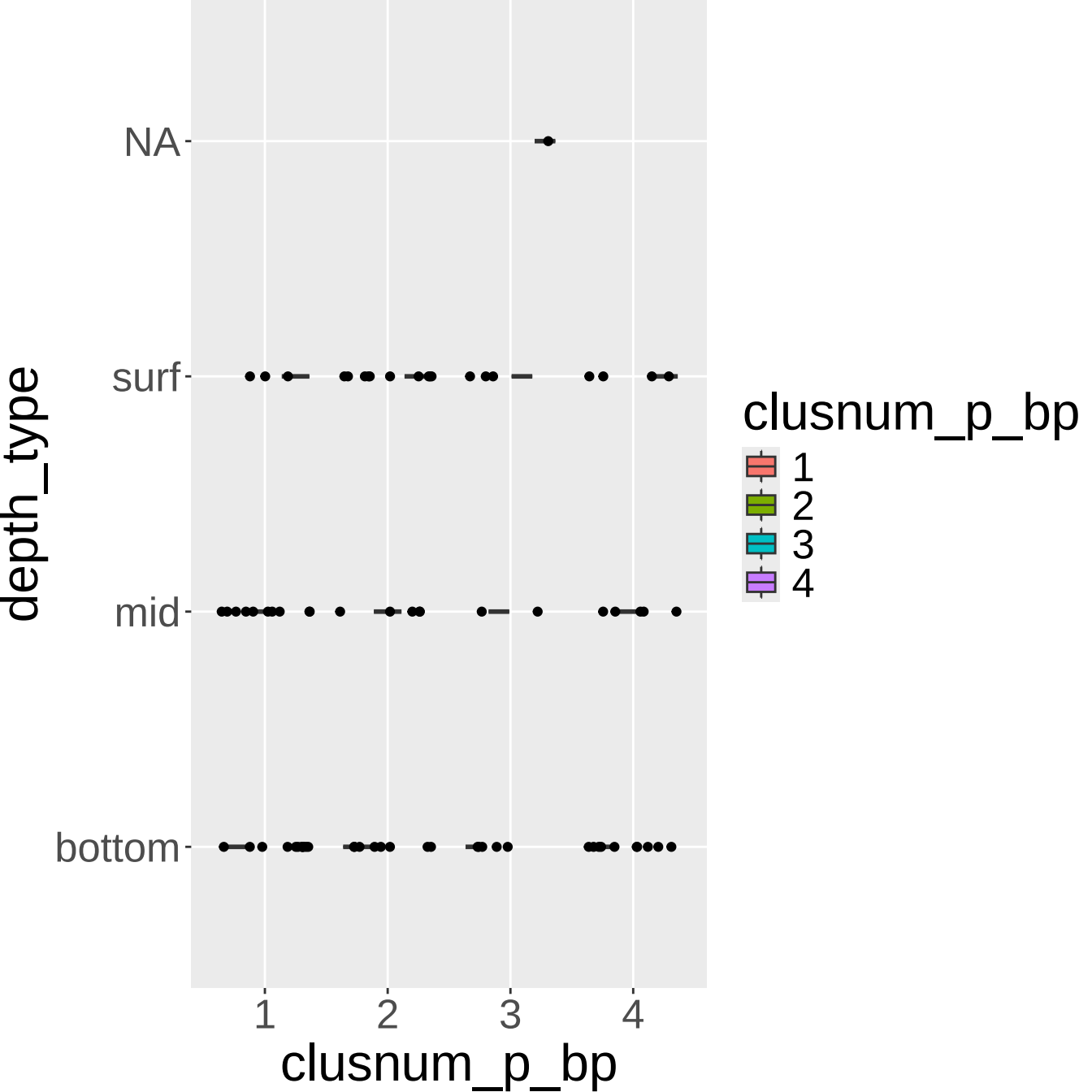

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
CTD_number
25
50
75
100
125
clusnum_p_bp
1
2
3
4

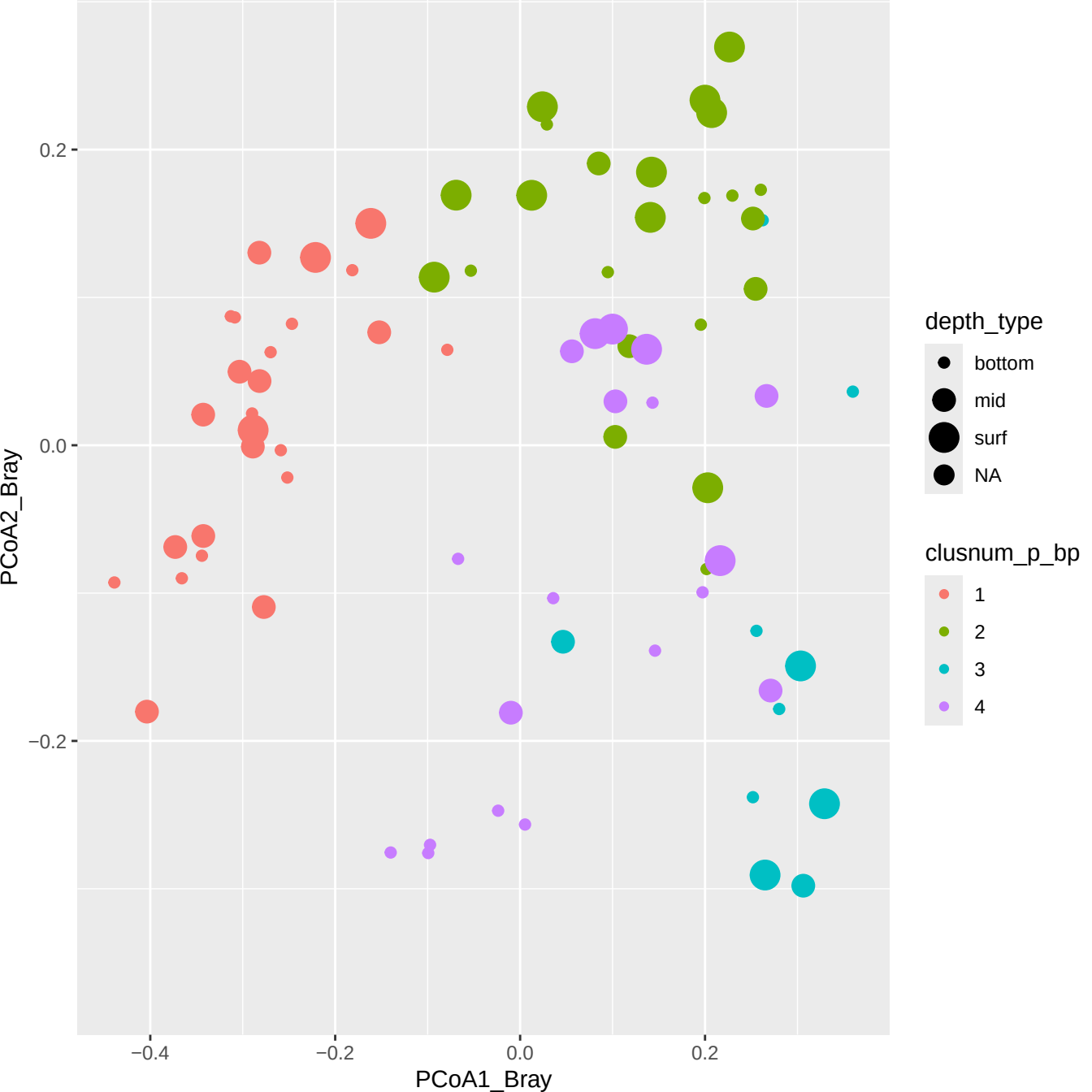

20 µm
3 µm
0.2 µm
filter
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

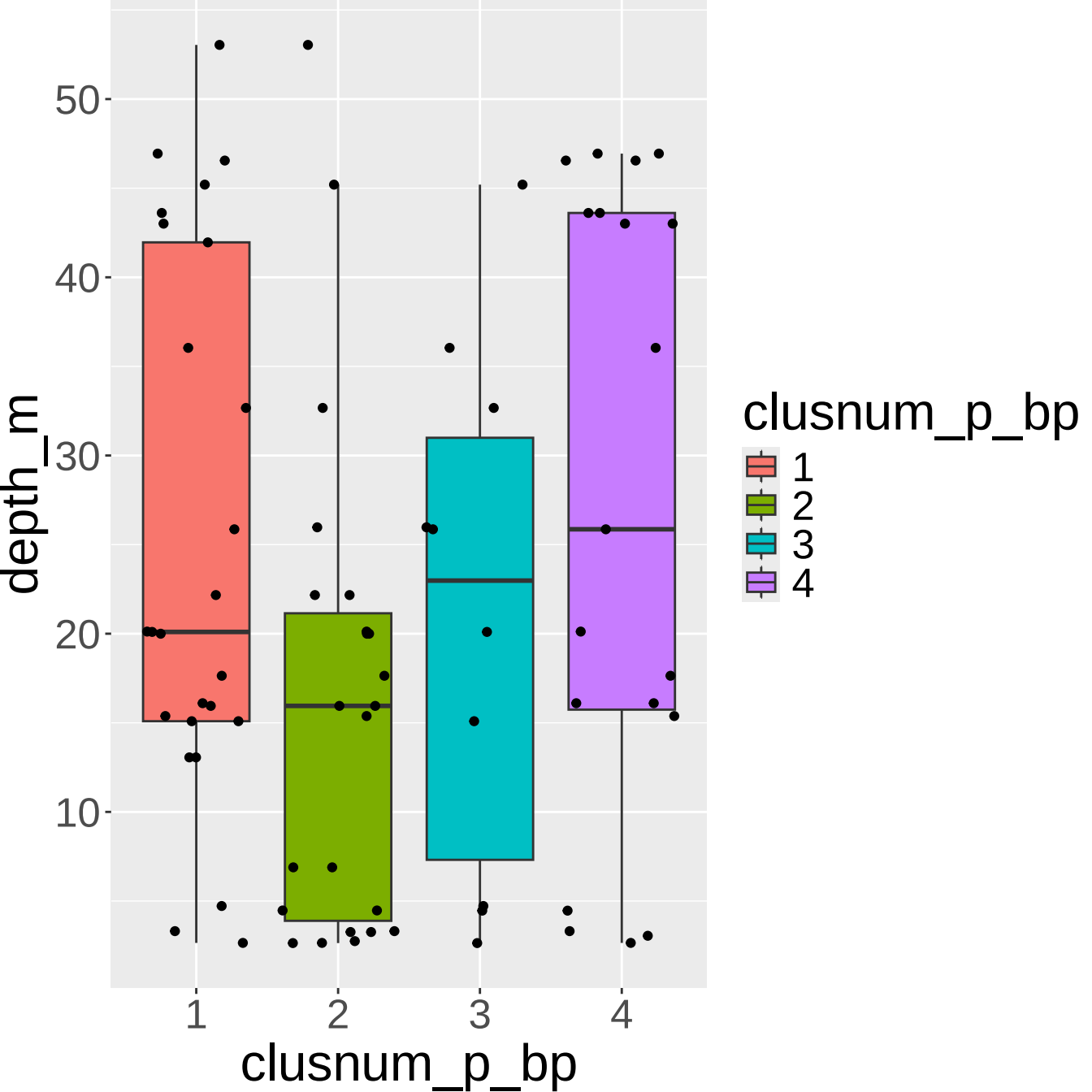

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
filter
0.2 µm
3 µm
20 µm

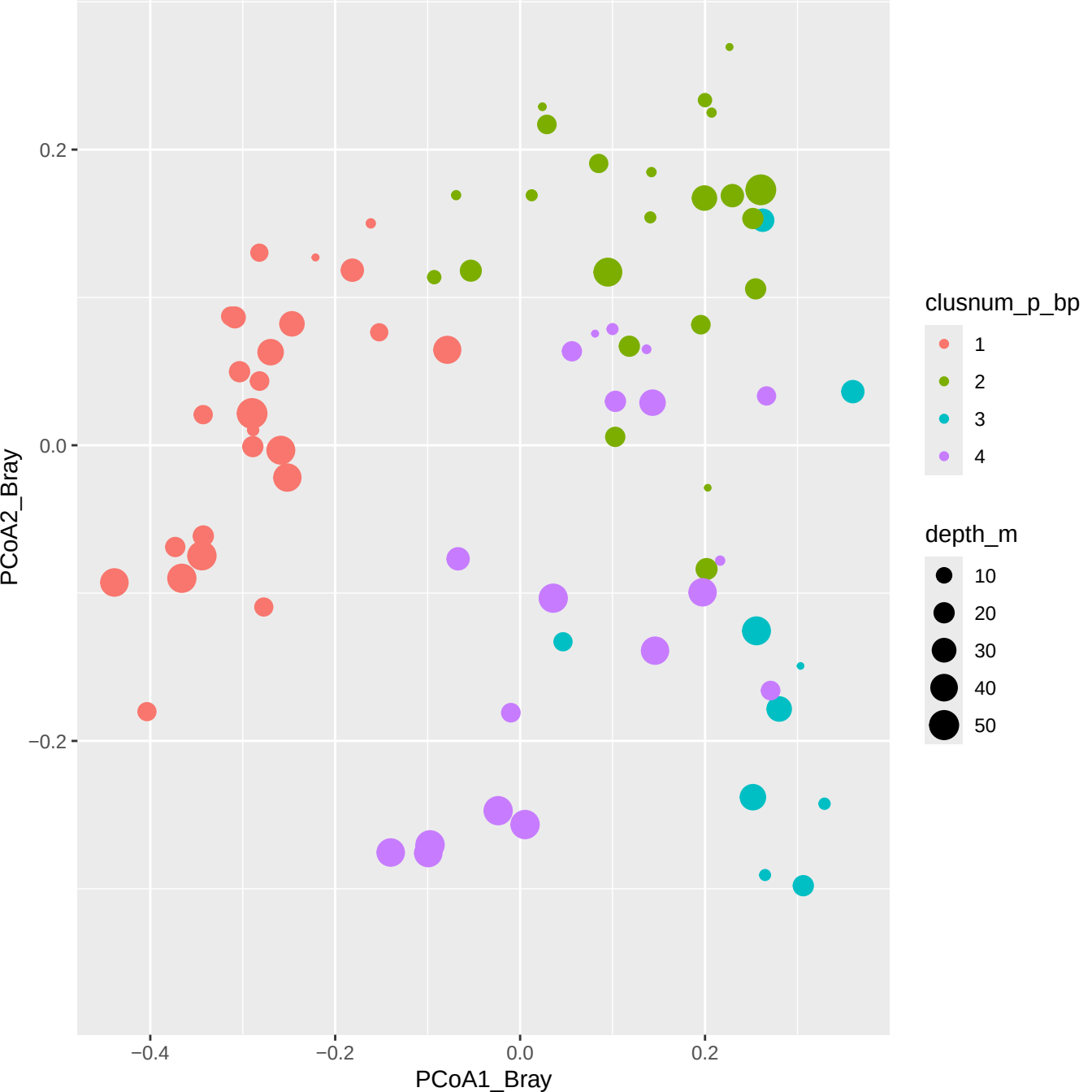

NA
thermo*
Surf
surf
pycno/thermo
pycno
mid
deep/chl
Deep
deep
Chl m/pycnocline
Chl m
chl
depth_original
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

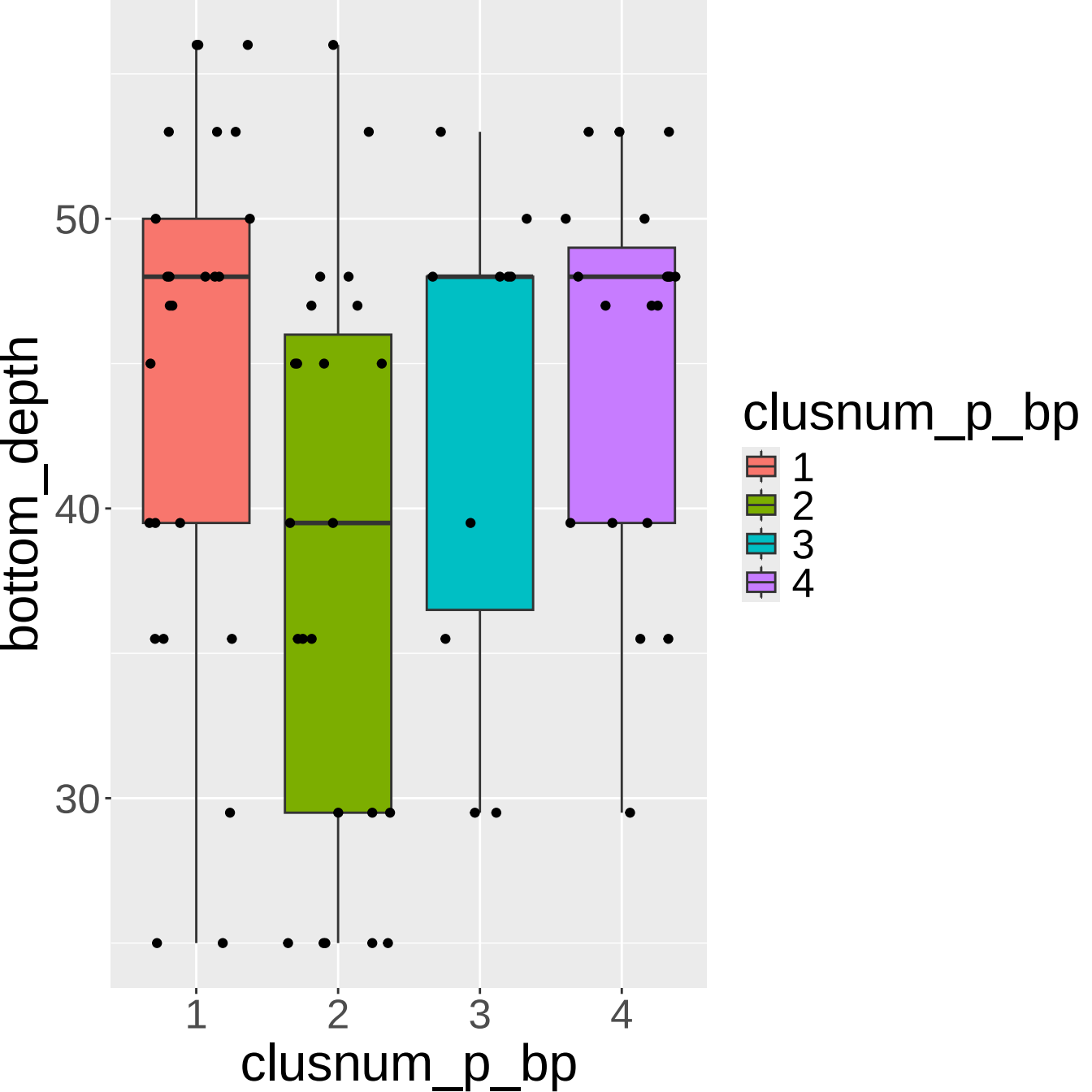

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
depth_original
chl
Chl m
Chl m/pycnocline
deep
Deep
deep/chl
mid
pycno
pycno/thermo
surf
Surf
thermo*
NA

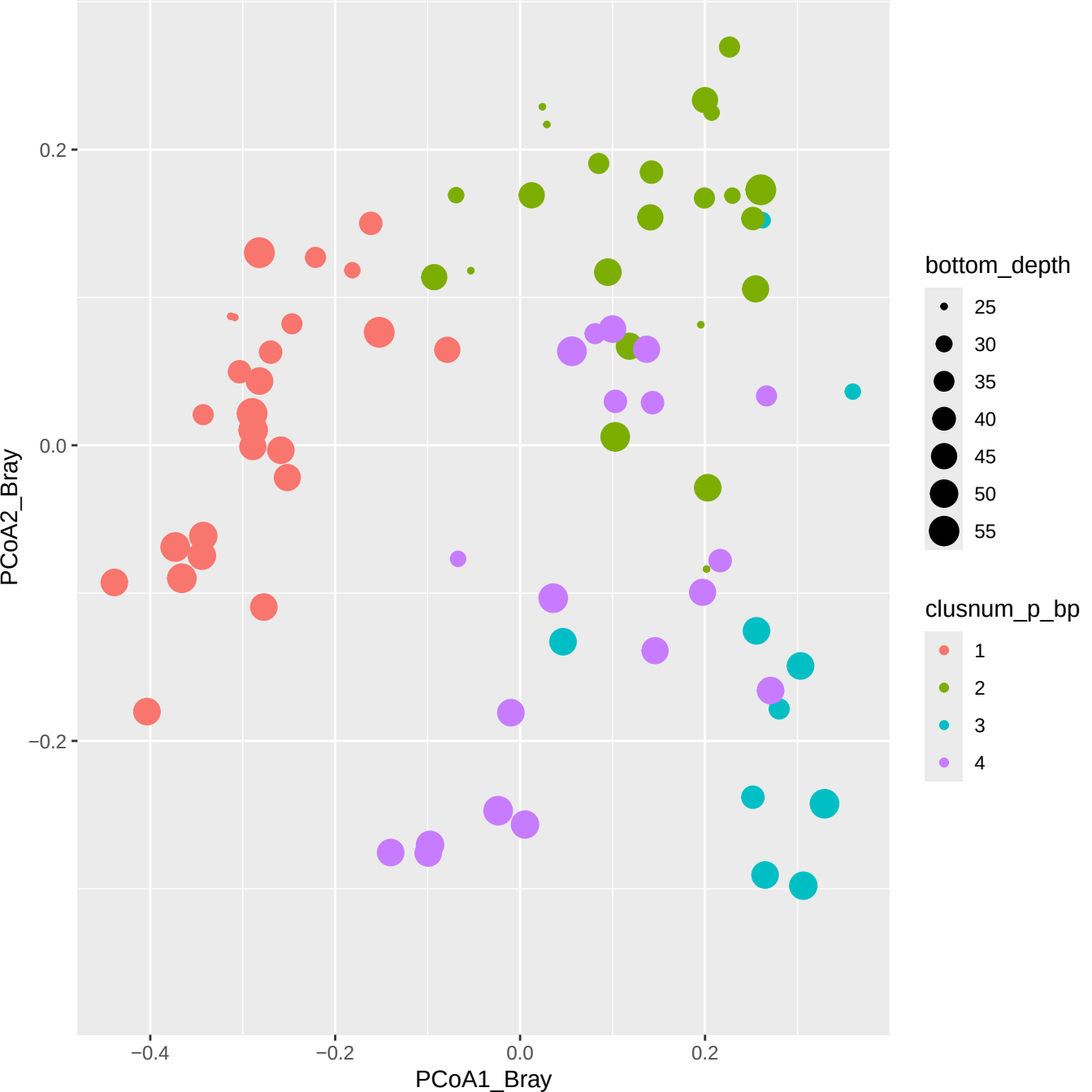

NA
surf
mid
bottom
depth_type
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

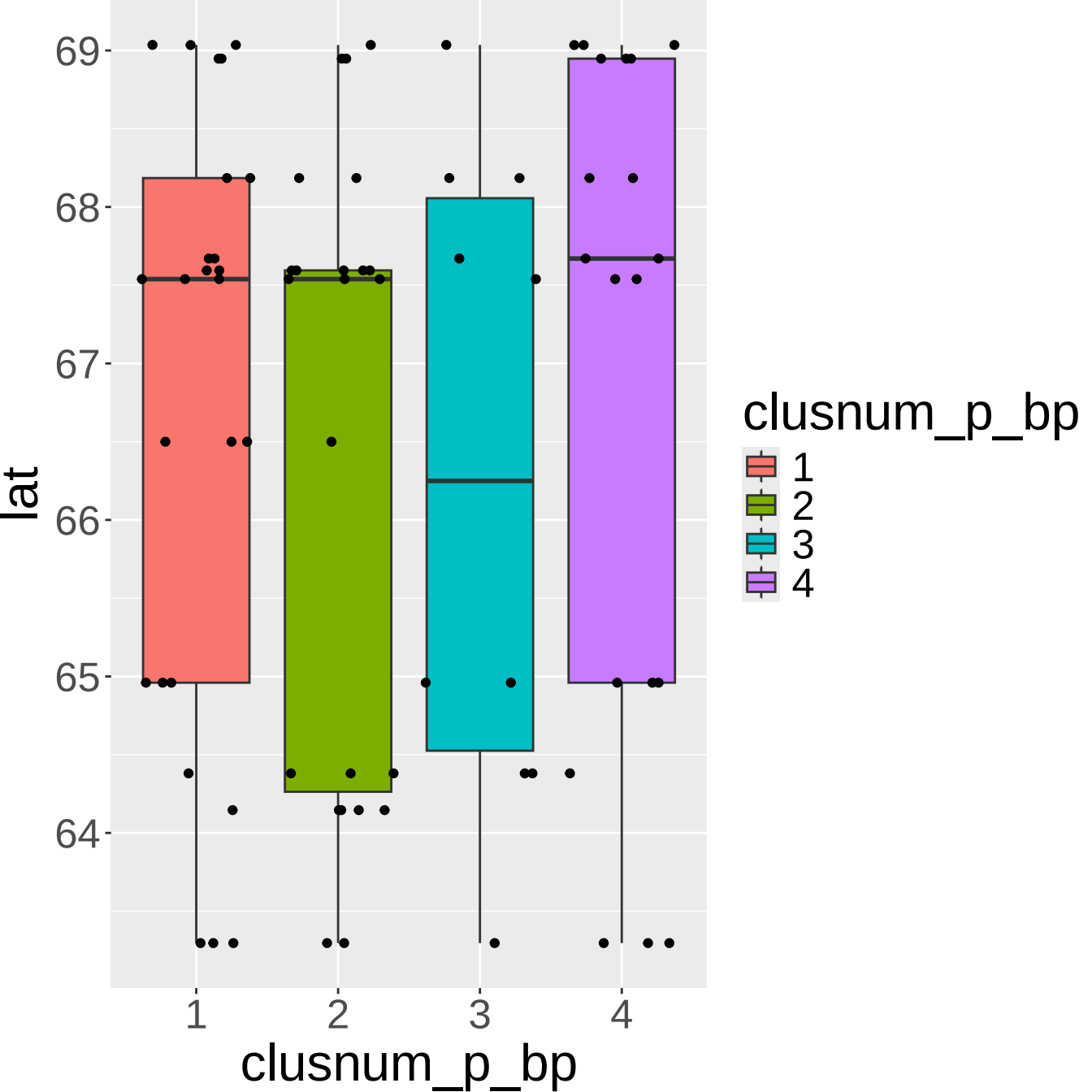

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
depth_type
bottom
mid
surf
NA
clusnum_p_bp
1
2
3
4

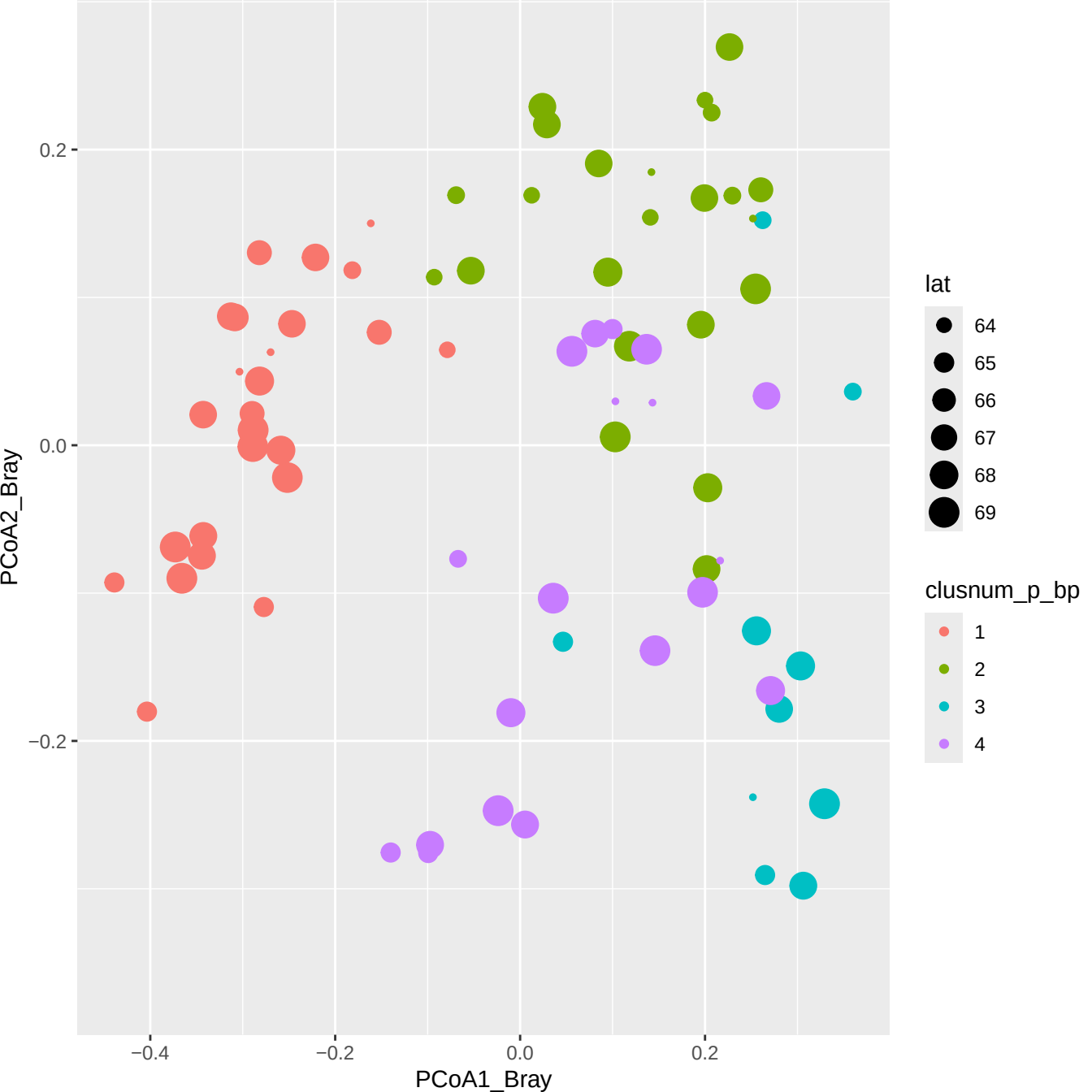

depth_m
50
40
30
20
10
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

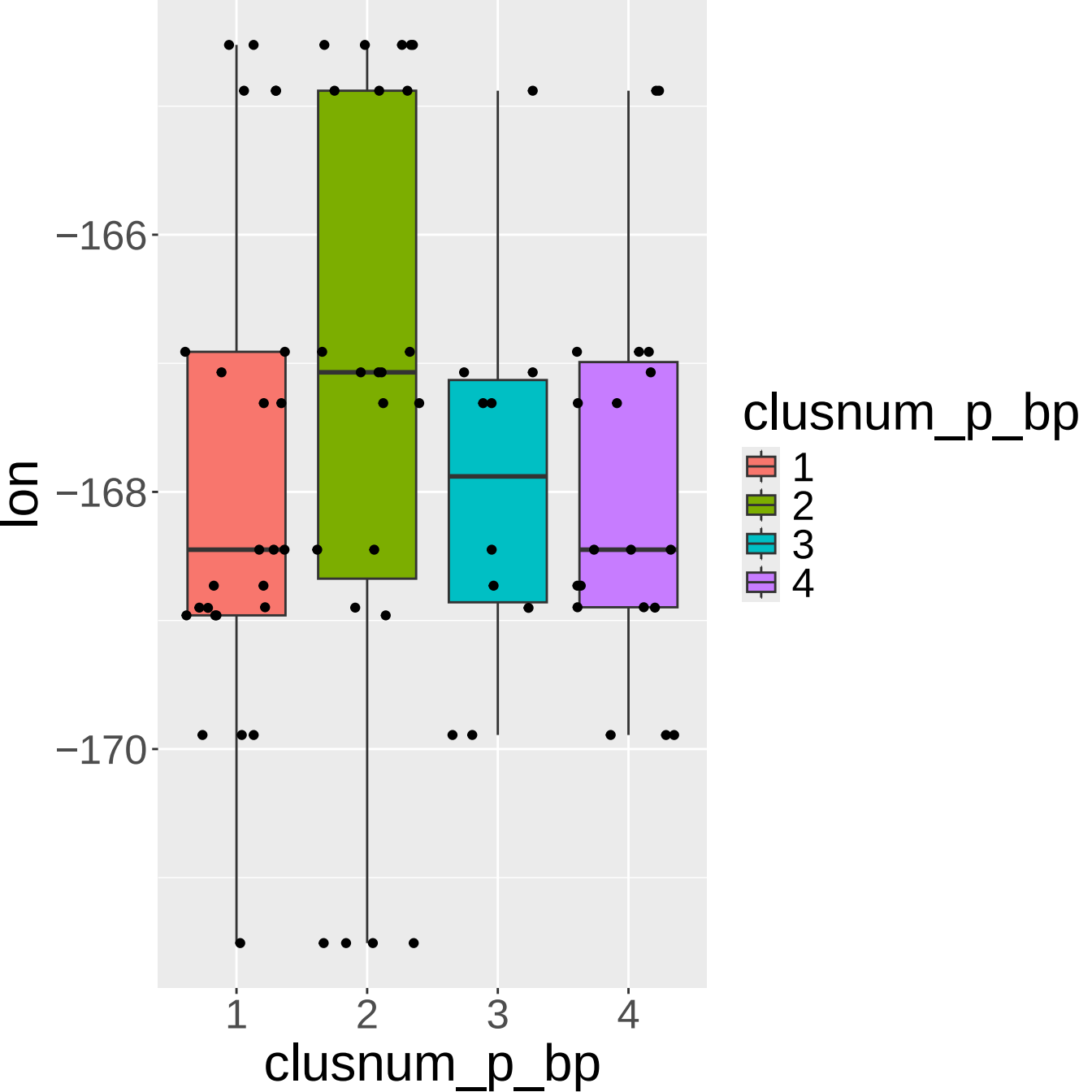

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
depth_m
10
20
30
40
50

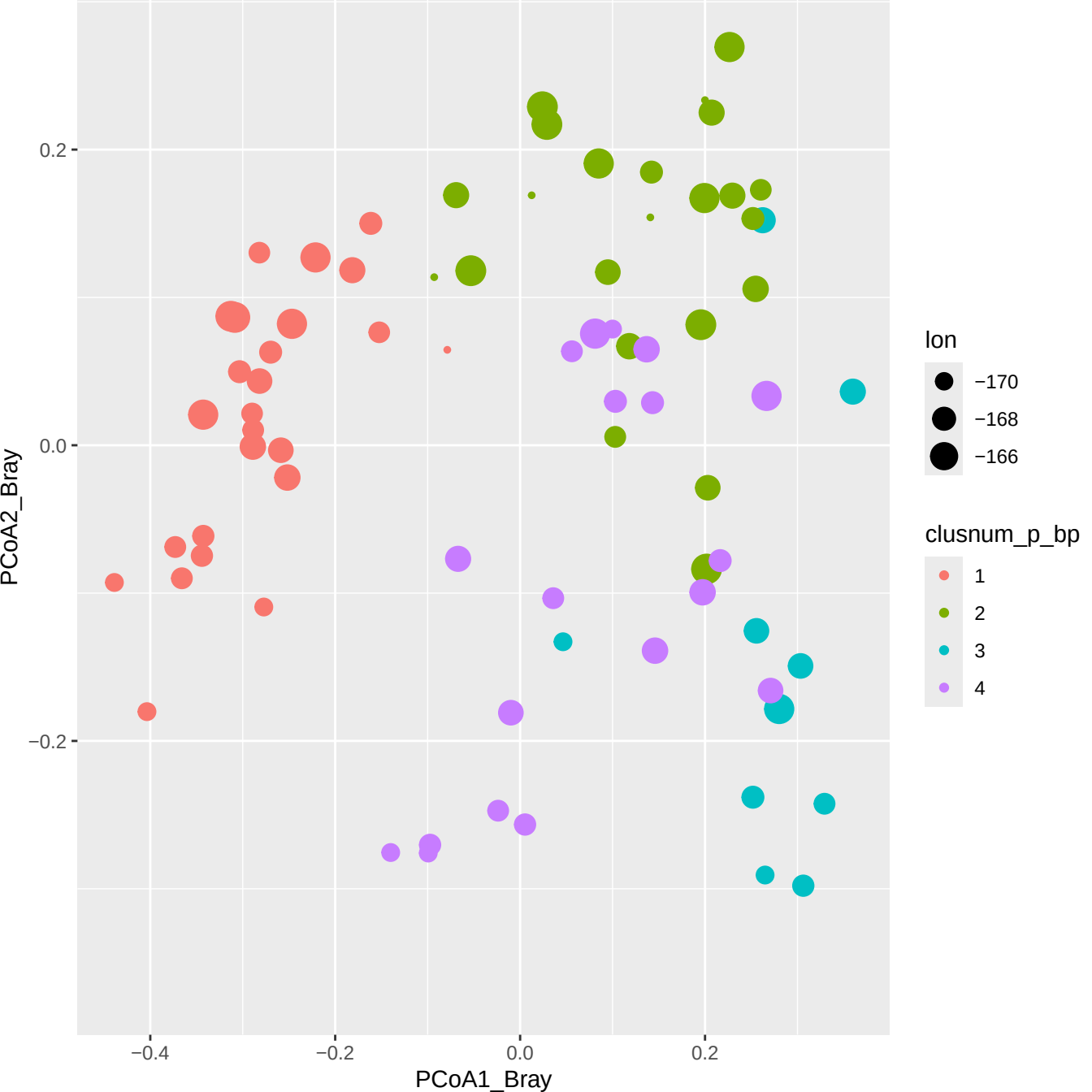

bottom_depth
50
40
30
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

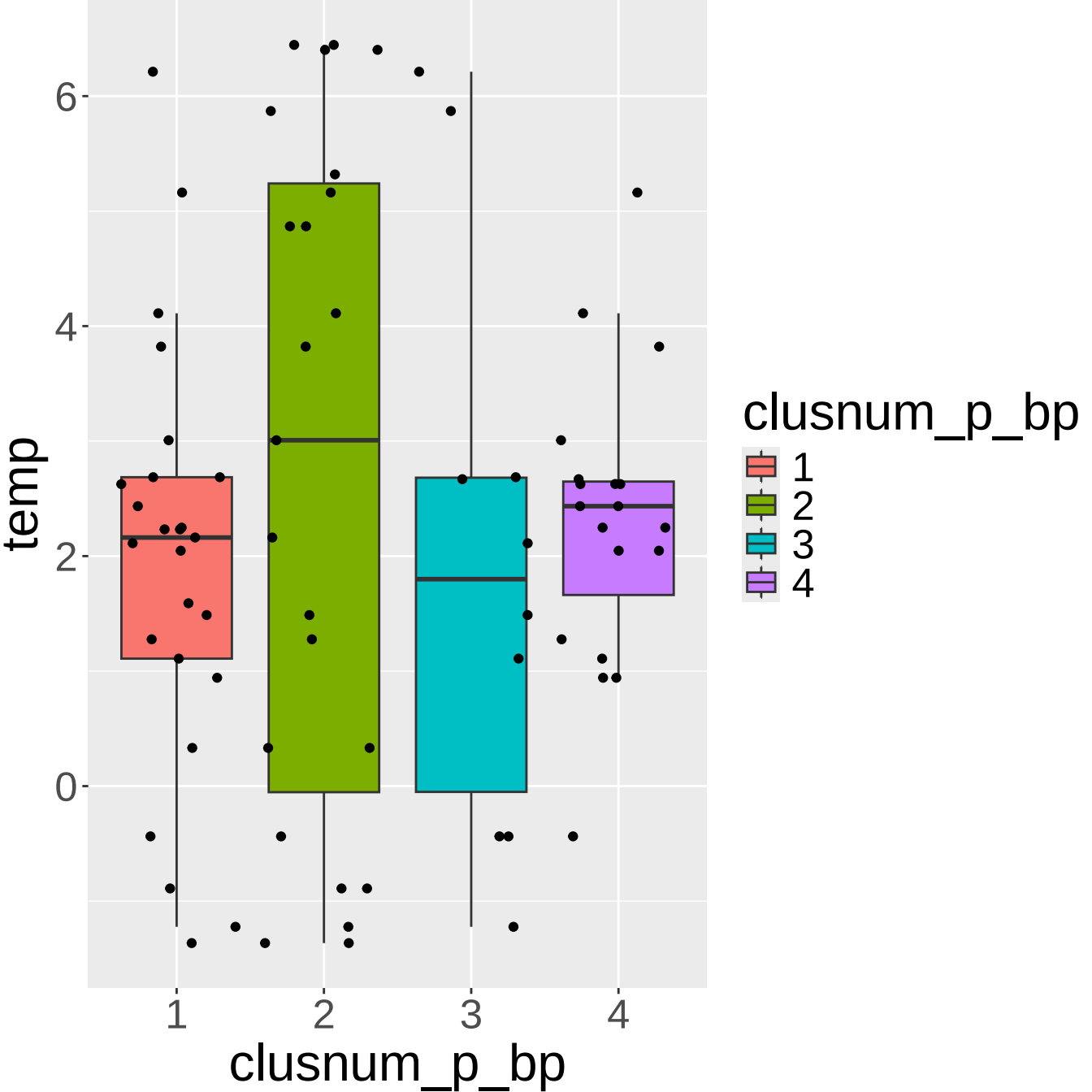

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
bottom_depth
25
30
35
40
45
50
55
clusnum_p_bp
1
2
3
4

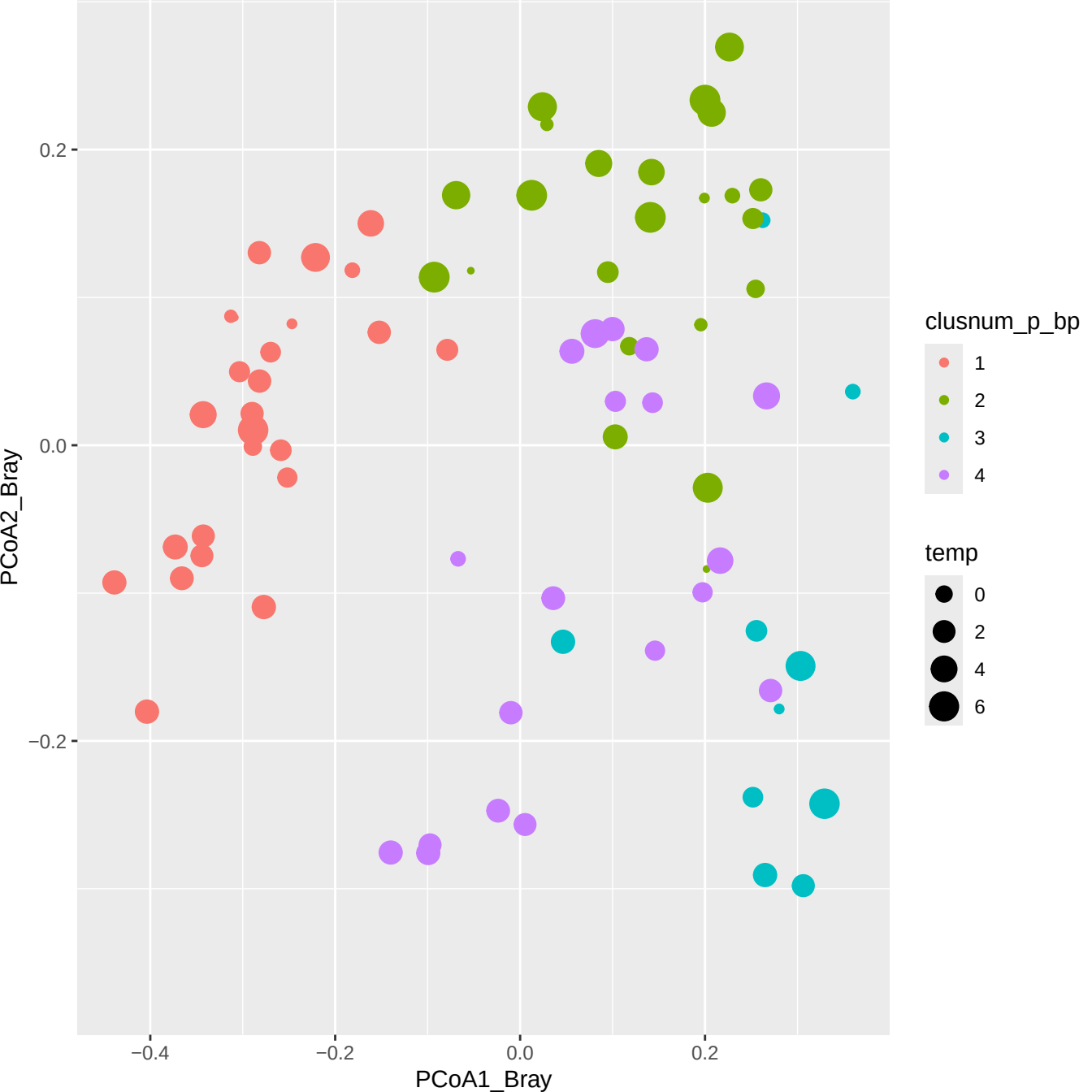

69
68
67
66
65
64
lat
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

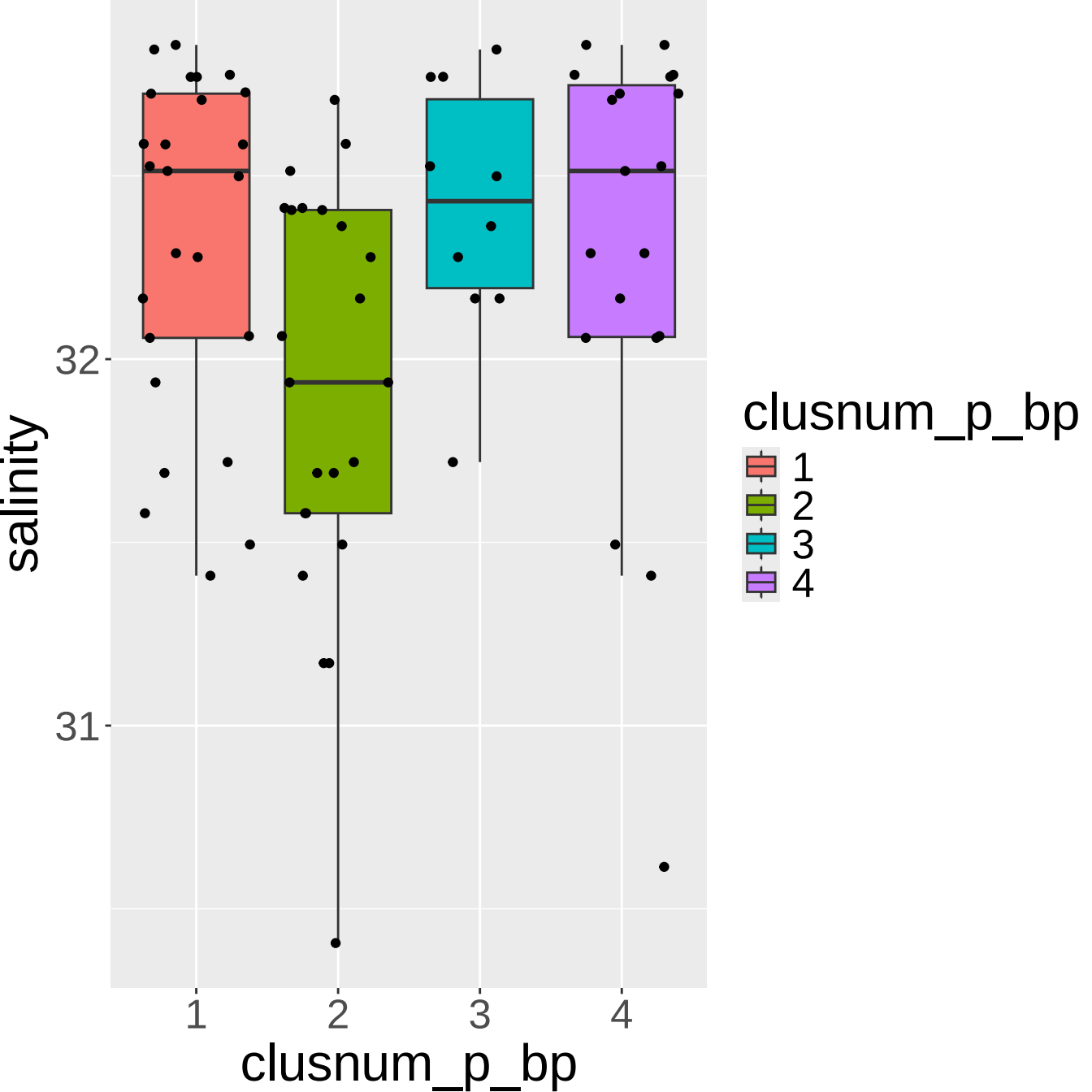

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
lat
64
65
66
67
68
69
clusnum_p_bp
1
2
3
4

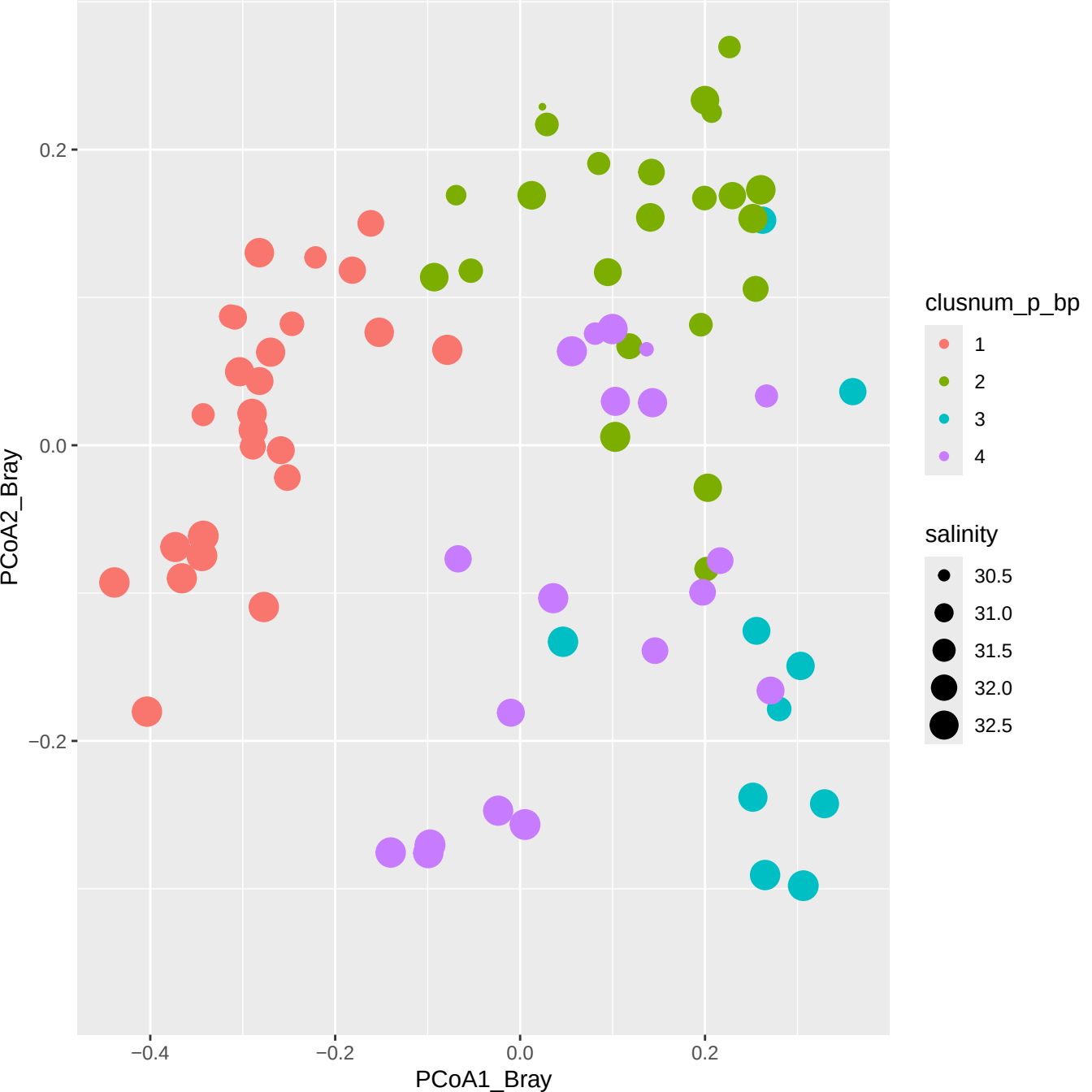

lon
−166
−168
−170
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

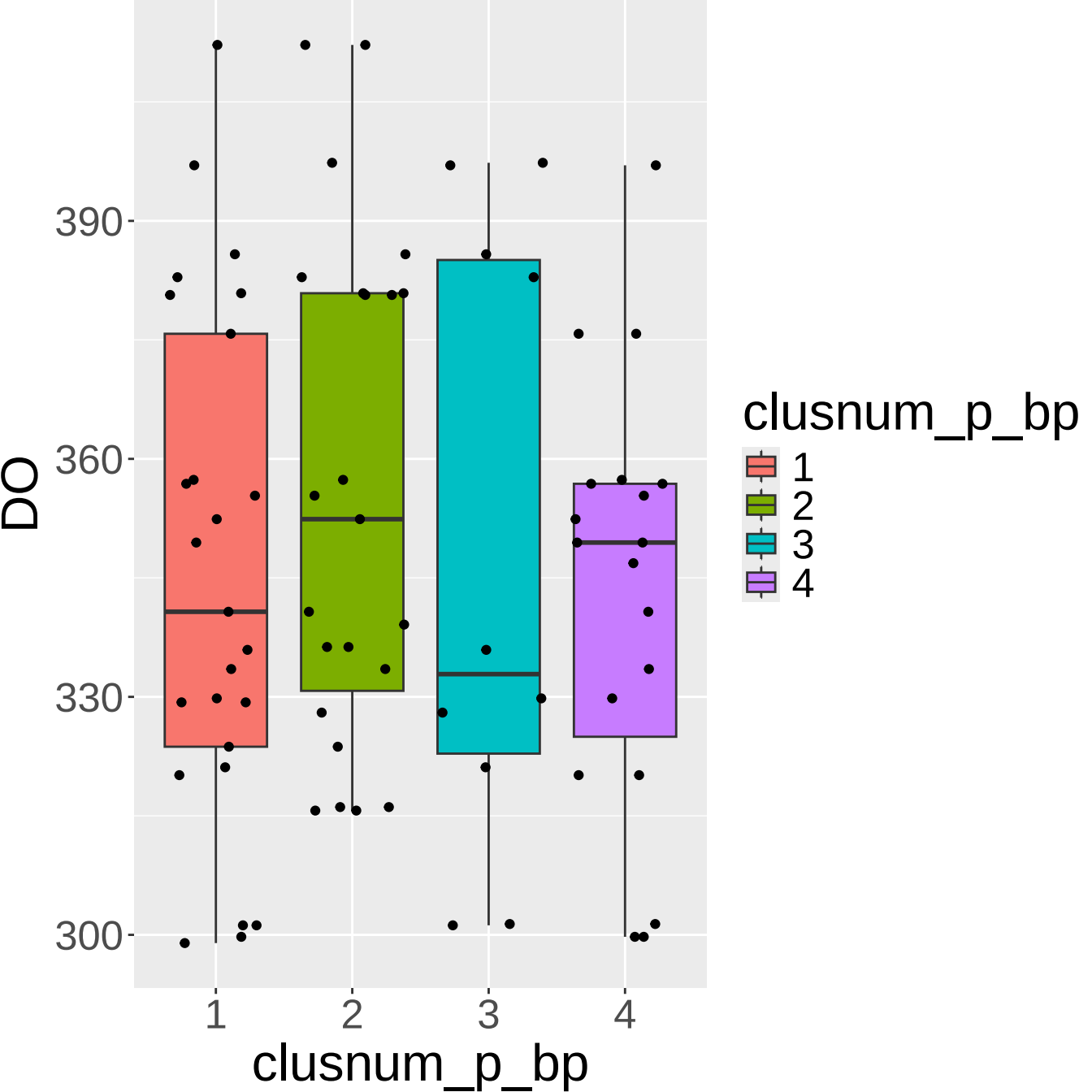

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
lon
−170
−168
−166
clusnum_p_bp
1
2
3
4

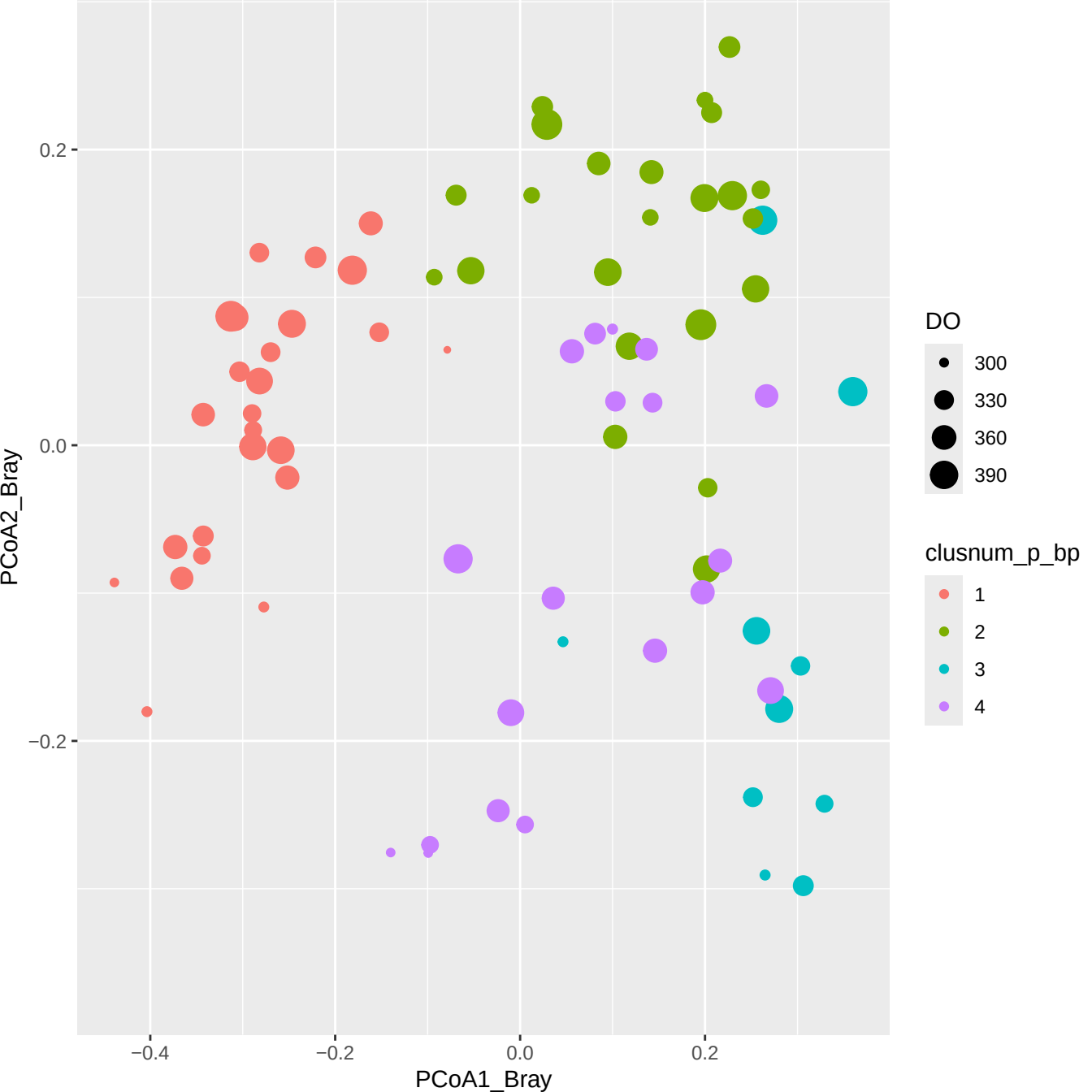

temp
6
4
2
0
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

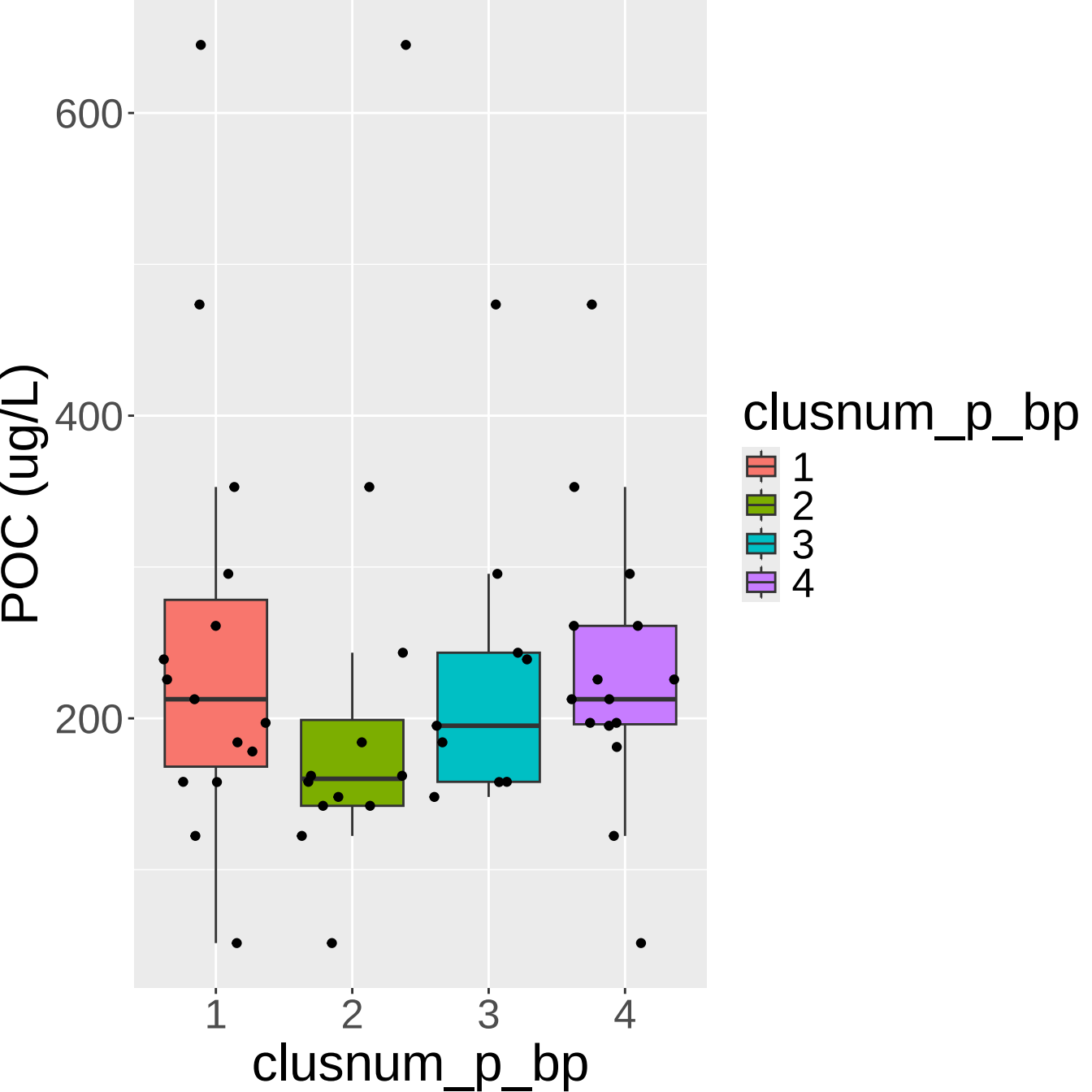

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
temp
0
2
4
6

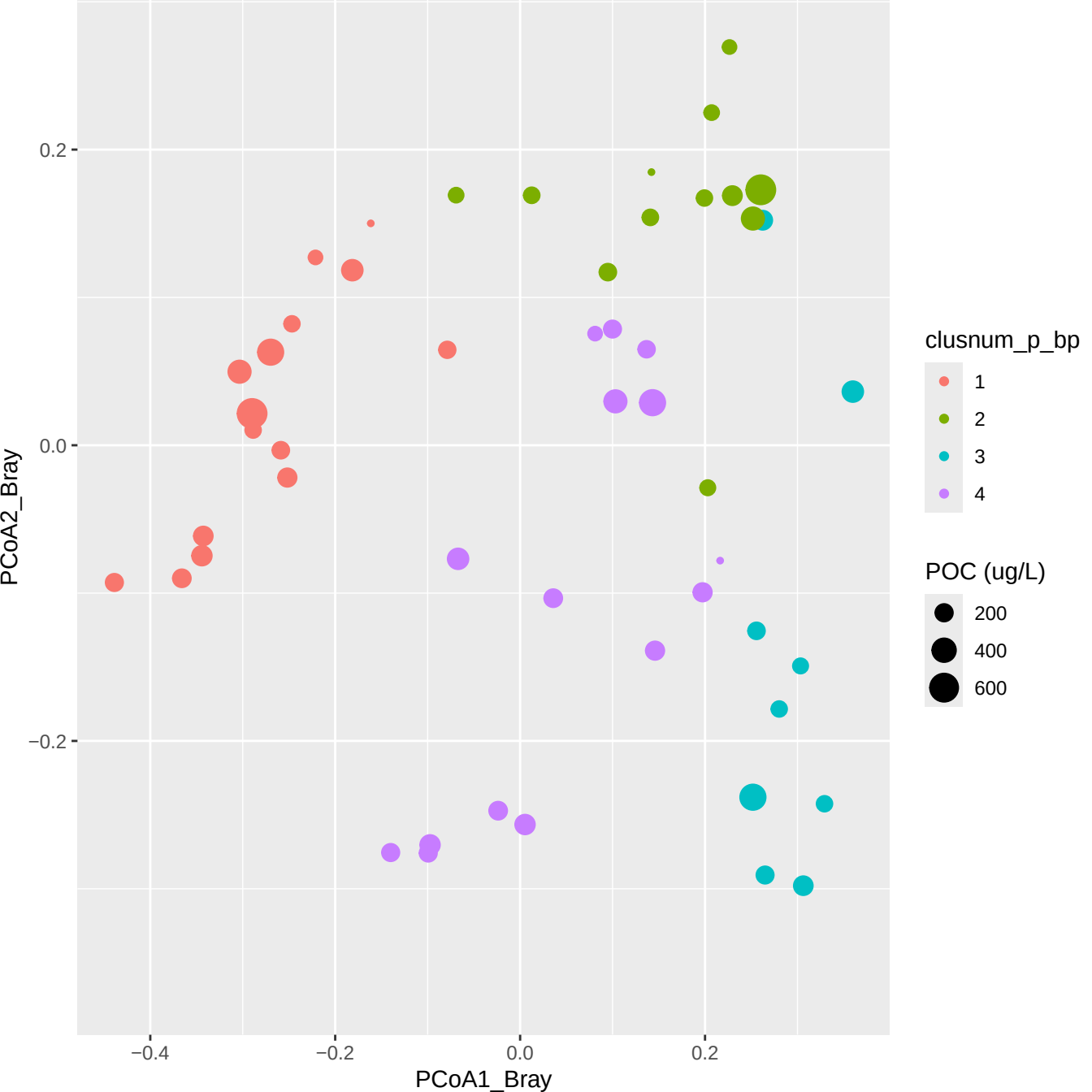

salinity
32
31
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

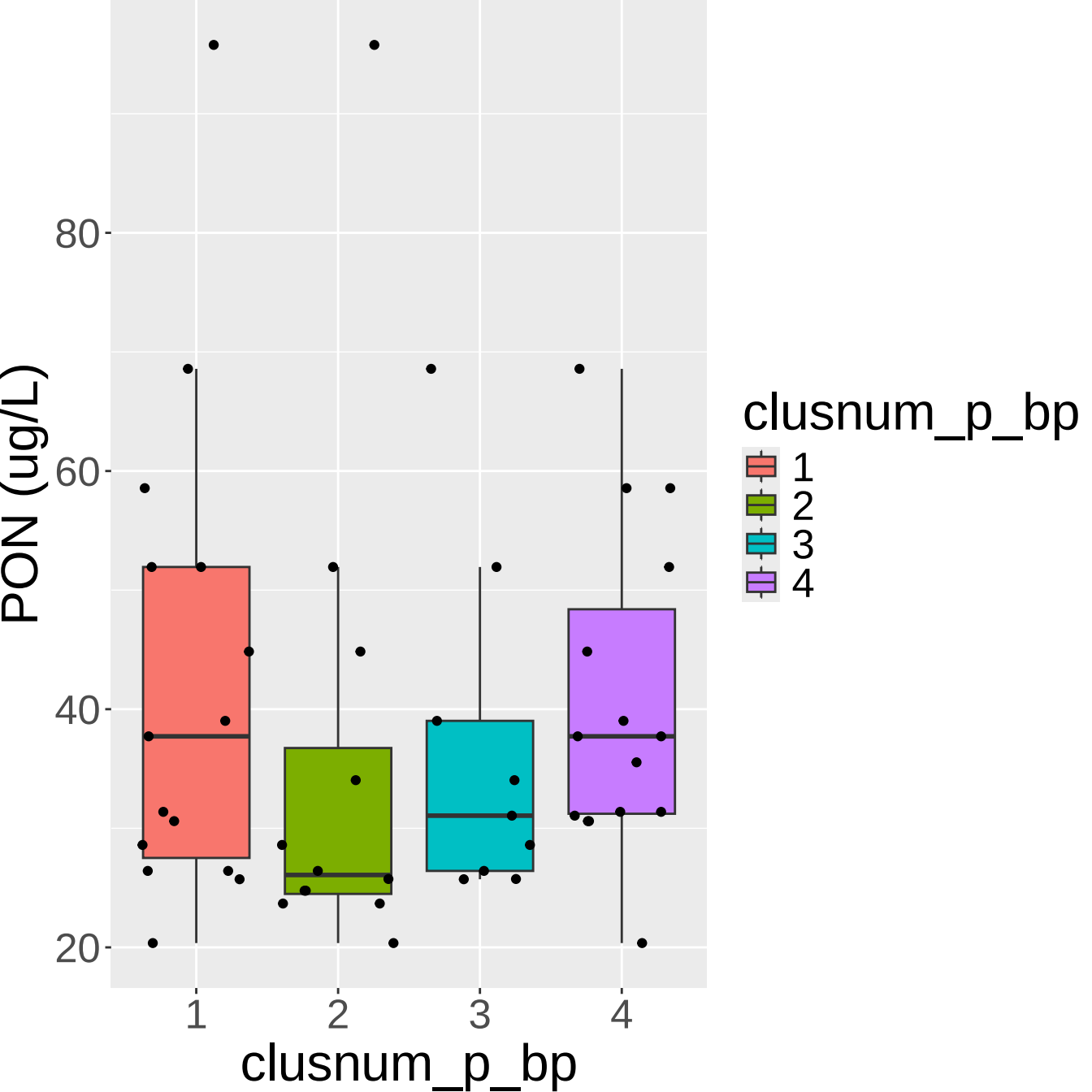

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
salinity
30.5
31.0
31.5
32.0
32.5

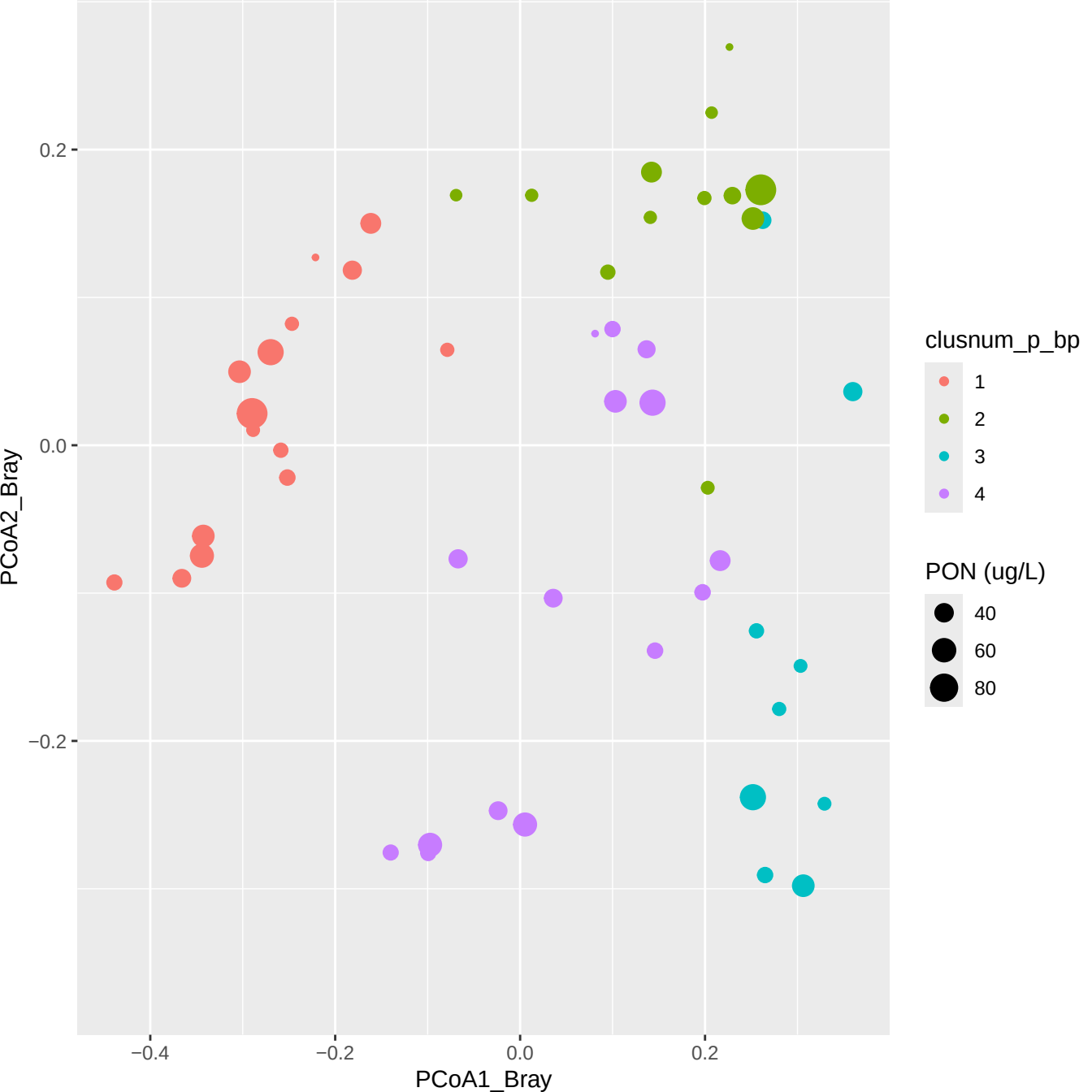

DO
390
360
330
300
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

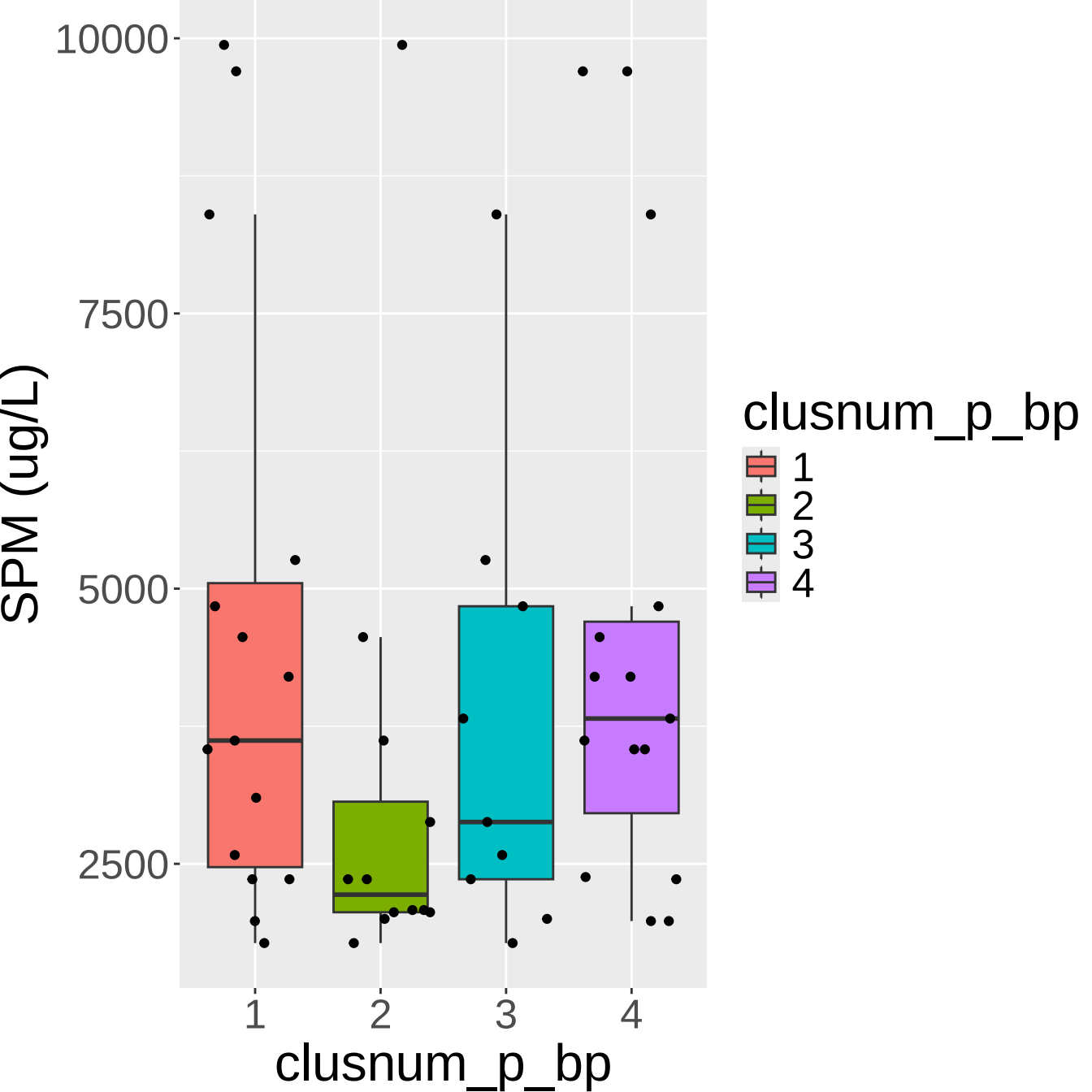

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
DO
300
330
360
390
clusnum_p_bp
1
2
3
4

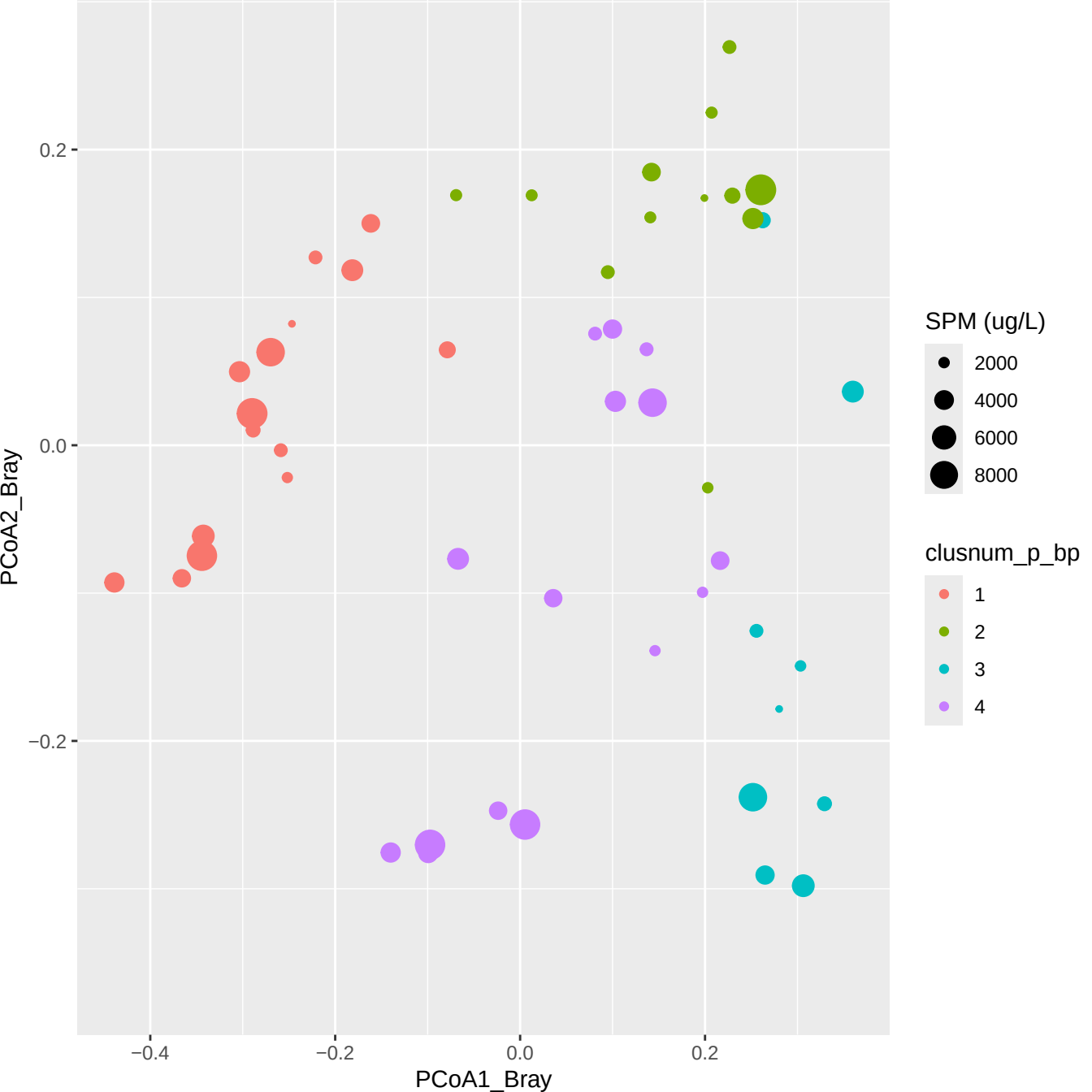

600
400
200
POC (ug/L)
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

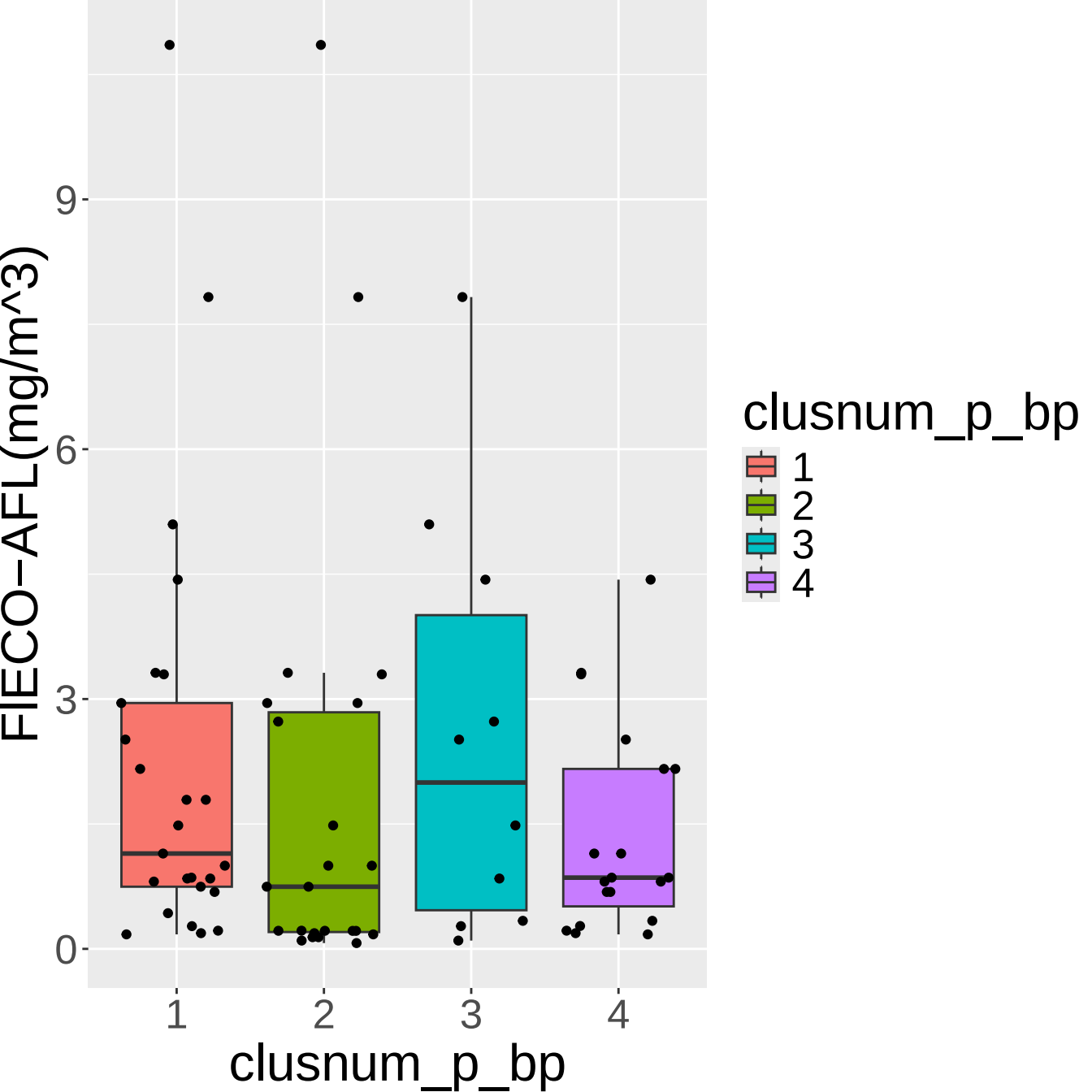

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
POC (ug/L)
200
400
600

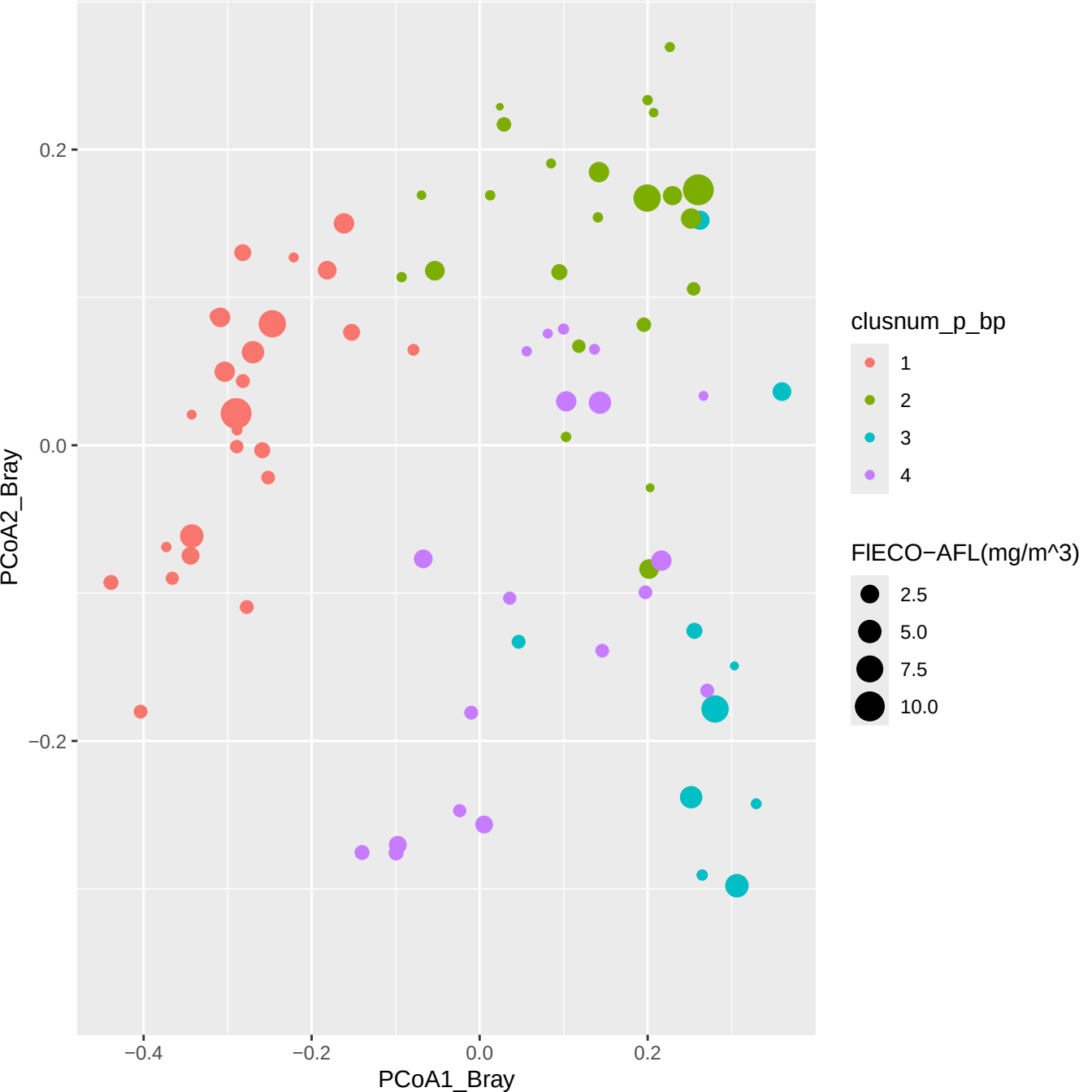

PON (ug/L)
80
60
40
20
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

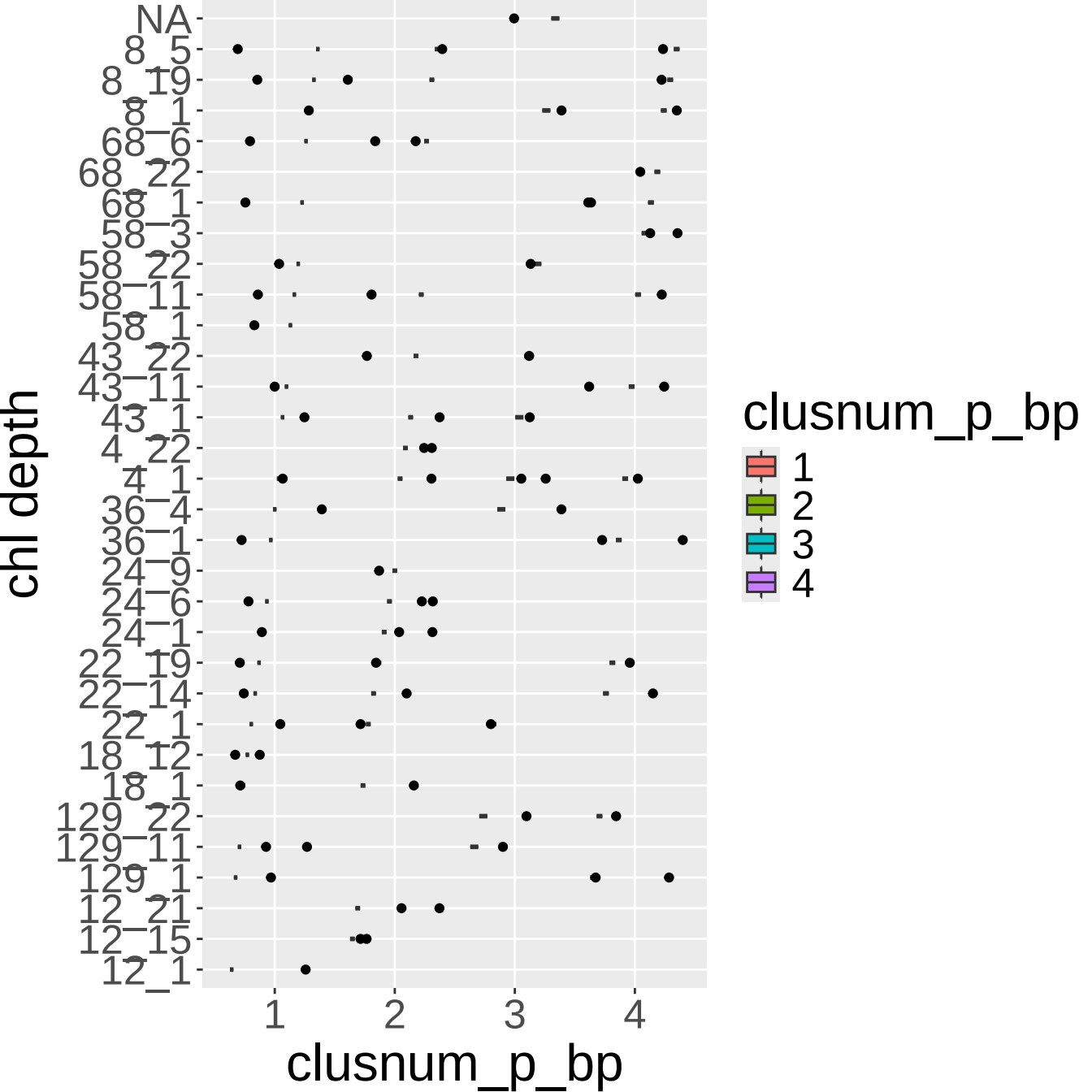

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
PON (ug/L)
40
60
80

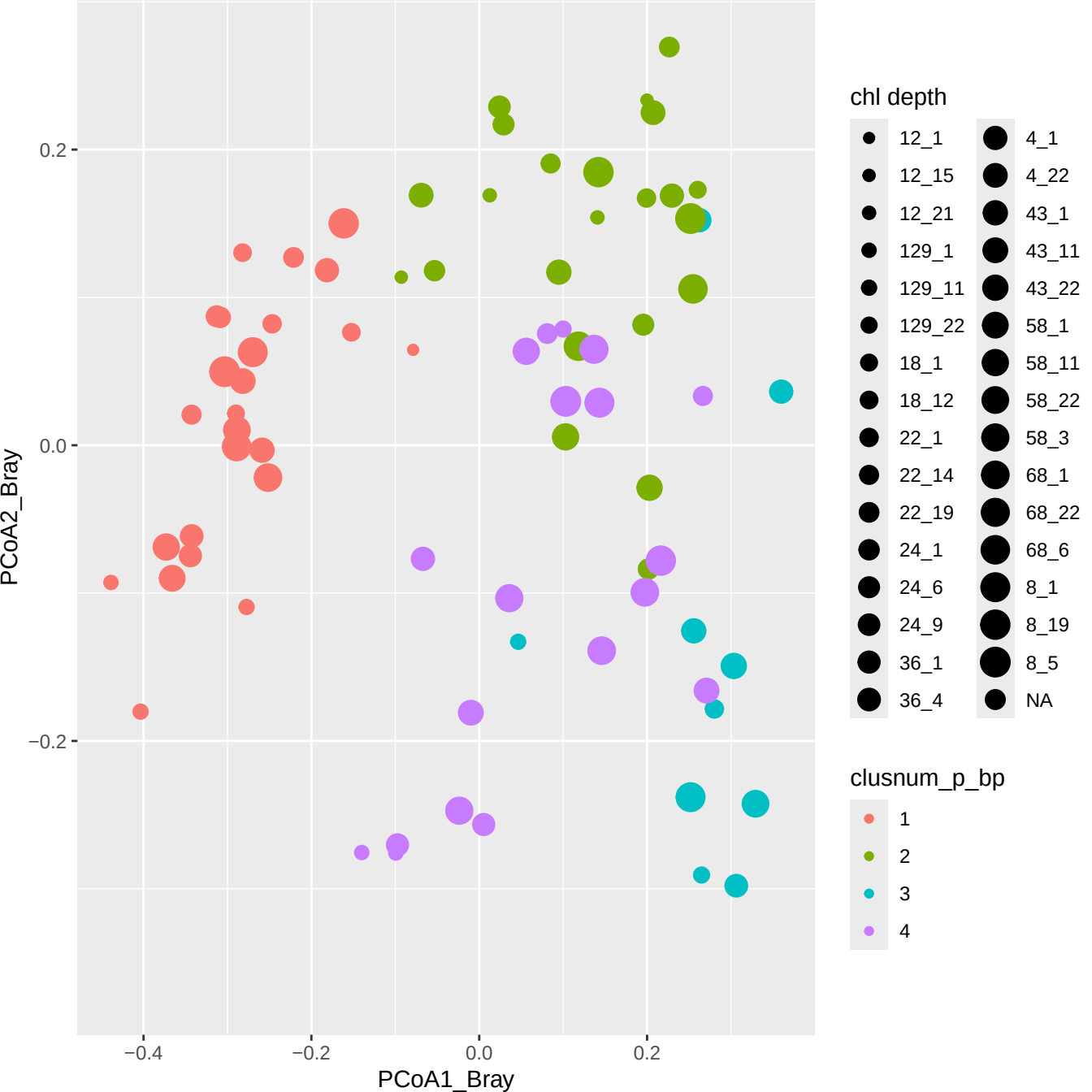

10000
7500
5000
2500
SPM (ug/L)
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

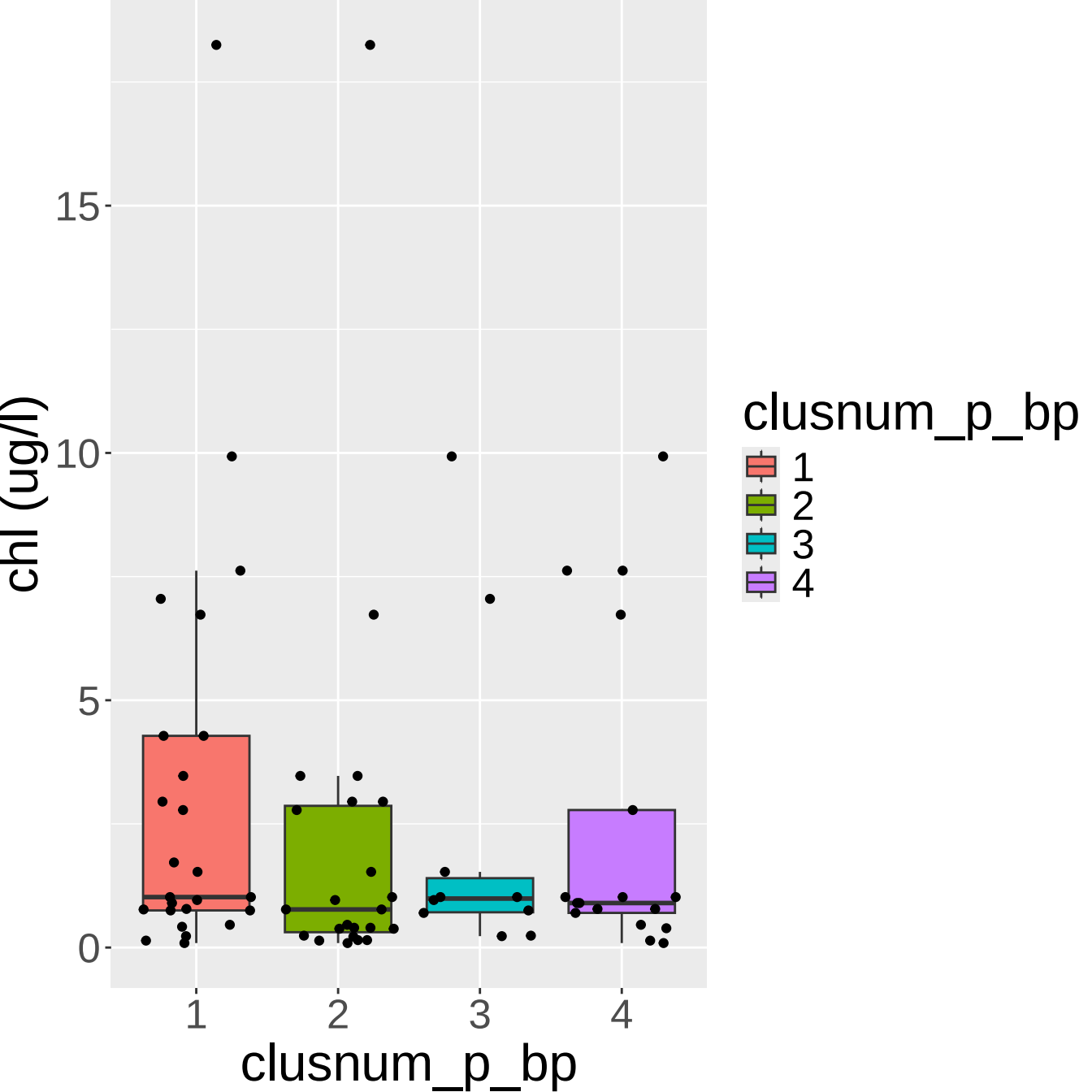

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
SPM (ug/L)
2000
4000
6000
8000
clusnum_p_bp
1
2
3
4

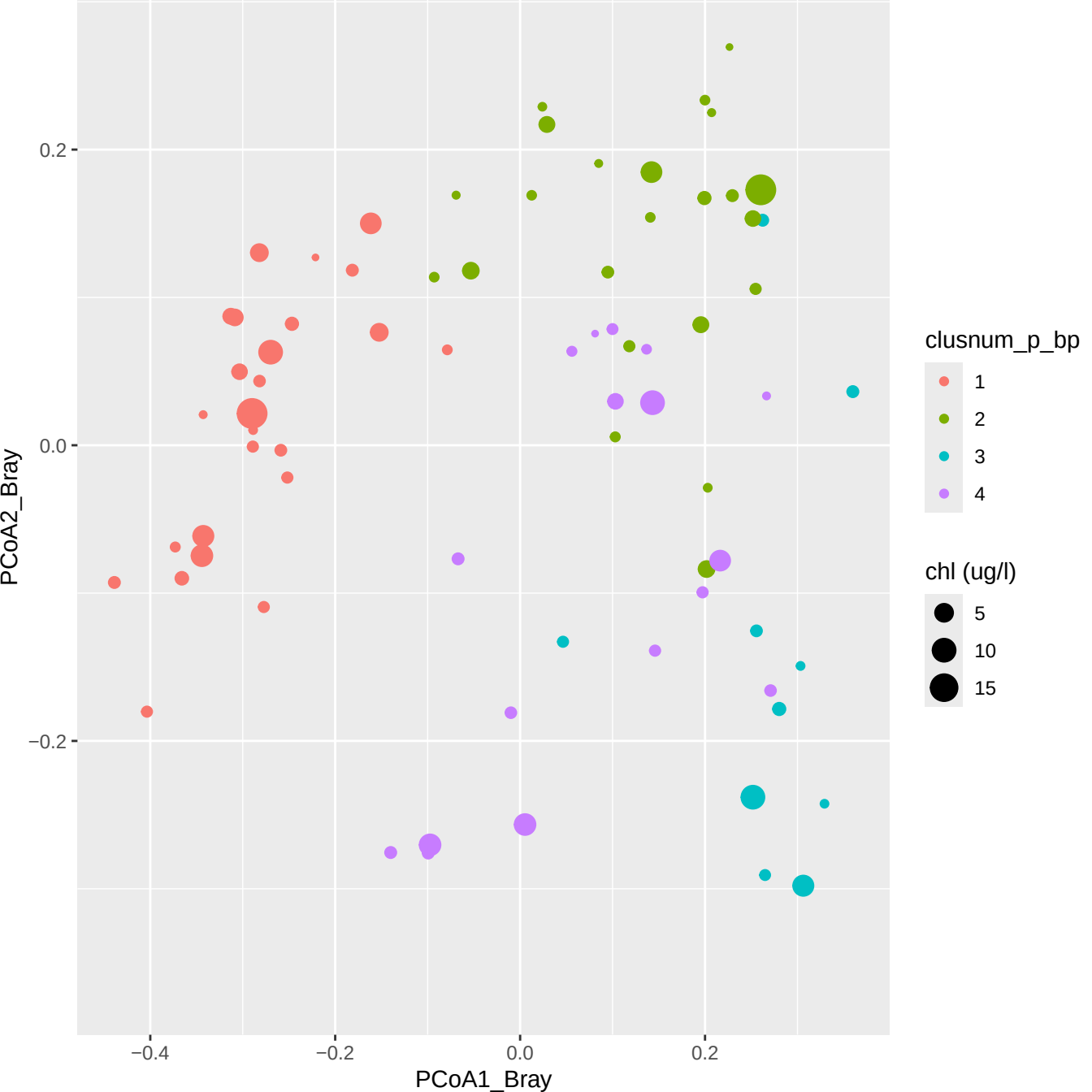

FIECO−AFL(mg/m^3)
9
6
3
0
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

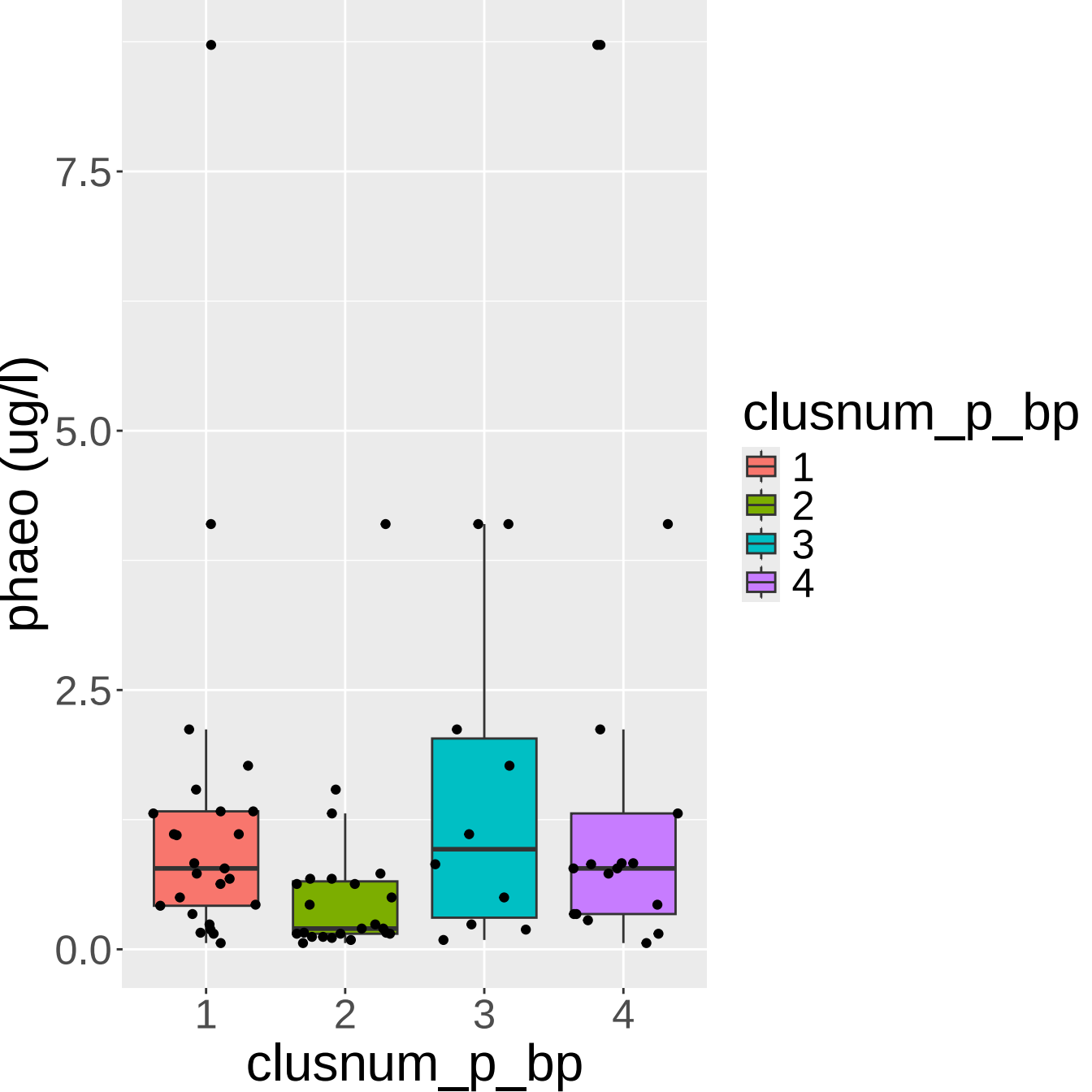

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
FlECO−AFL(mg/m^3)
2.5
5.0
7.5
10.0

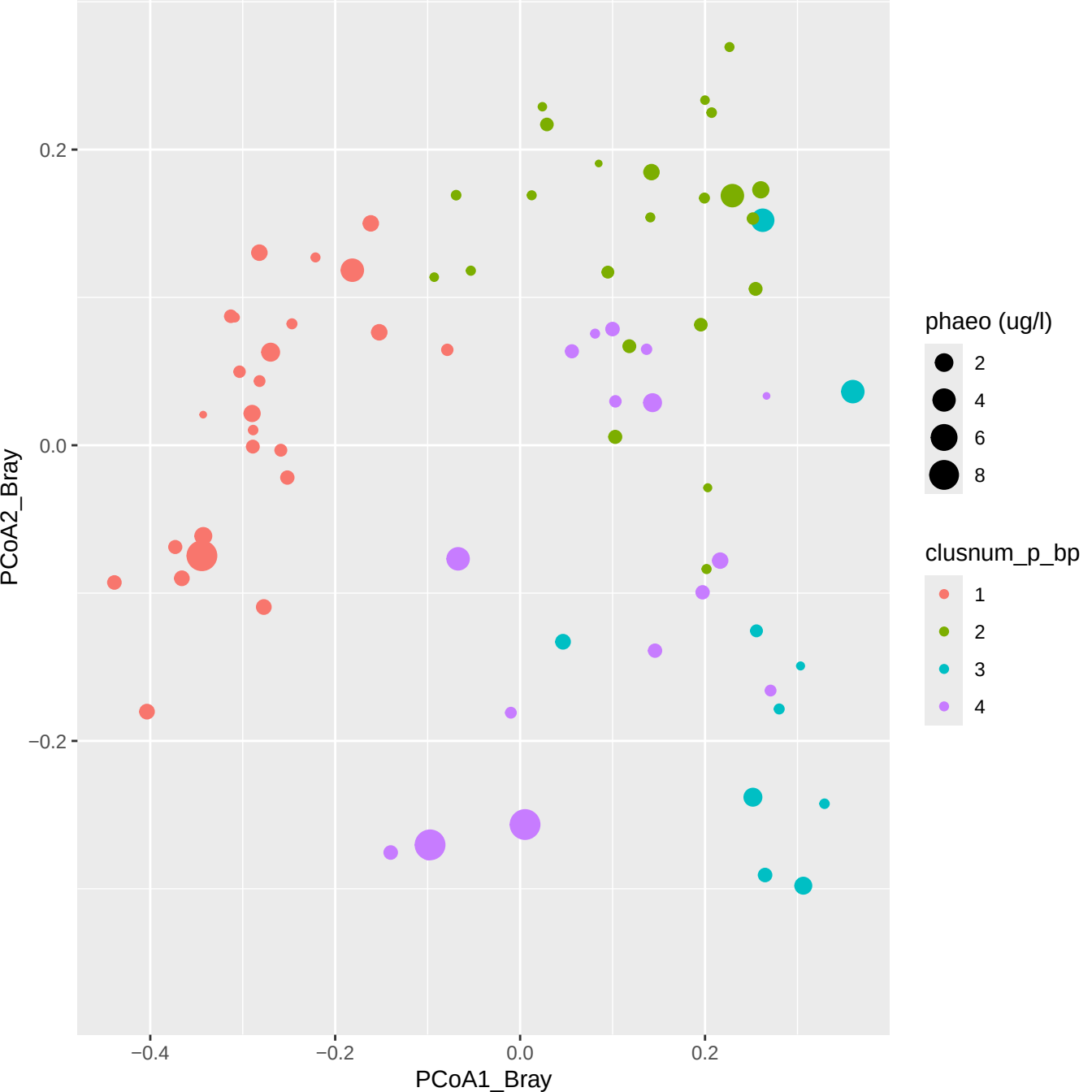

chl depth
NA
8_5
8_19
8_1
68_6
68_22
68_1
58_3
58_22
58_11
58_1
43_22
43_11
43_1
4_22
4_1
36_4
36_1
24_9
24_6
24_1
22_19
22_14
22_1
18_12
18_1
129_22
129_11
129_1
12_21
12_15
12_1
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

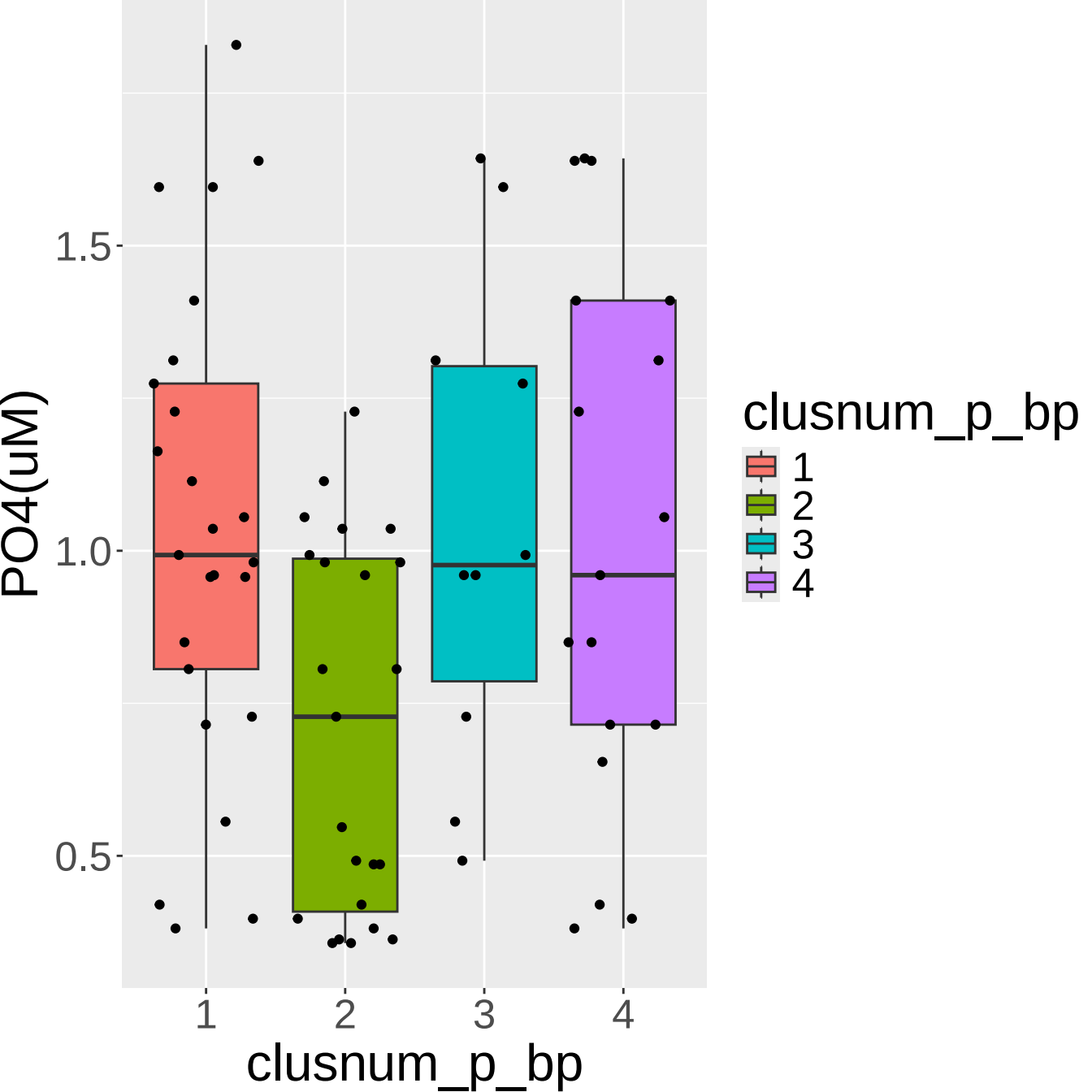

chl depth
12_1
12_15
12_21
129_1
129_11
129_22
18_1
18_12
22_1
22_14
22_19
24_1
24_6
24_9
36_1
36_4
4_1
4_22
43_1
43_11
43_22
58_1
58_11
58_22
58_3
68_1
68_22
68_6
8_1
8_19
8_5
NA
clusnum_p_bp
1
2
3
4
PCoA2_Bray
PCoA1_Bray
0.2
0.0
−0.2
−0.4
−0.2
0.0
0.2

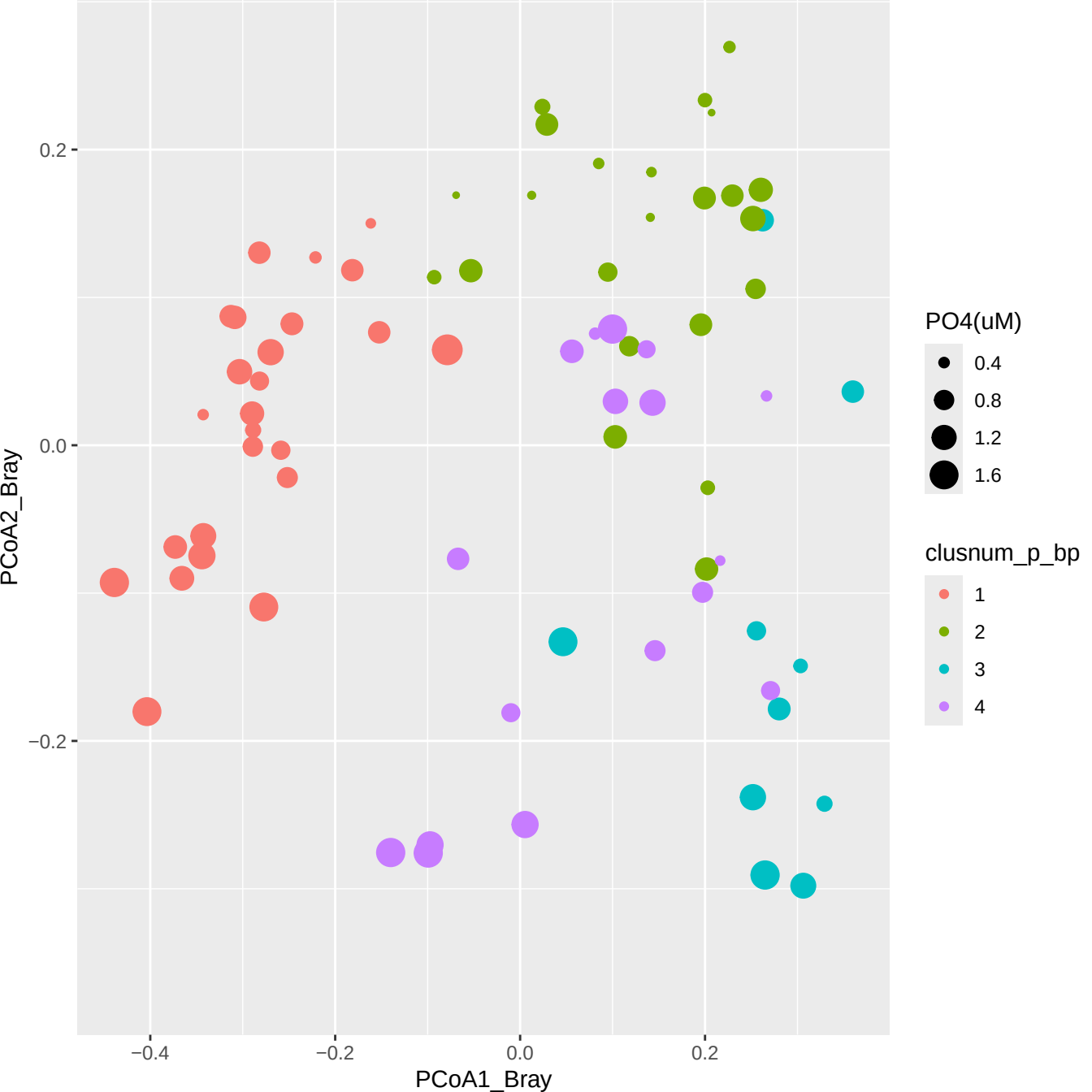

chl (ug/l)
15
10
5
0
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

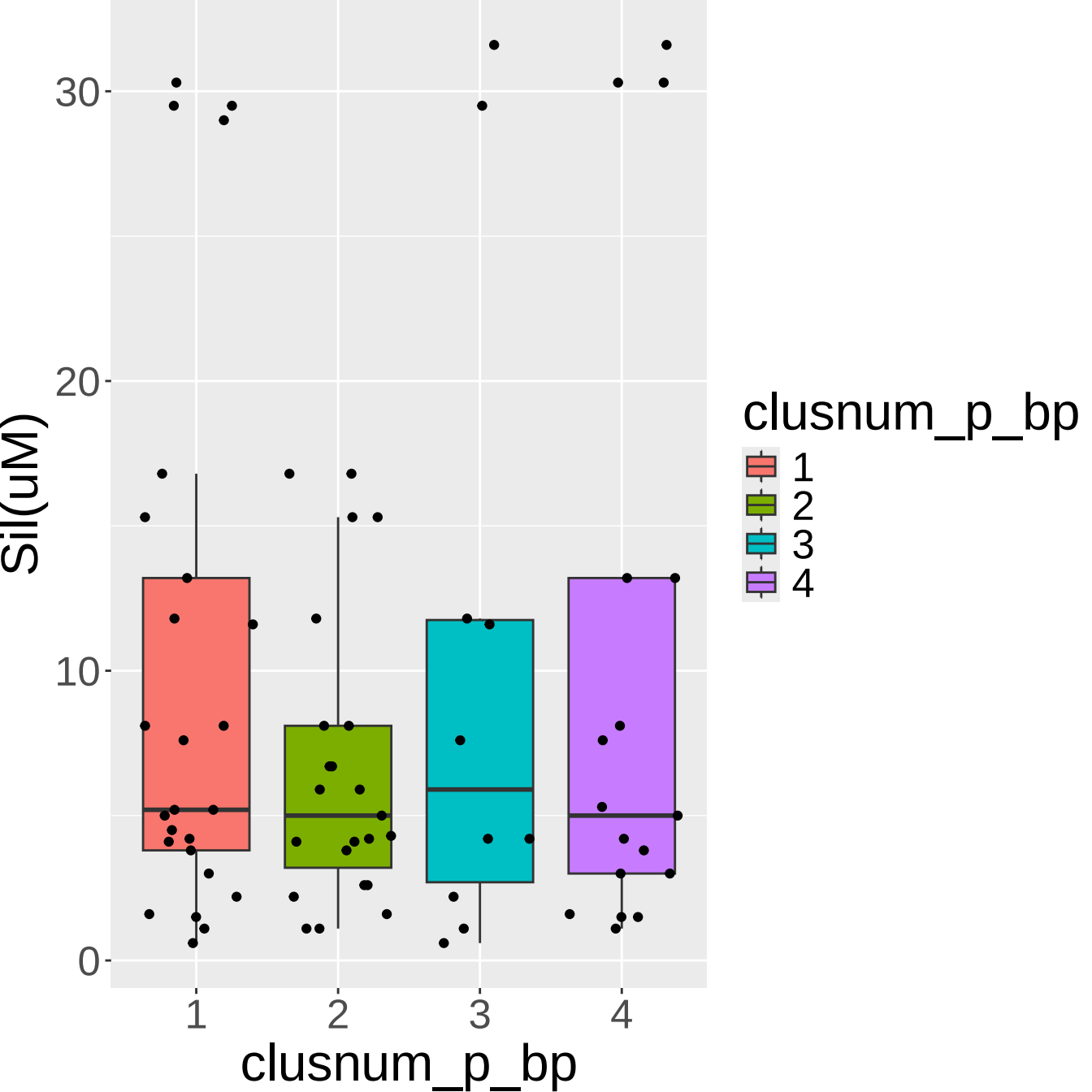

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
chl (ug/l)
5
10
15

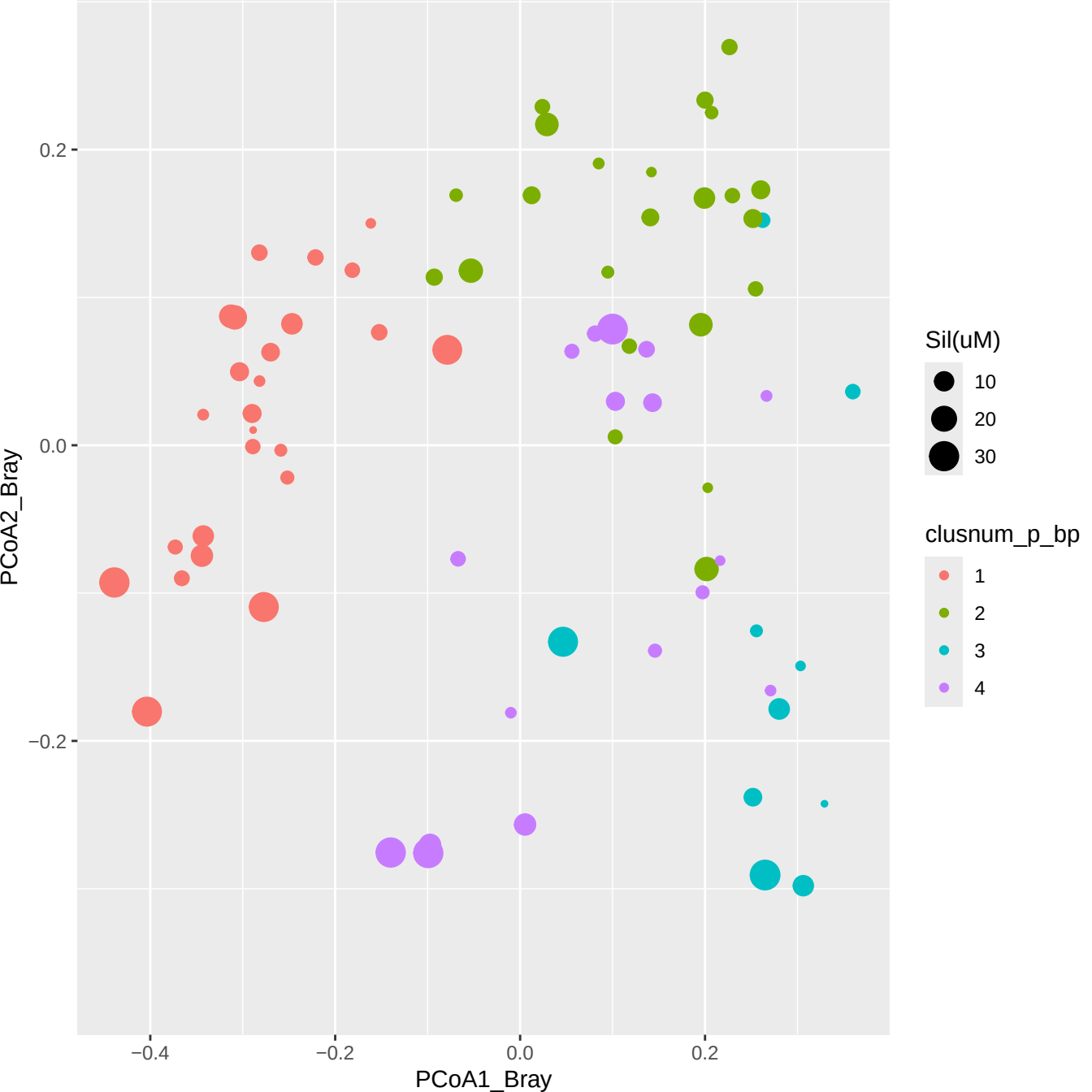

phaeo (ug/l)
7.5
5.0
2.5
0.0
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

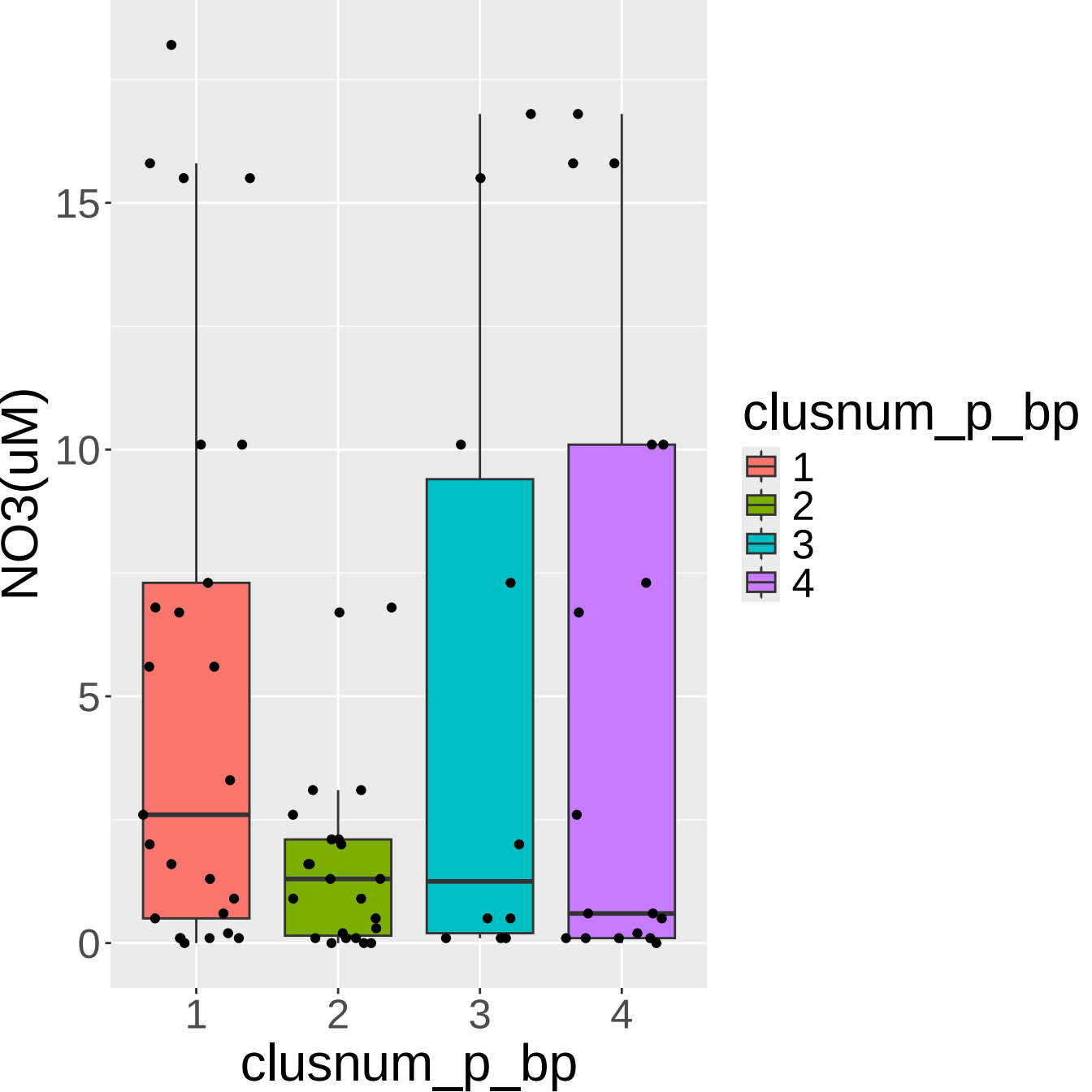

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
phaeo (ug/l)
2
4
6
8
clusnum_p_bp
1
2
3
4

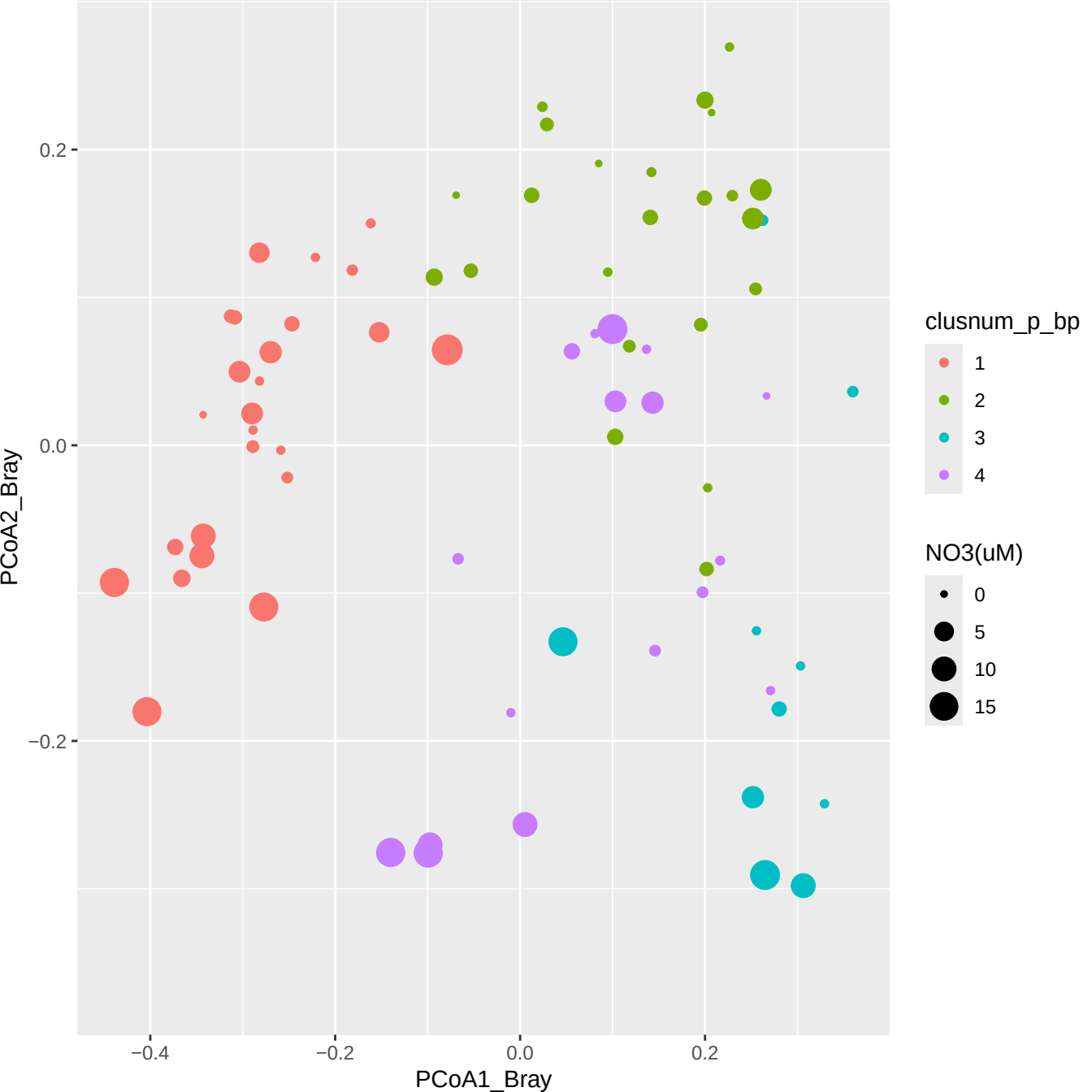

PO4(uM)
1.5
1.0
0.5
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

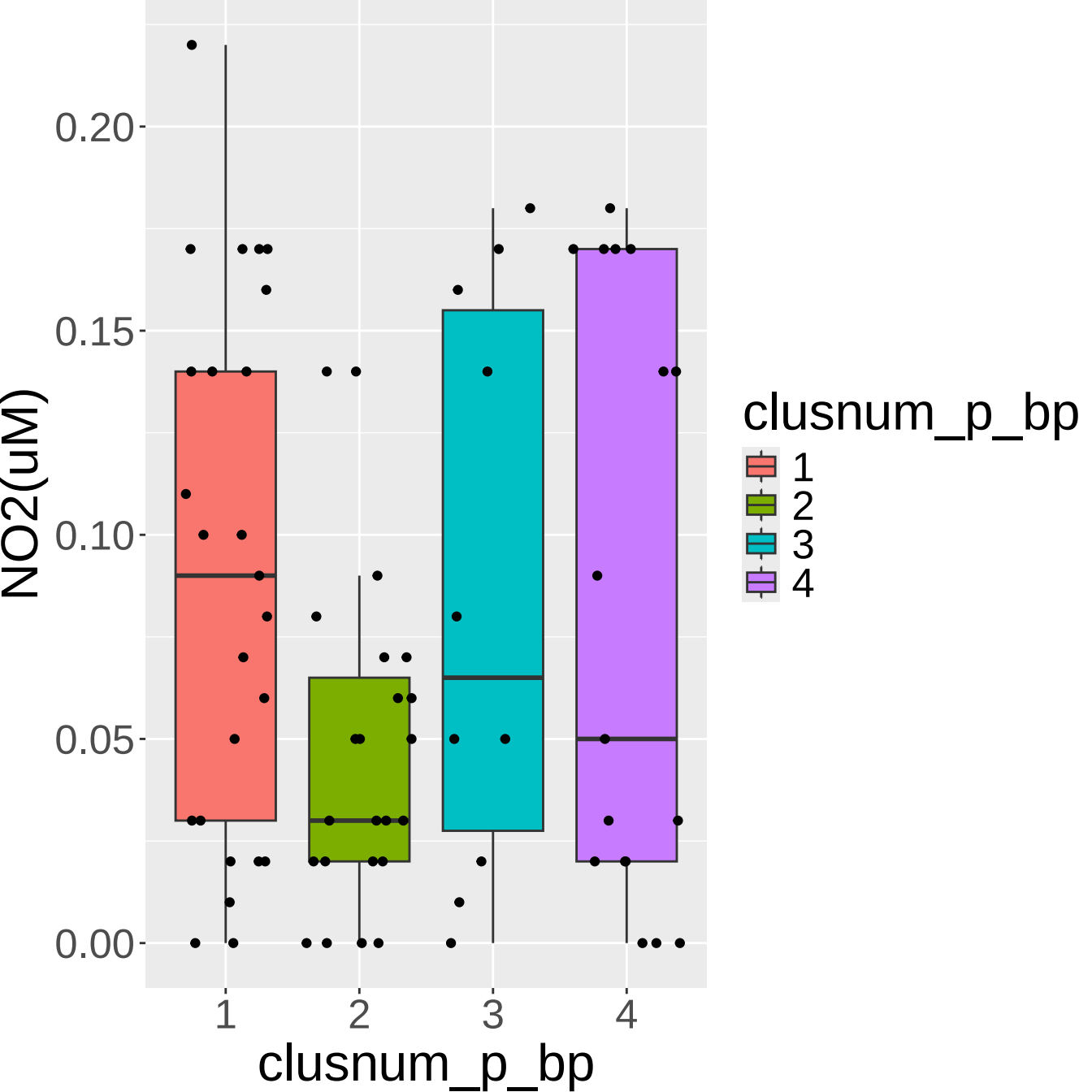

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
PO4(uM)
0.4
0.8
1.2
1.6
clusnum_p_bp
1
2
3
4

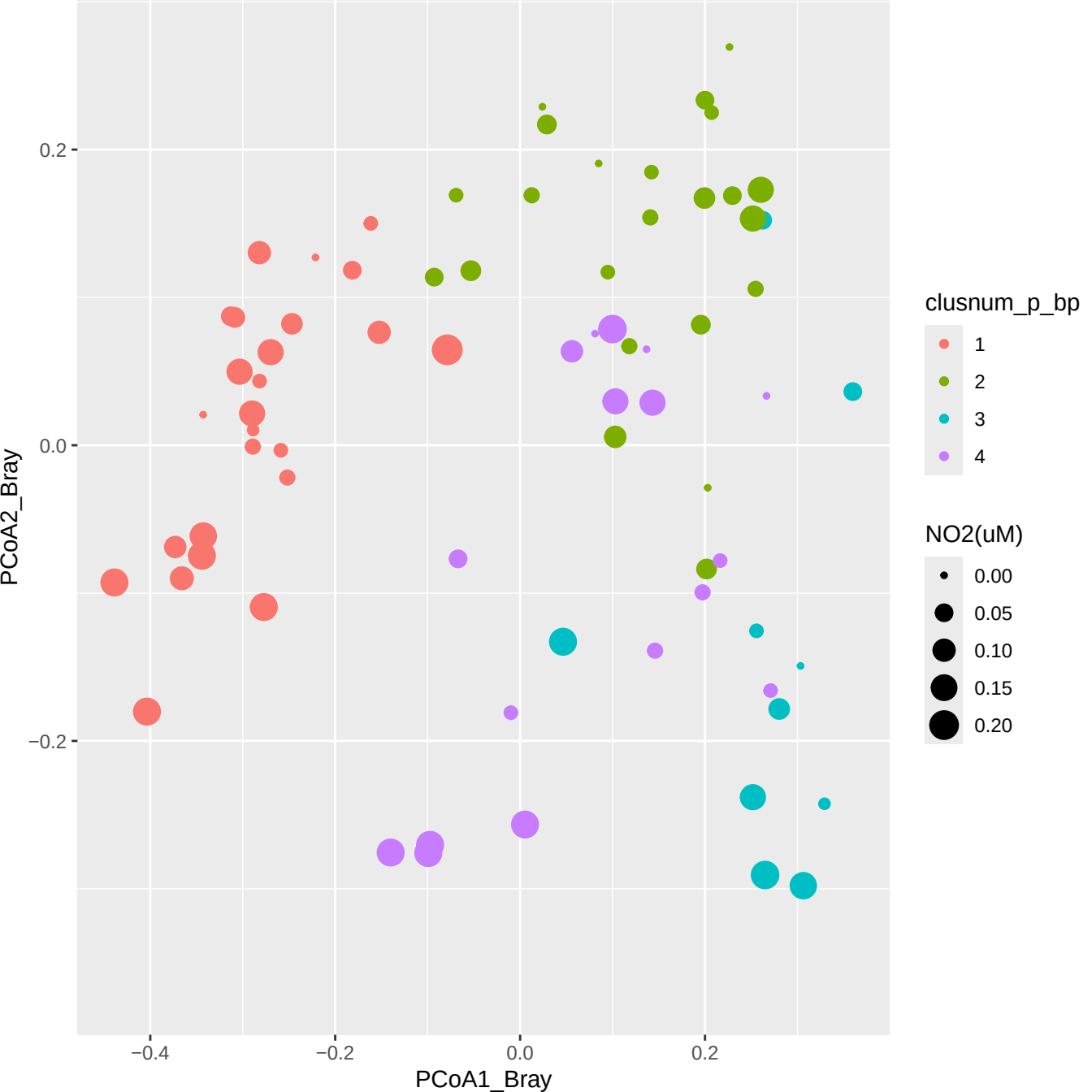

Sil(uM)
30
20
10
0
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

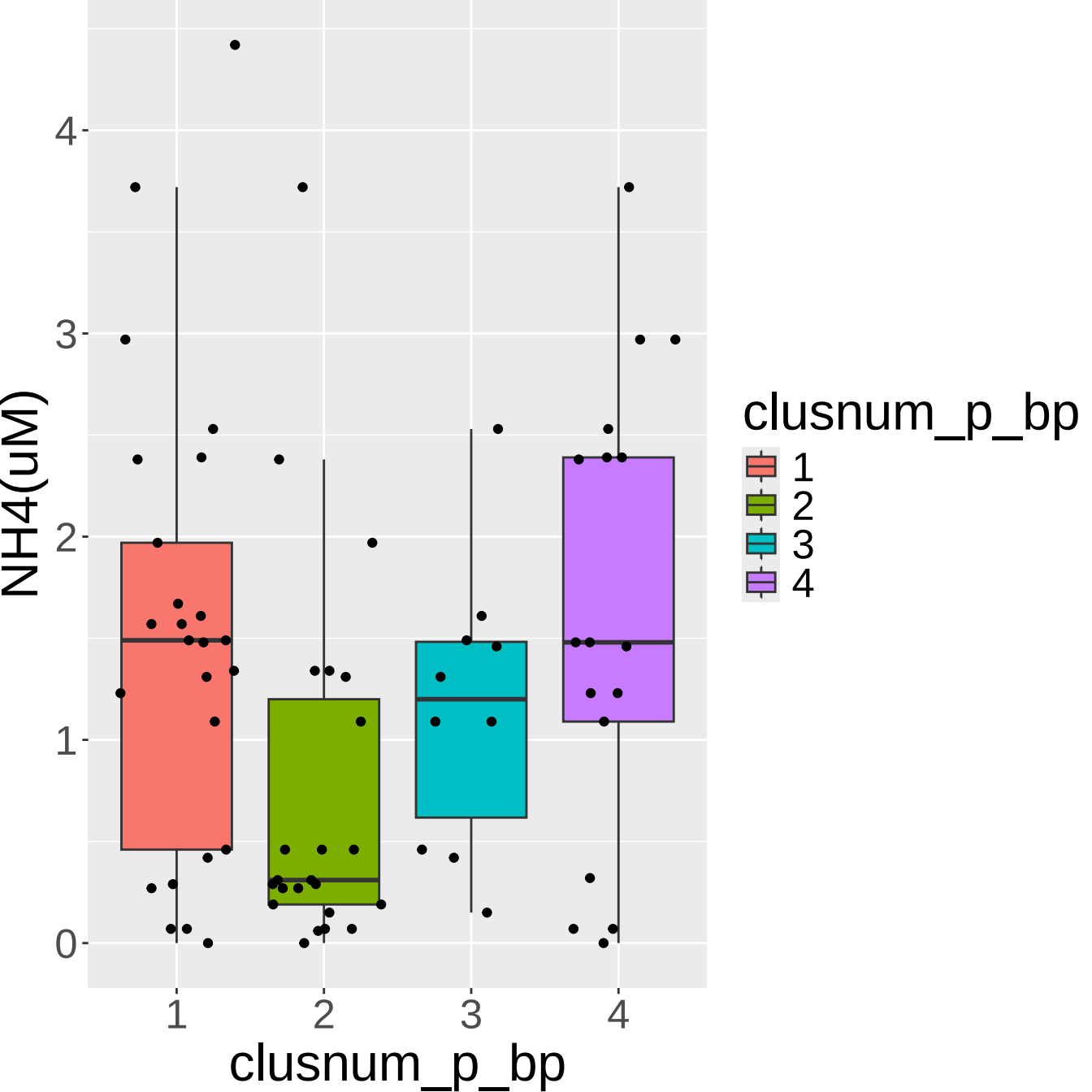

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
Sil(uM)
10
20
30
clusnum_p_bp
1
2
3
4

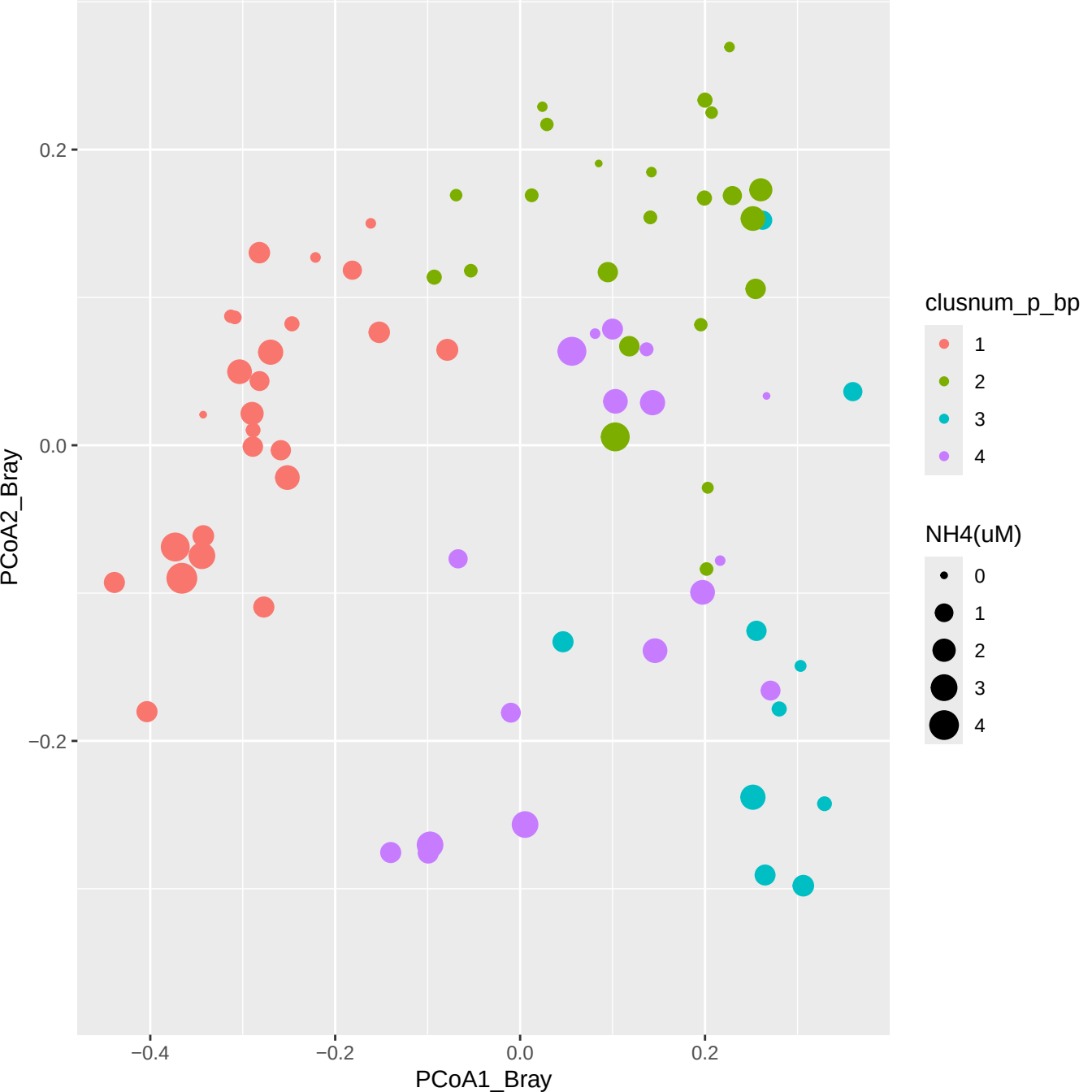

NO3(uM)
15
10
5
0
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

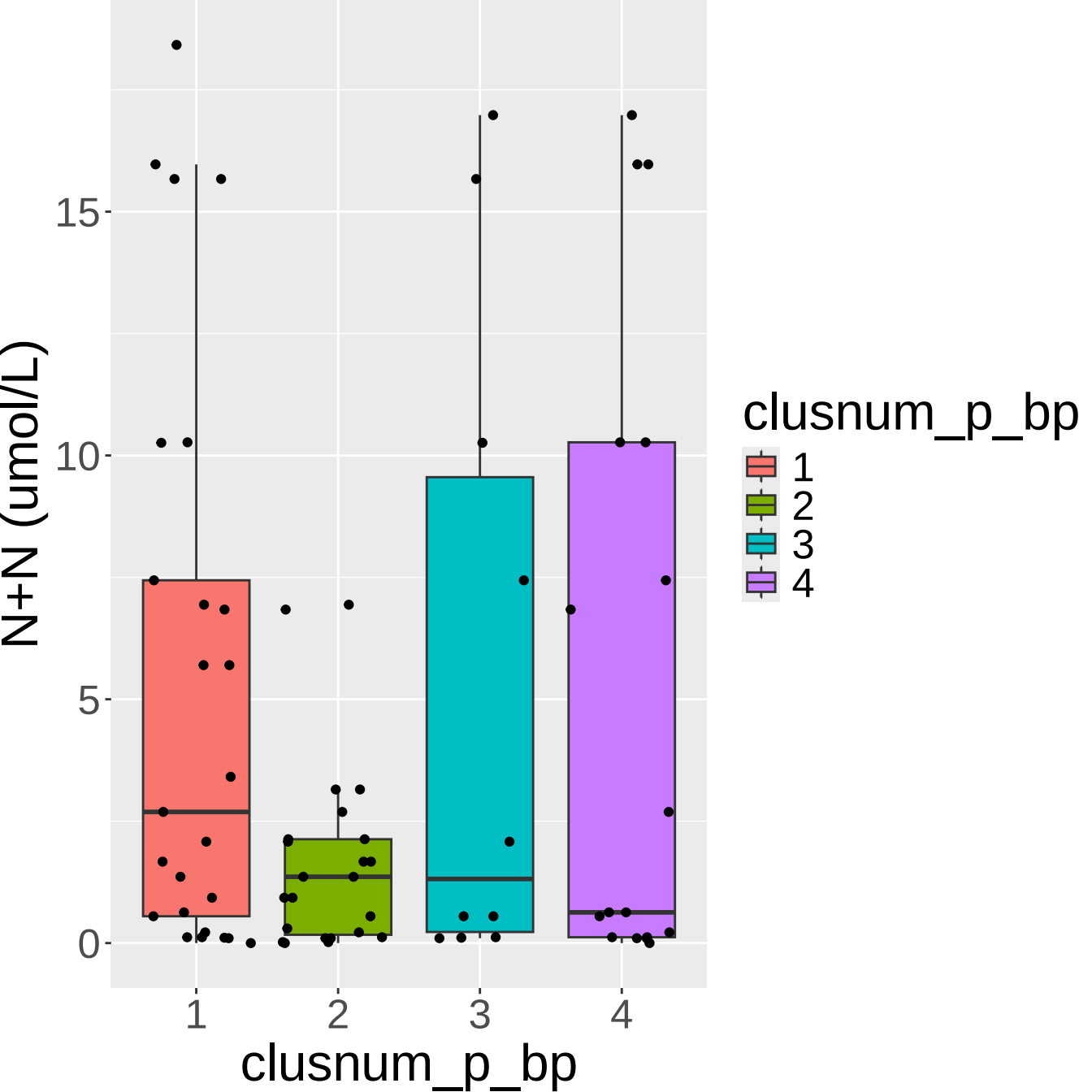

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
NO3(uM)
0
5
10
15

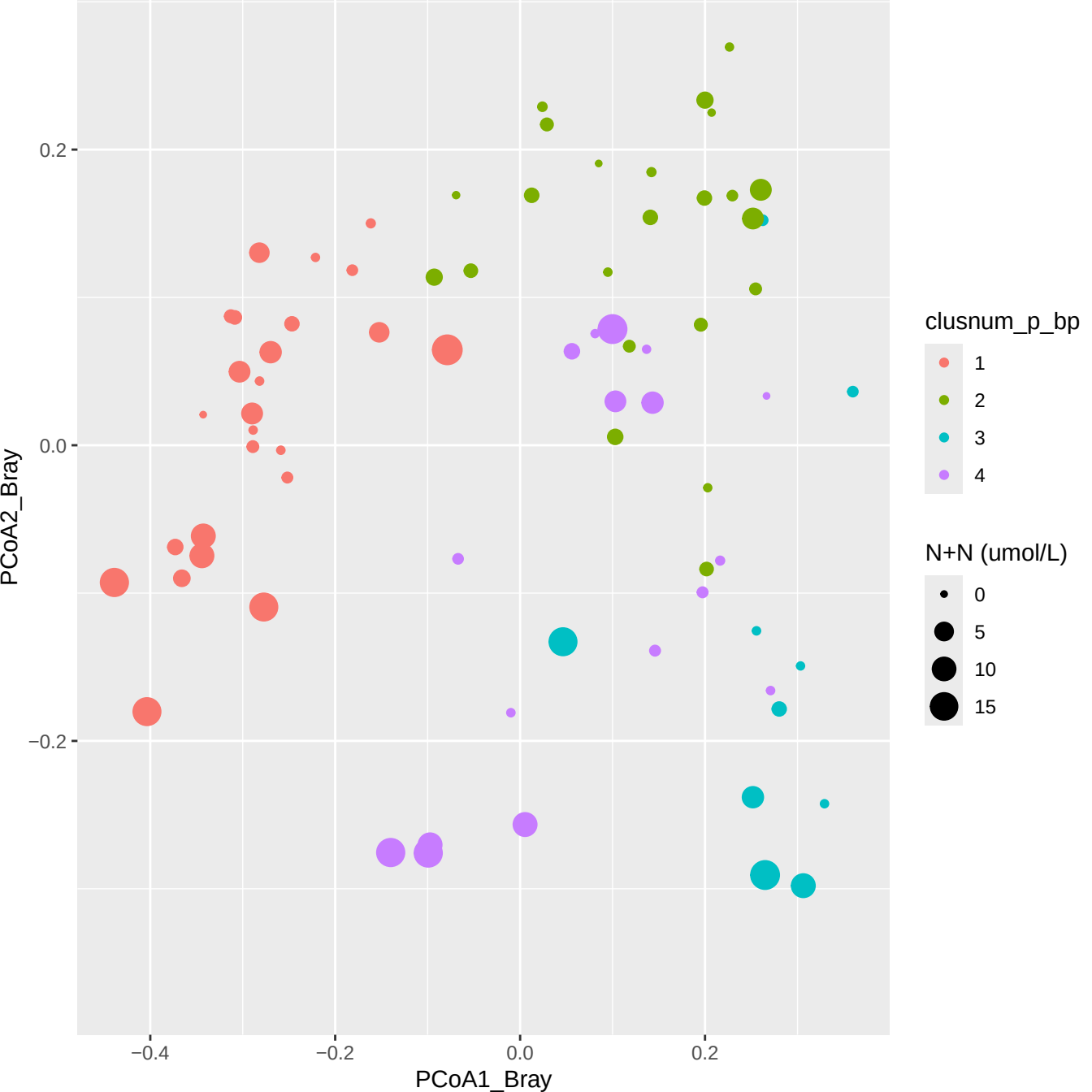

NO2(uM)
0.20
0.15
0.10
0.05
0.00
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

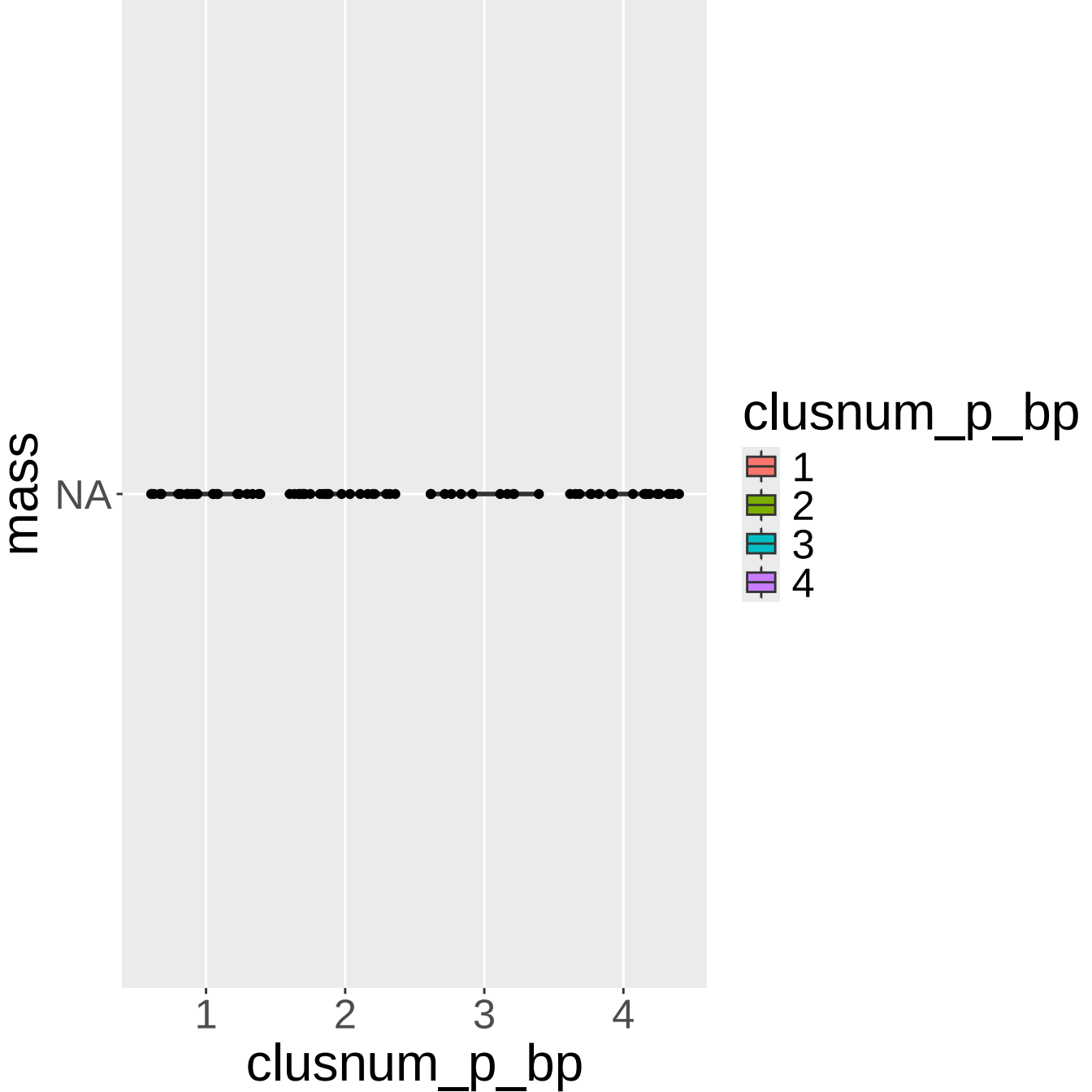

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
NO2(uM)
0.00
0.05
0.10
0.15
0.20

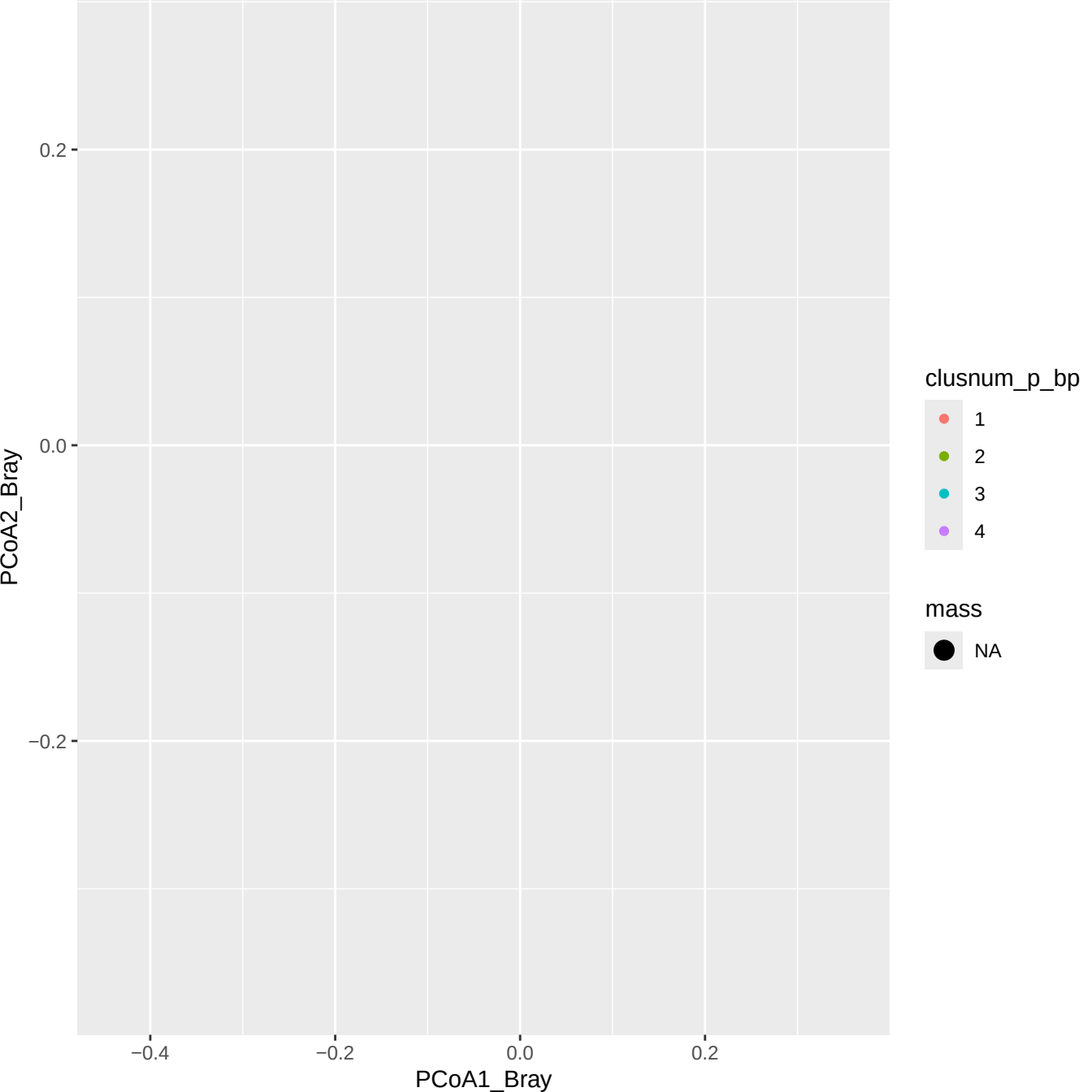

NH4(uM)
0
1
2
3
4
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

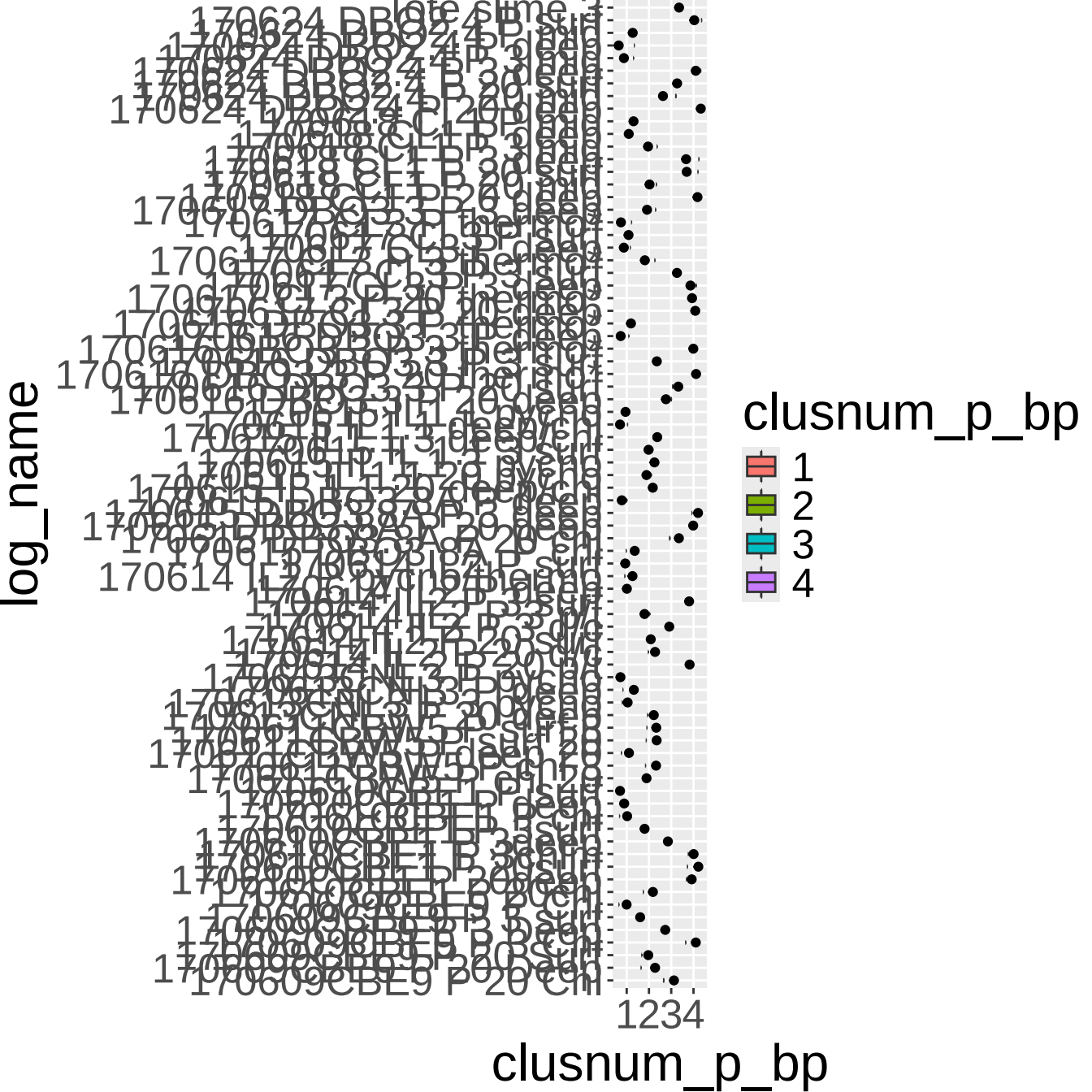

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
NH4(uM)
0
1
2
3
4


N+N (umol/L)
15
10
5
0
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4


0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
N+N (umol/L)
0
5
10
15


mass
NA
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4


0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
mass
NA


log_name
clusnum_p_bp
1
2
3
4
clusnum_p_bp

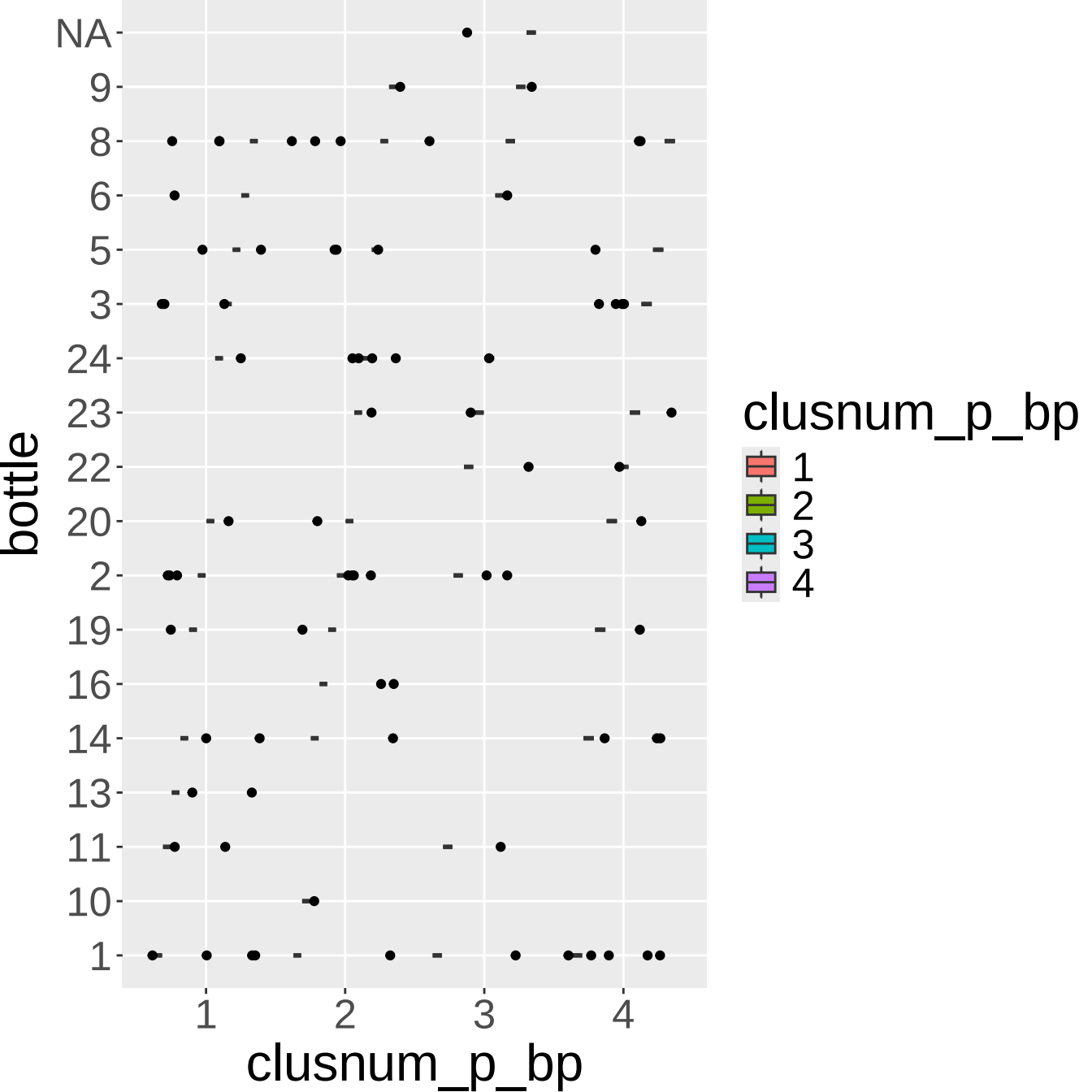
log_name

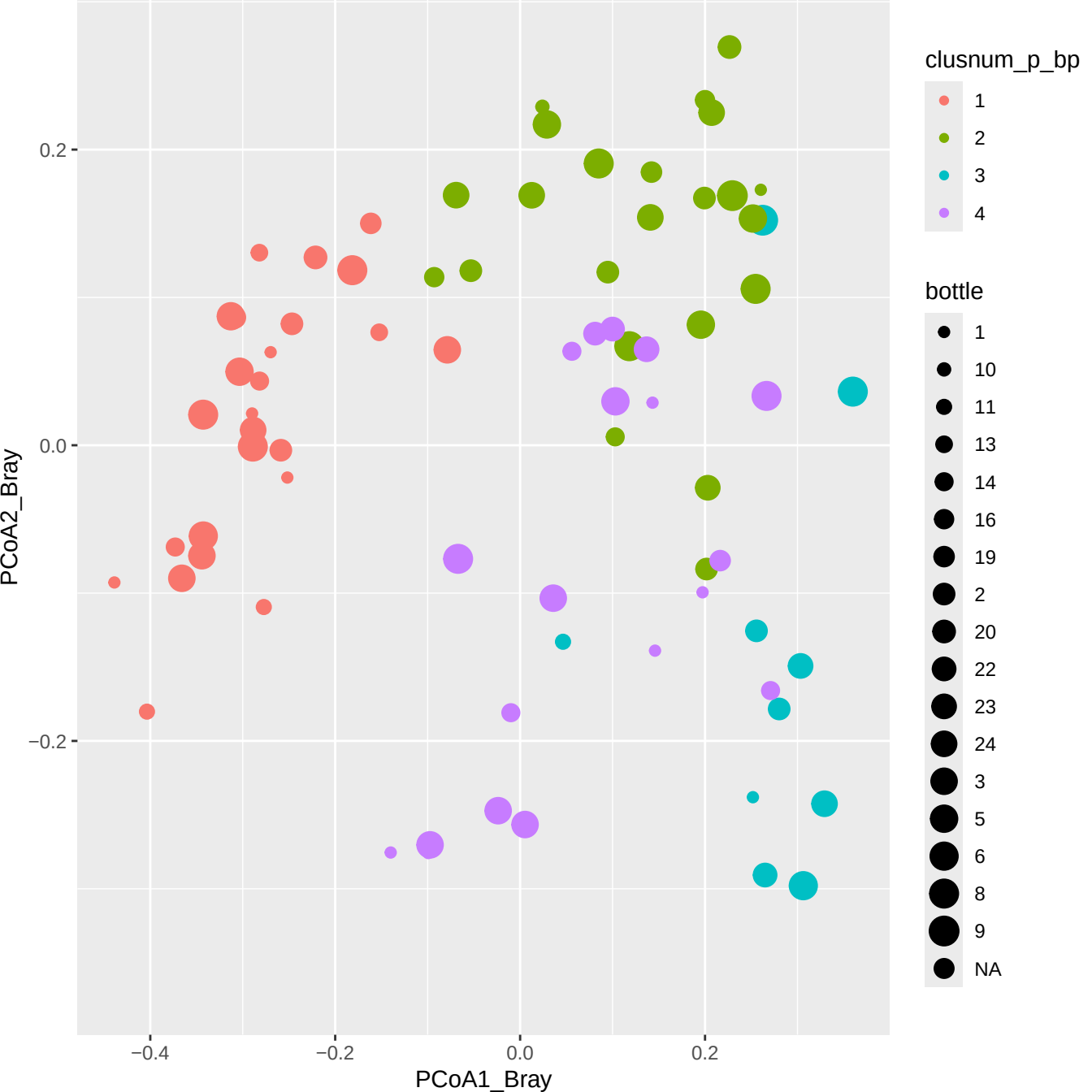
- 170609CBE9 P 20 Chl
- 170609CBE9 P 20 Deep
- 170609CBE9 P 20 Surf
- 170609CBE9 P 3 Chl
- 170609CBE9 P 3 Deep
- 170609CBE9 P 3 surf
- 170609CBE9 P Chl
- 170610CBE1 P 20chl
- 170610CBE1 P 20deep
- 170610CBE1 P 20surf
- 170610CBE1 P 3chlm
- 170610CBE1 P 3deep
- 170610CBE1 P 3surf
- 170610CBE1 P chl
- 170610CBE1 P deep
- 170610CBE1 P surf
- 170611CBW5 P chl 20
- 170611CBW5 P chl 3
- 170611CBW5 P deep 20
- 170611CBW5 P surf 20
- 170611CBW5 P surf 3
- 170613CNL3 P 20 deep
- 170613CNL3 P 3 pycno
- 170613CNL3 P deep
- 170613CNL3 P pycno
- 170614 IL 2 P 20 p/t
- 170614 IL2 P 20 d/c
- 170614 IL2 P 20 surf
- 170614 IL2 P 3 d/c
- 170614 IL2 P 3 p/t
- 170614 IL2 P 3 surf
- 170614 IL2 P deep
- 170614 IL2 P pycno/thermo
- 170614 IL2 P surf
- 170615 DBO3.8A P chl
- 170615 DBO3.8A P 20 chl
- 170615 DBO3.8A P 20 deep
- 170615 DBO3.8A P 3 deep
- 170615 DBO3.8A P deep
- 170615 IL 1.1 20 deep/chl
- 170615 IL 1.1 20 pycno
- 170615 IL 1.1 3 pycno
- 170615 IL 1.1 3 surf
- 170615 IL1.1 3 deep/chl
- 170615 IL1.1 deep/chl
- 170615 IL1.1 pycno
- 170616 DBO3.3 P 20 deep
- 170616 DBO3.3 P 20 surf
- 170616 DBO3.3 P 20 thermo*
- 170616 DBO3.3 P 3 surf
- 170616 DBO3.3 P 3 thermo*
- 170616 DBO3.3 P deep
- 170616 DBO3.3 P thermo*
- 170617 CL3 P 20 deep
- 170617 CL3 P 20 thermo*
- 170617 CL3 P 3 deep
- 170617 CL3 P 3 surf
- 170617 CL3 P 3 thermo*
- 170617 CL3 P deep
- 170617 CL3 P surf
- 170617 CL3 P t
- 170617 DBO3.3
- 170618 CL1 P 2
- 170618 CL1 P 2
- 170618 CL1 P 2
- 170618 CL1 P 3
- 170618 CL1 P 3
- 170618 CL1 P c
- 170618 CL1 P r
- 170624 DBO2.4
- 170624 DBO2.4
- 170624 DBO2.4
- 170624 DBO2.4
- 170624 DBO2.4
- 170624 DBO2.4
- 170624 DBO2.4
- Tote slime 3

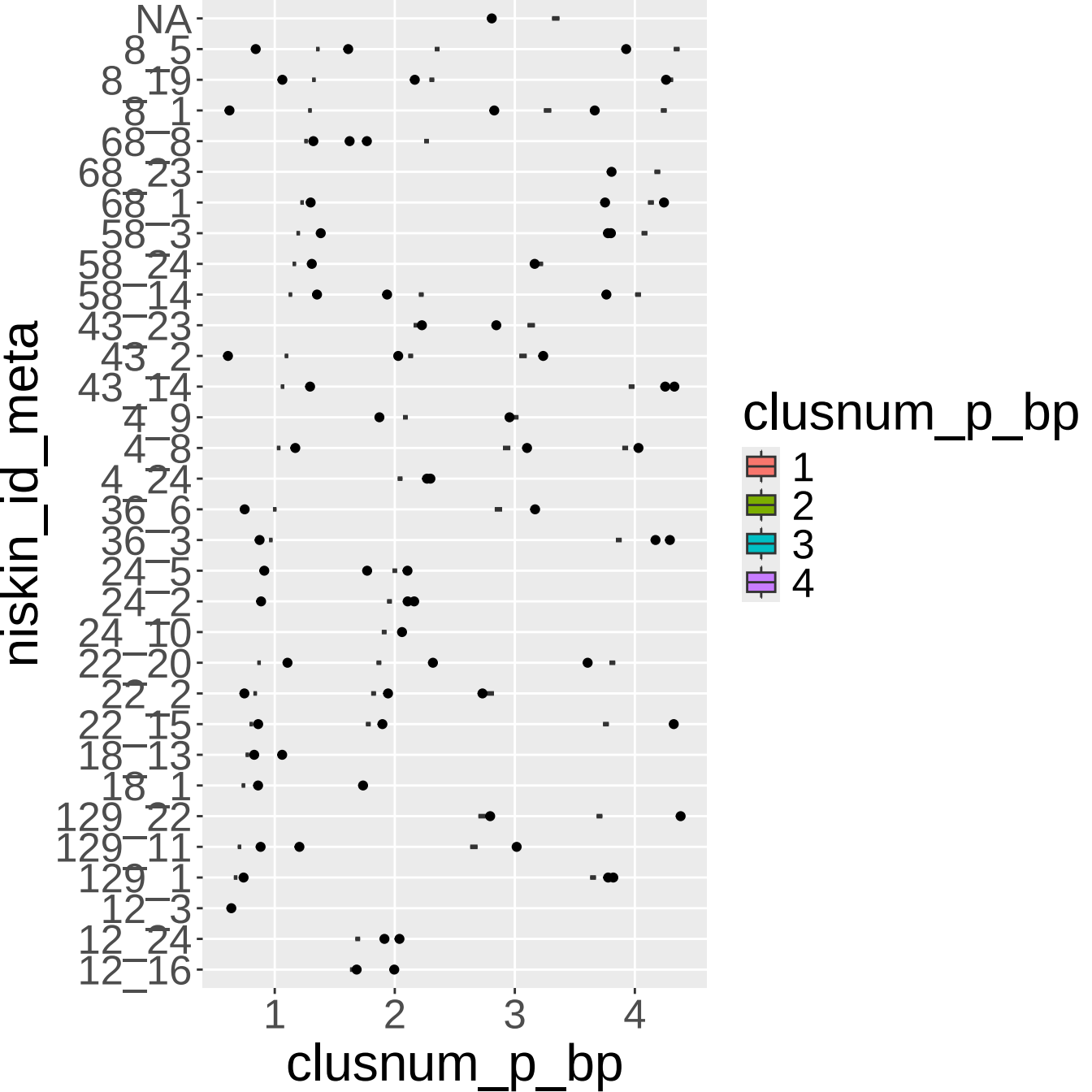
clusnum_p_bp

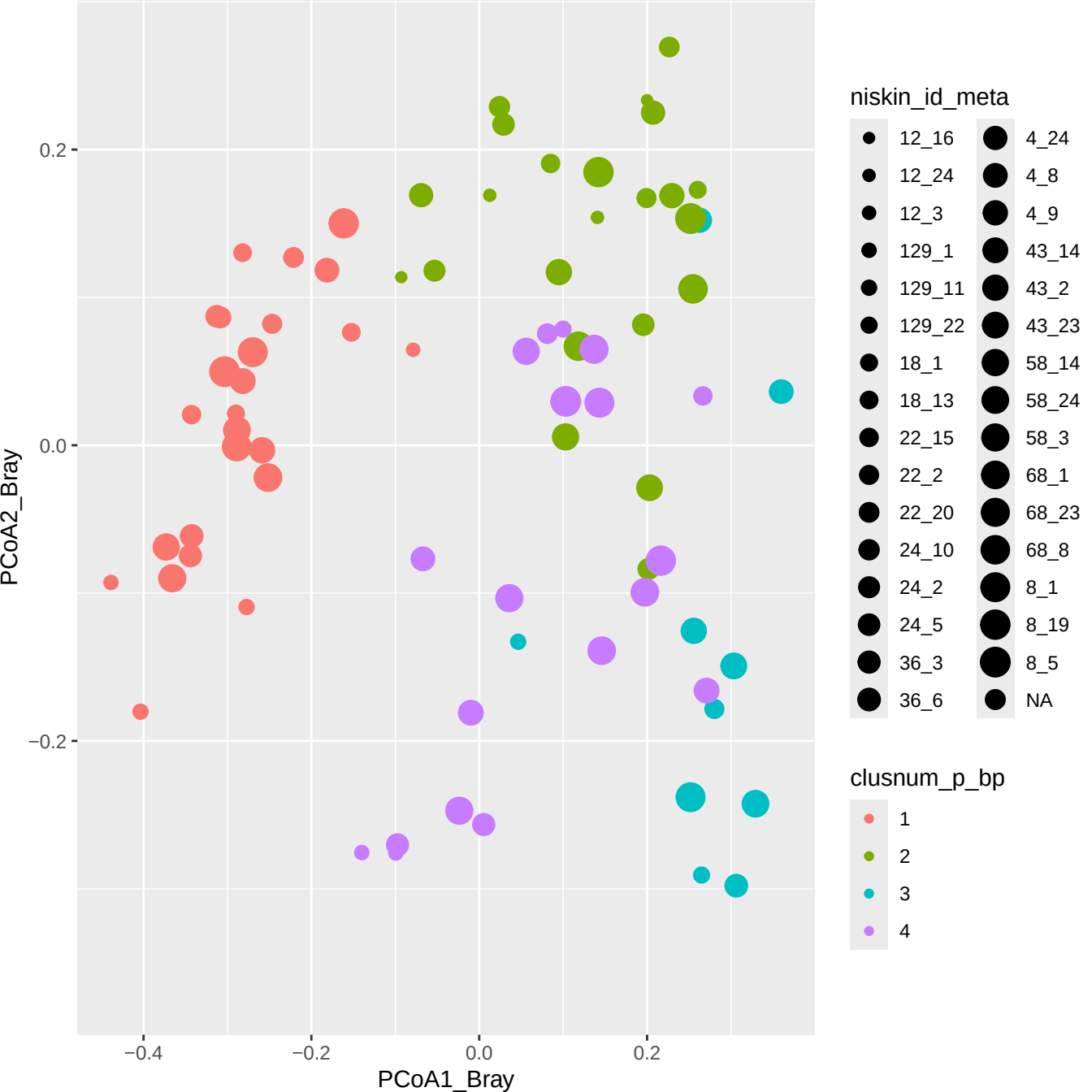
- 1
- 2
- 3
- 4

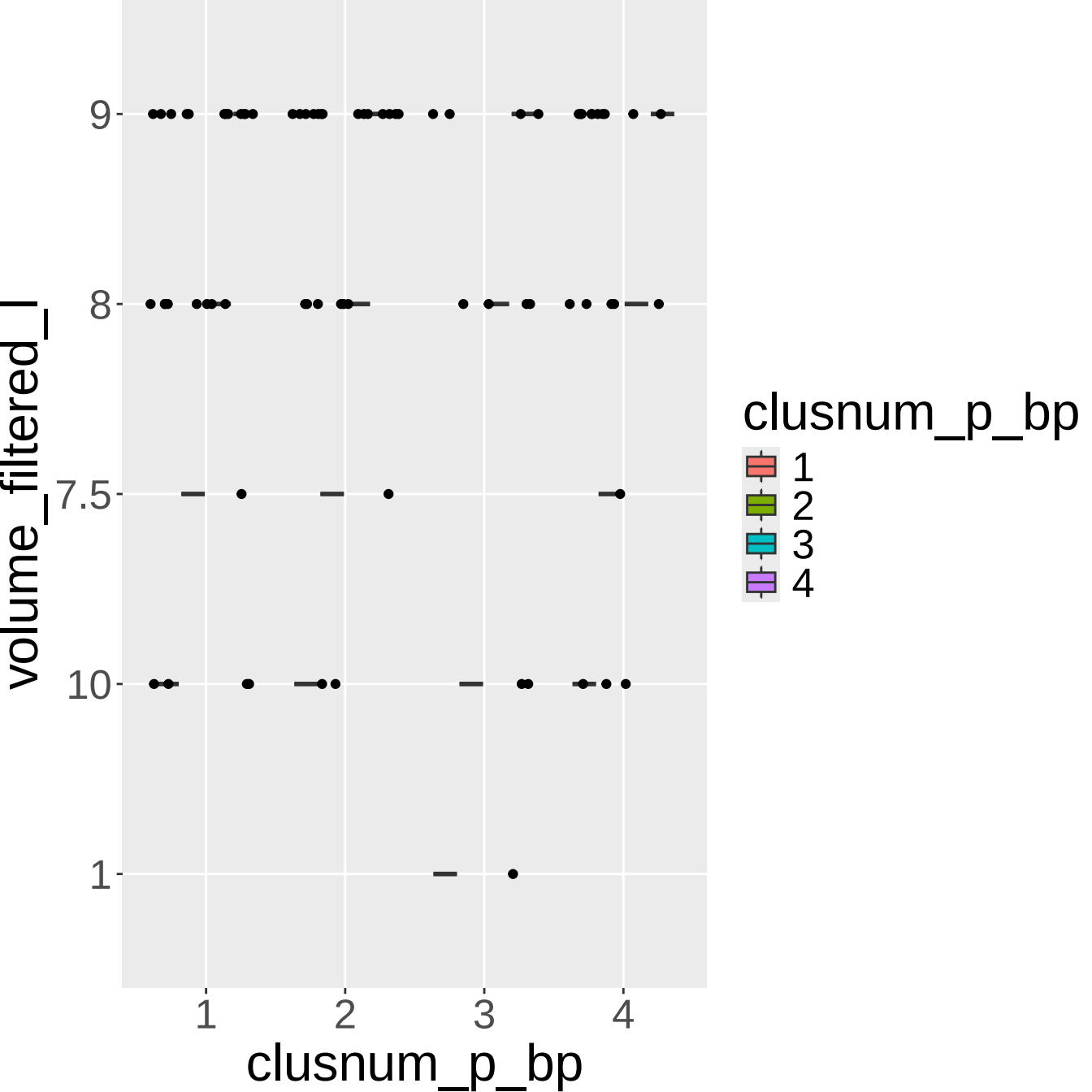
Bray

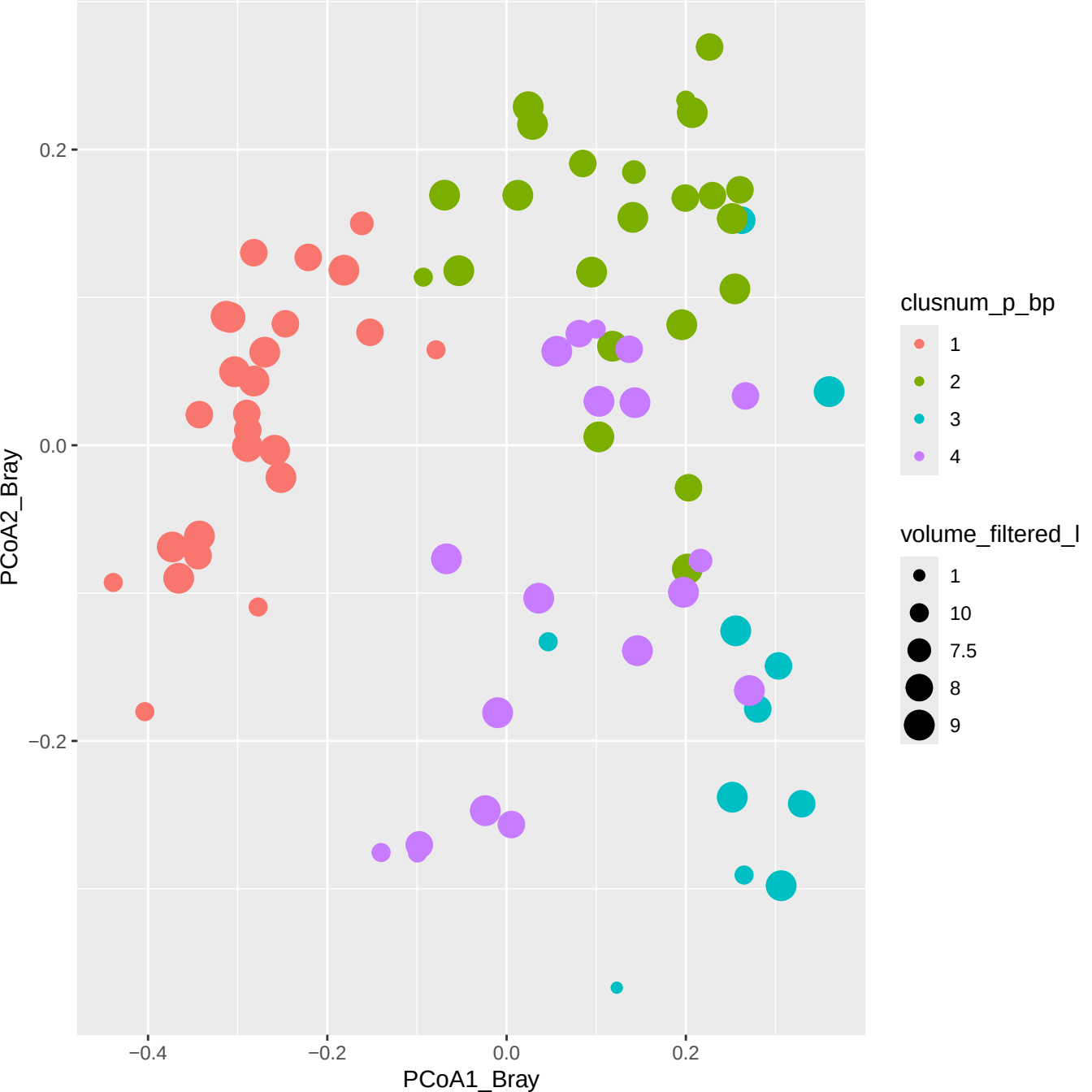

bottle
NA
9
8
6
5
3
24
23
22
20
2
19
16
14
13
11
10
1
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

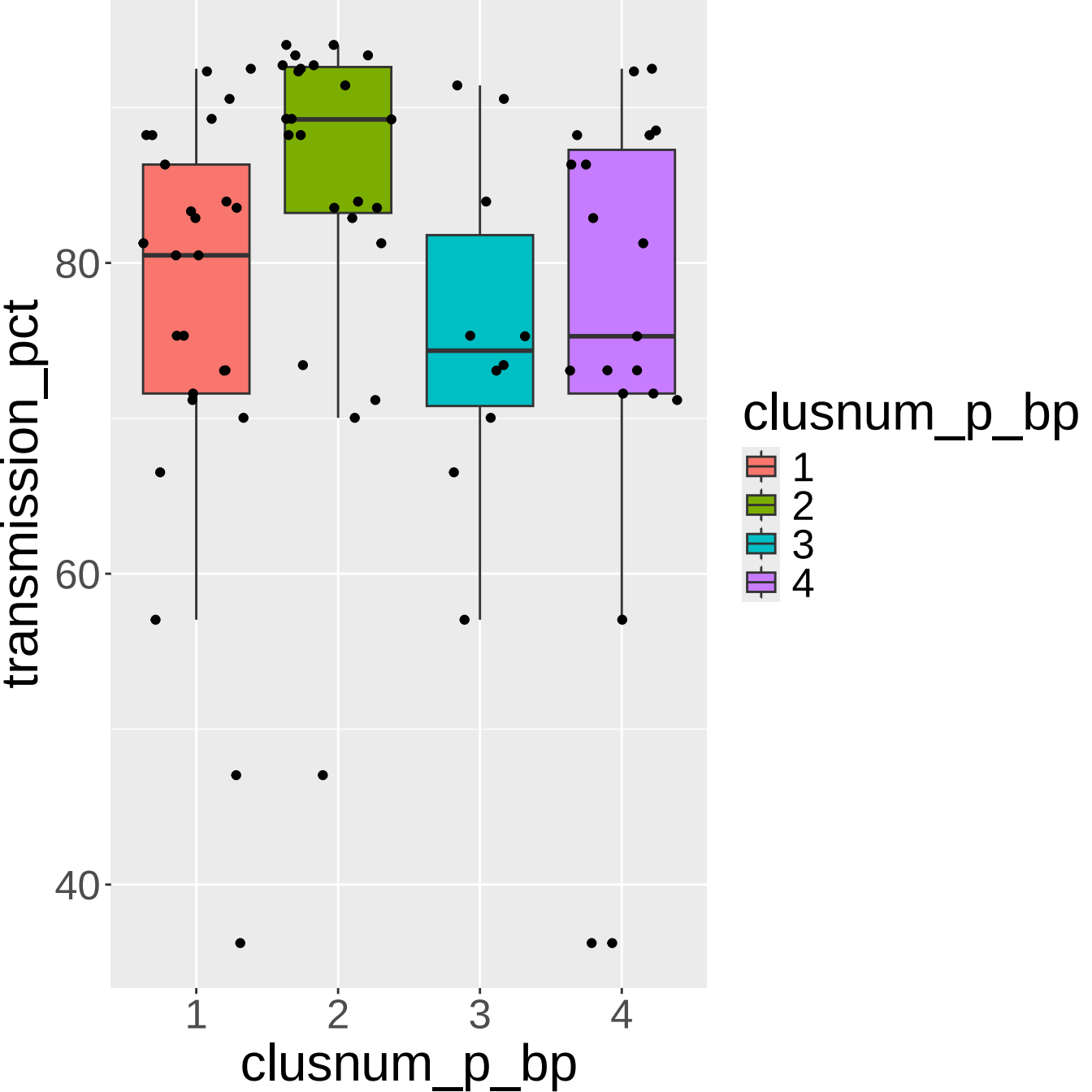

clusnum_p_bp
1
2
3
4
bottle
1
10
11
13
14
16
19
2
20
22
23
24
3
5
6
8
9
NA
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray

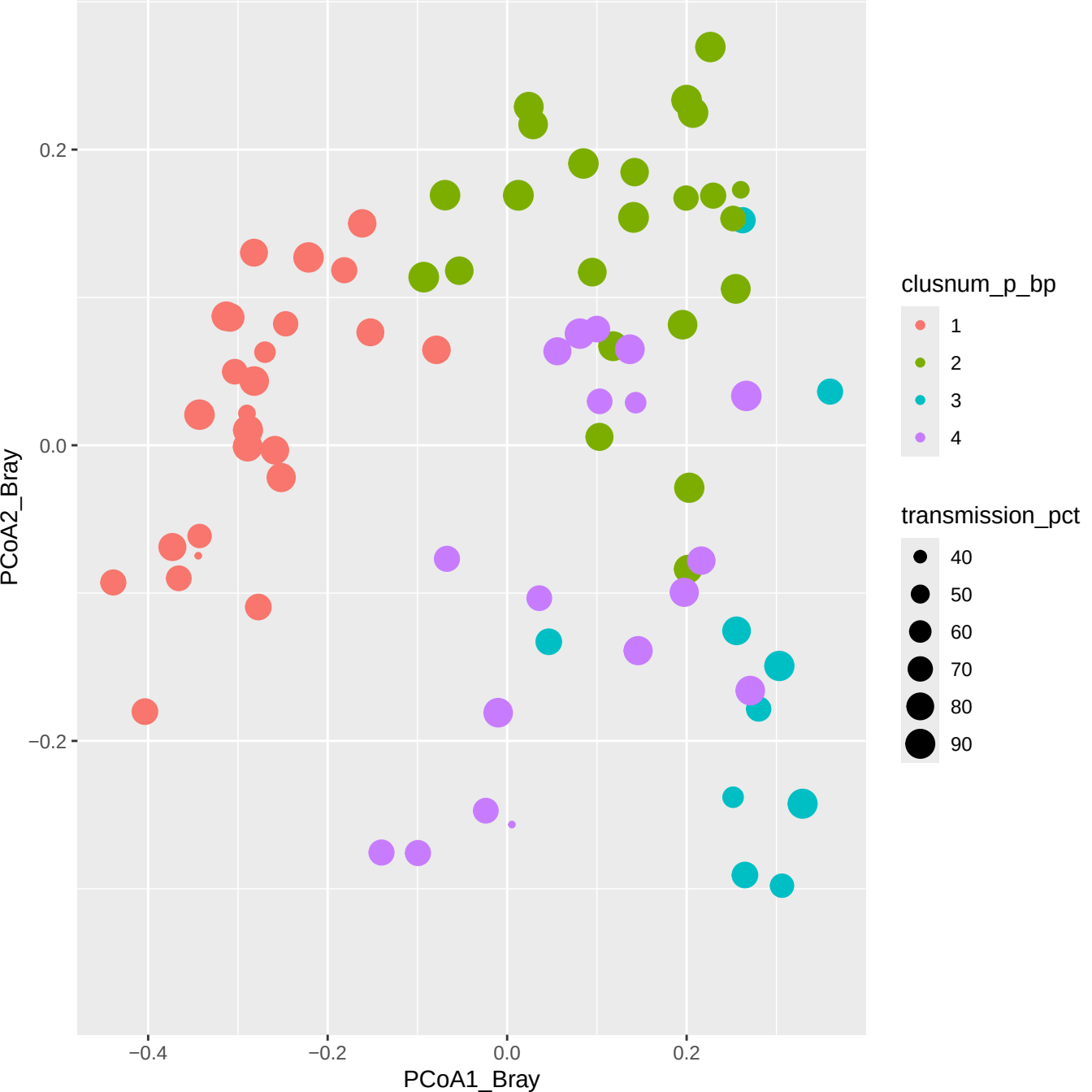

NA
8_5
8_19
8_1
68_8
68_23
68_1
58_3
58_24
58_14
43_23
43_2
43_14
4_9
4_8
4_24
36_6
36_3
24_5
24_2
24_10
22_20
22_2
22_15
18_13
18_1
129_22
129_11
129_1
12_3
12_24
12_16
niskin_id_meta
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

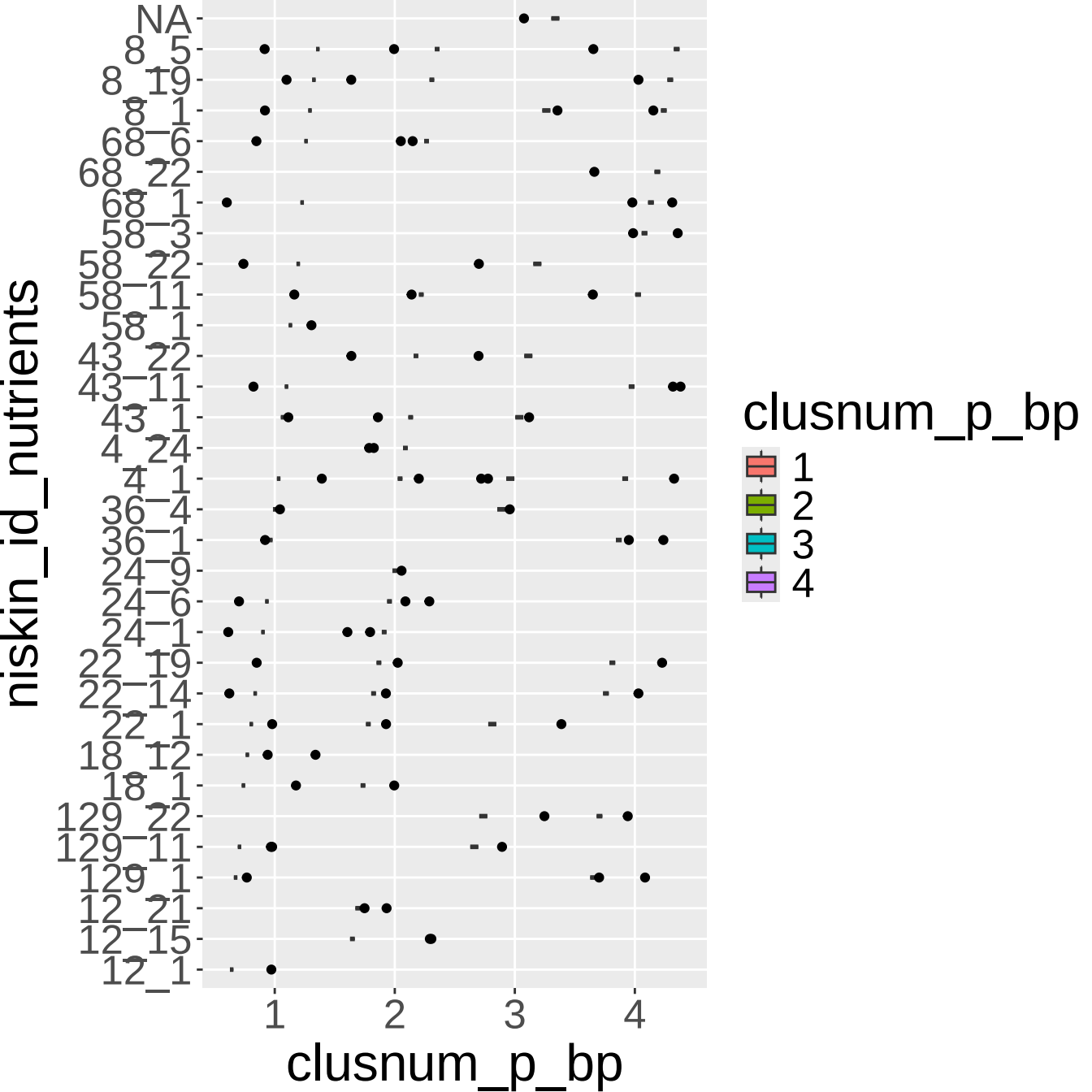

PCoA2_Bray
PCoA1_Bray
0.2
0.0
−0.2
−0.4
−0.2
0.0
0.2
niskin_id_meta
12_16
12_24
12_3
129_1
129_11
129_22
18_1
18_13
22_15
22_2
22_20
24_10
24_2
24_5
36_3
36_6
4_24
4_8
4_9
43_14
43_2
43_23
58_14
58_24
58_3
68_1
68_23
68_8
8_1
8_19
8_5
NA
clusnum_p_bp
1
2
3
4

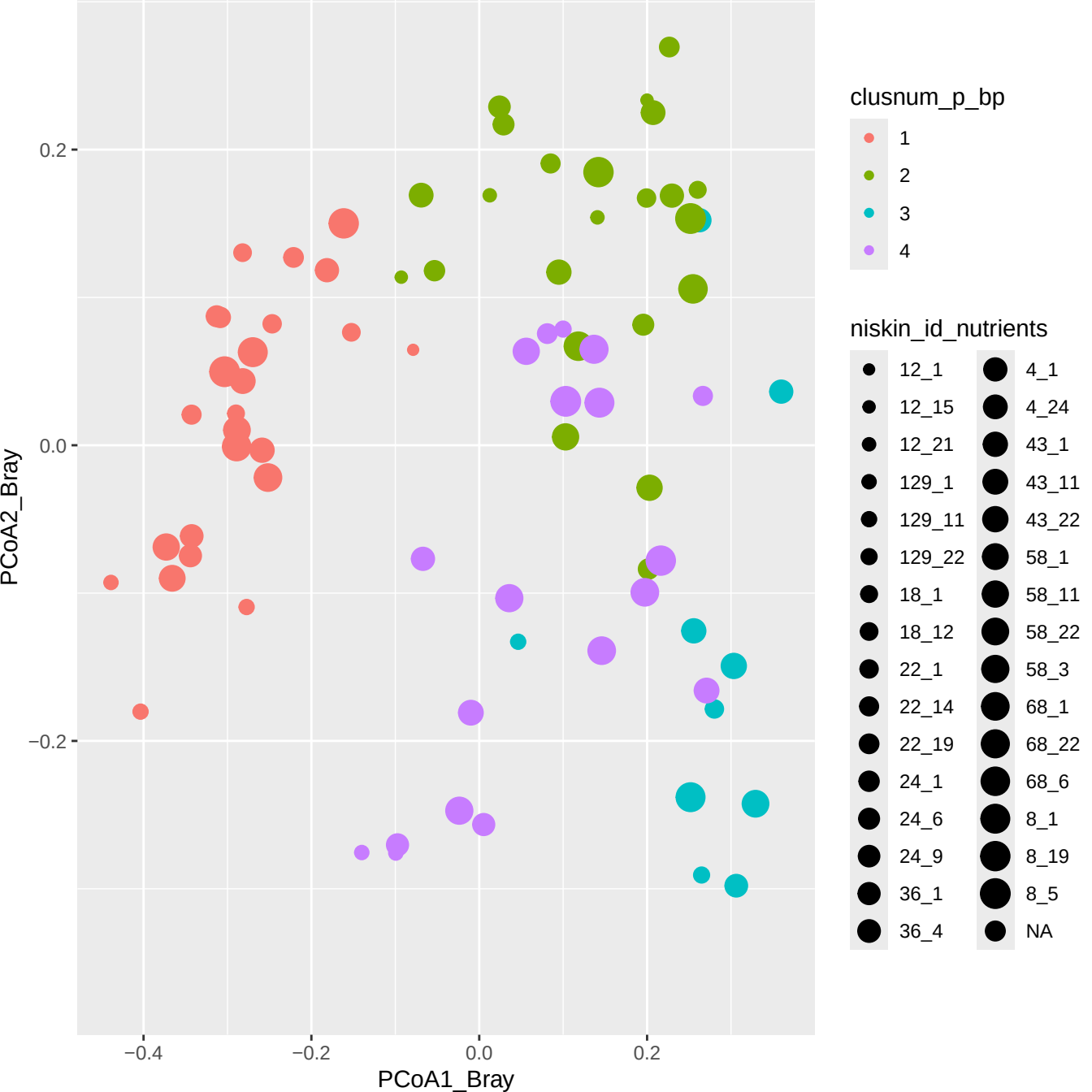

volume_filtered_l
9
8
7.5
10
1
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

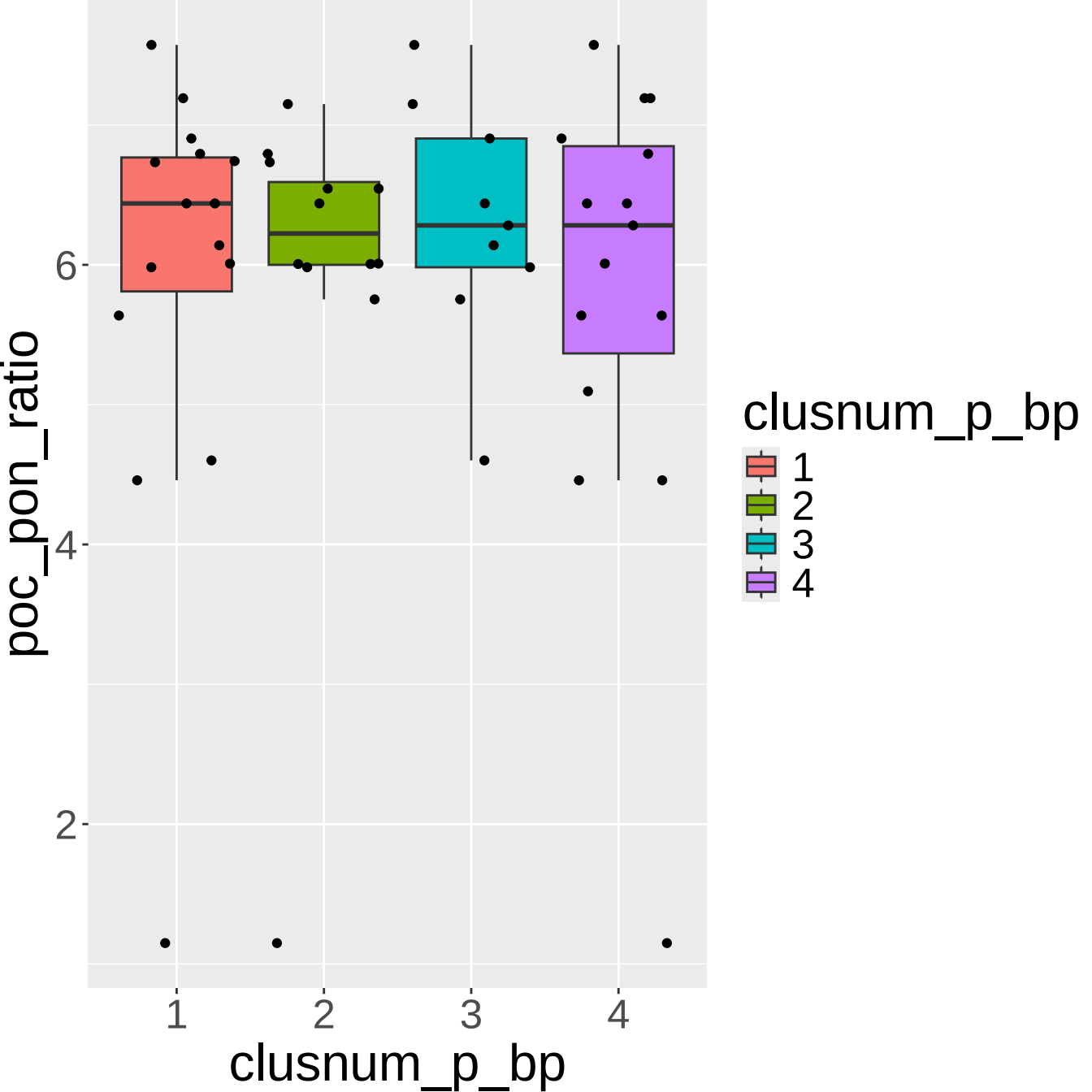

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
volume_filtered_l
1
10
7.5
8
9

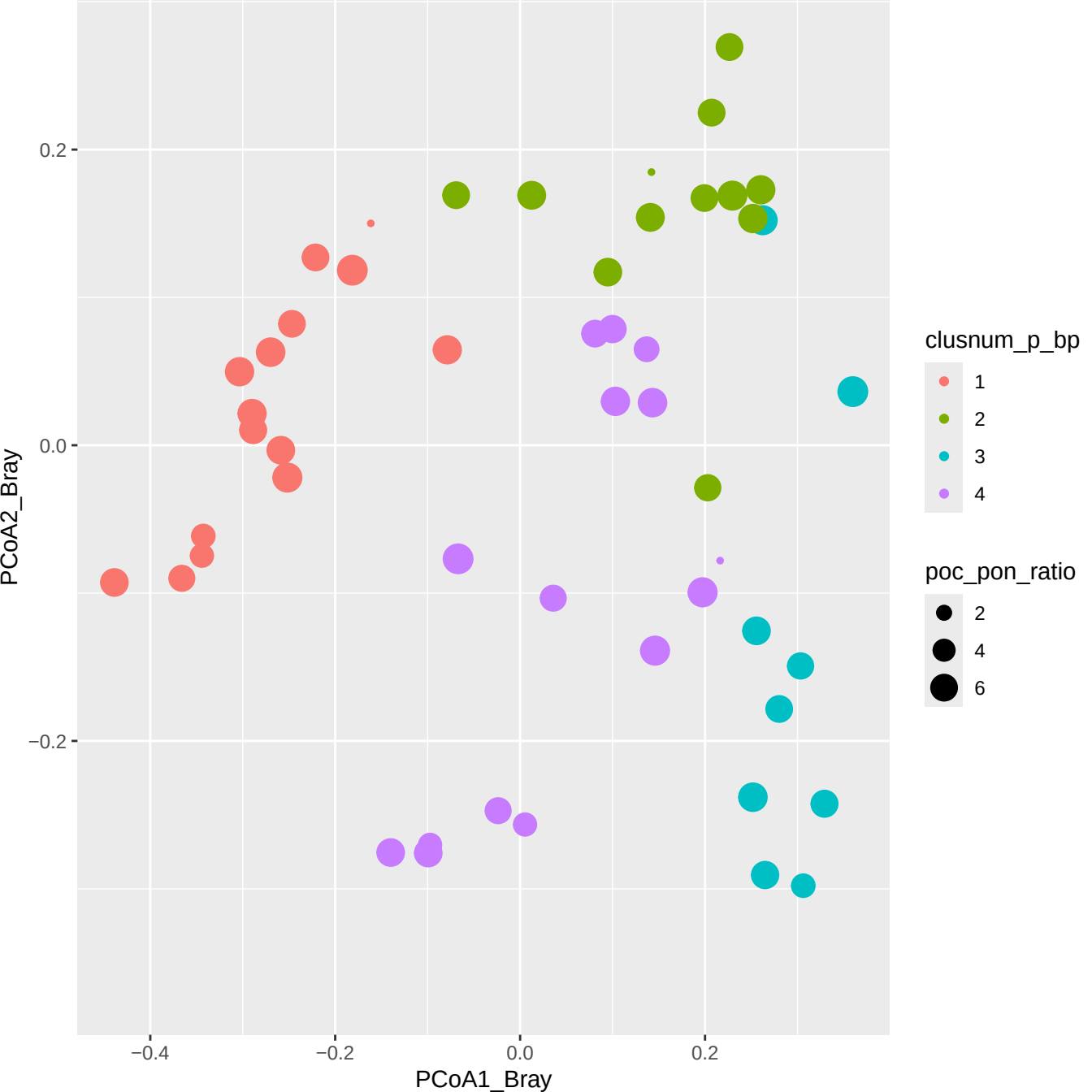

transmission_pct
80
60
40
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

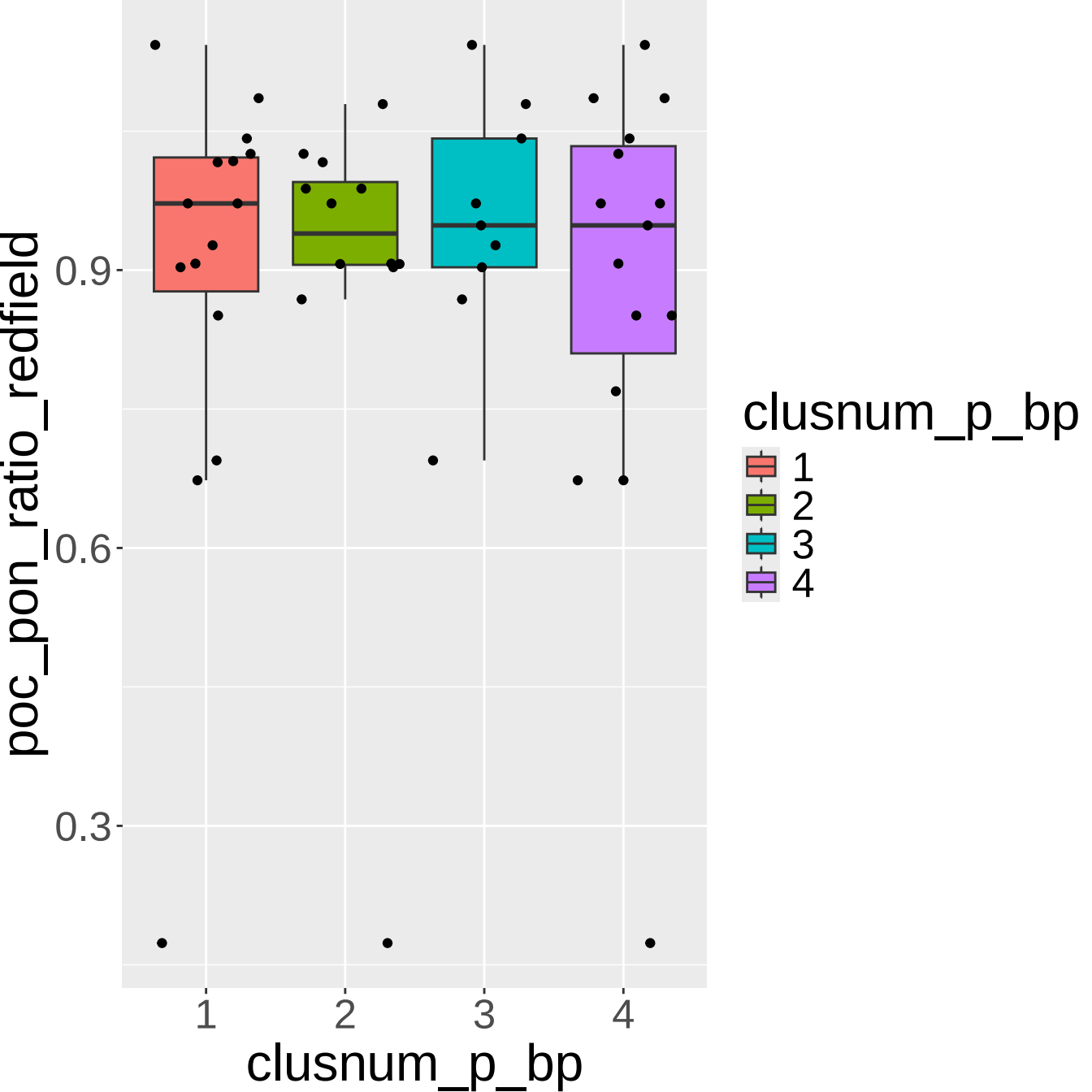

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
transmission_pct
40
50
60
70
80
90

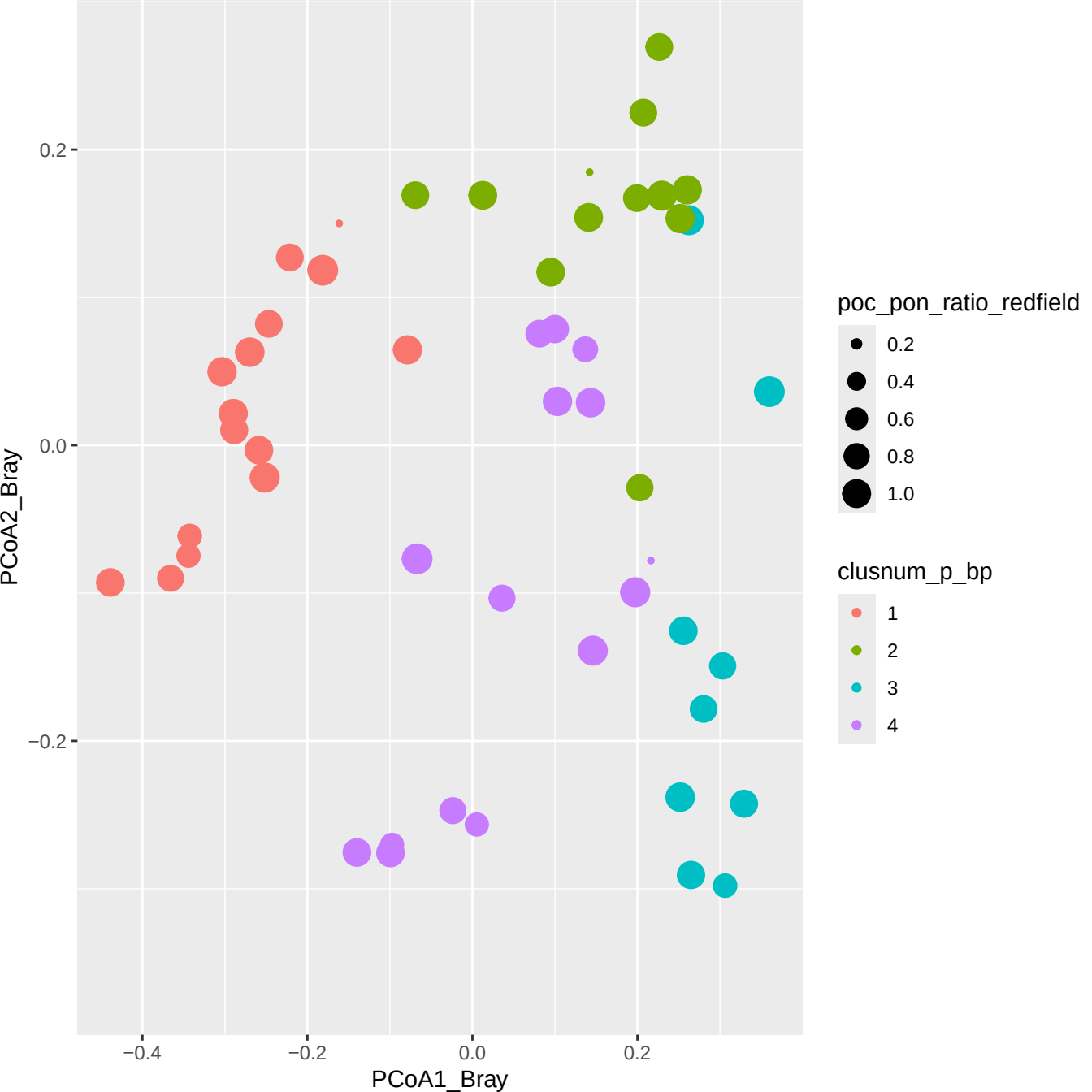

niskin_id_nutrients
NA
8_5
8_19
8_1
68_6
68_22
68_1
58_3
58_22
58_11
58_1
43_22
43_11
43_1
4_24
4_1
36_4
36_1
24_9
24_6
24_1
22_19
22_14
22_1
18_12
18_1
129_22
129_11
129_1
12_21
12_15
12_1
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

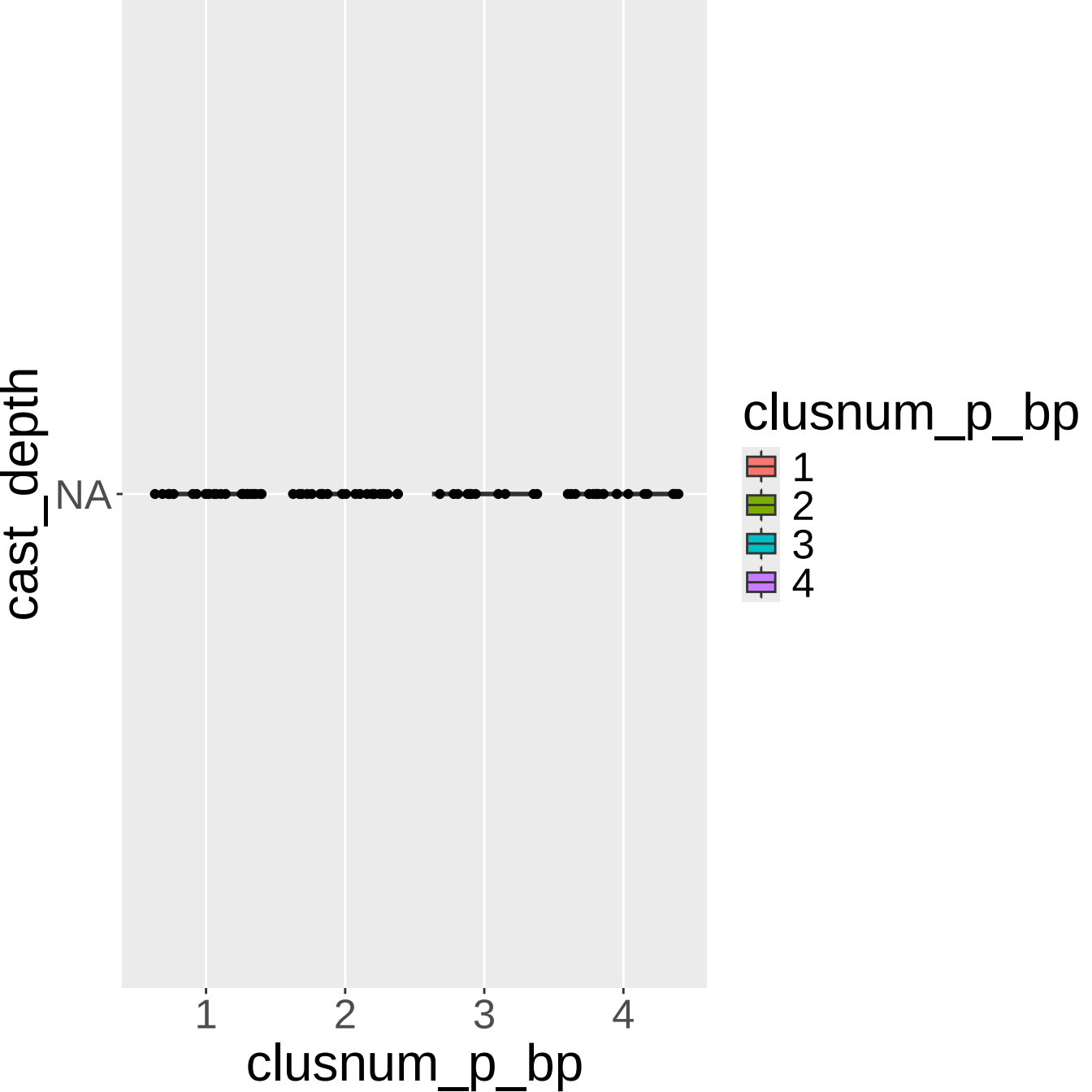

clusnum_p_bp
1
2
3
4
niskin_id_nutrients
12_1
12_15
12_21
129_1
129_11
129_22
18_1
18_12
22_1
22_14
22_19
24_1
24_6
24_9
36_1
36_4
4_1
4_24
43_1
43_11
43_22
58_1
58_11
58_22
58_3
68_1
68_22
68_6
8_1
8_19
8_5
NA
PCoA2_Bray
0.2
0.0
−0.2
PCoA1_Bray
−0.4
−0.2
0.0
0.2

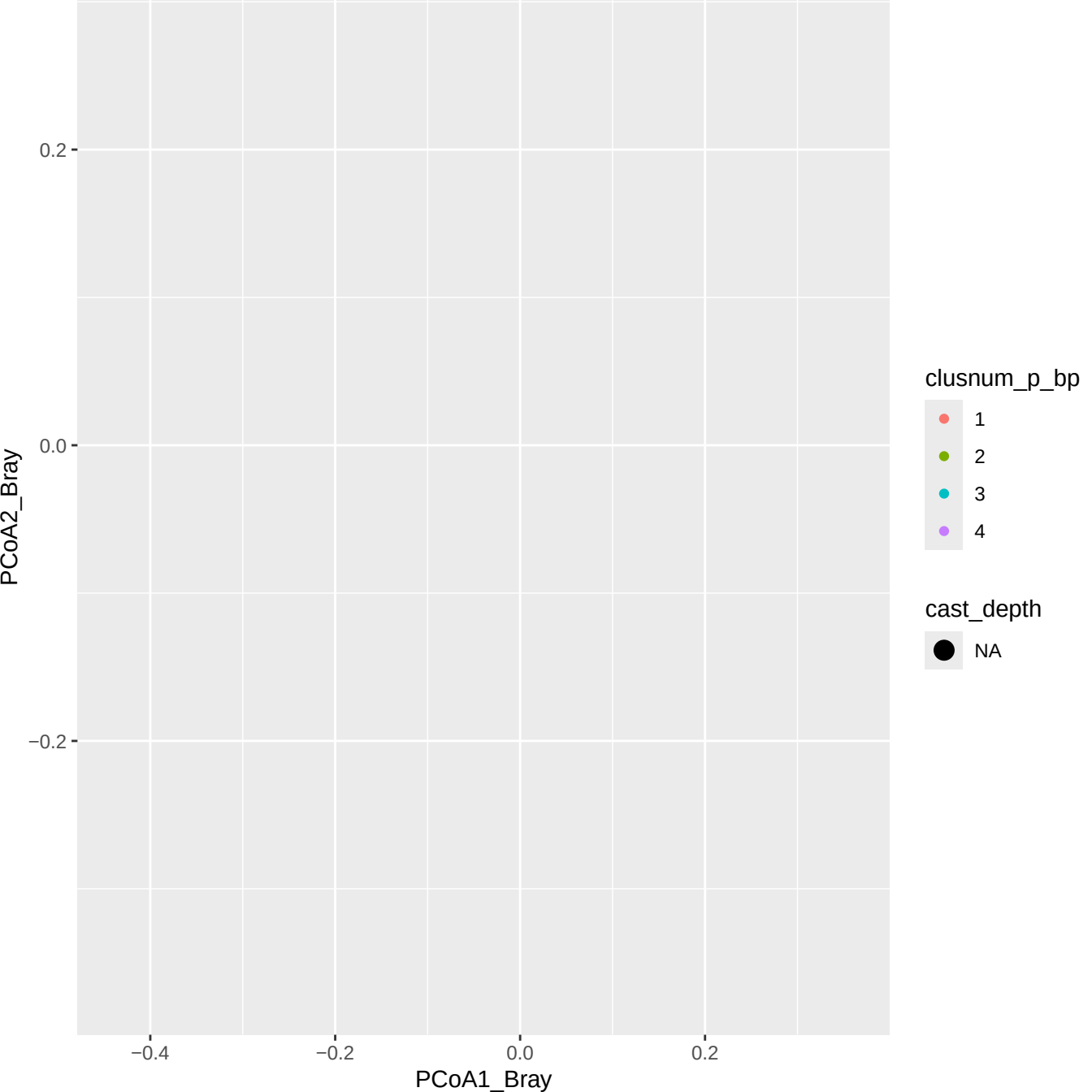

poc_pon_ratio
6
4
2
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

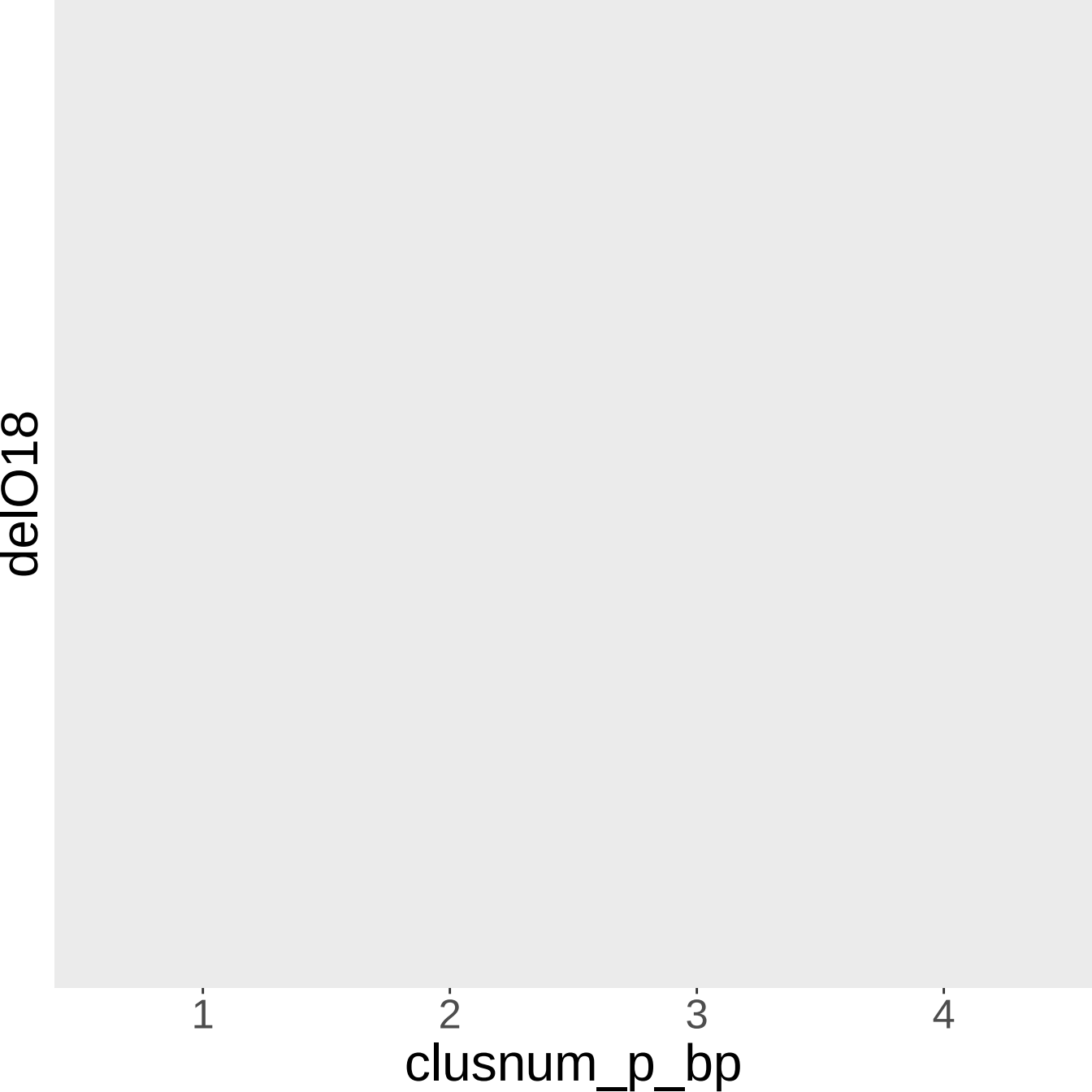

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
poc_pon_ratio
2
4
6

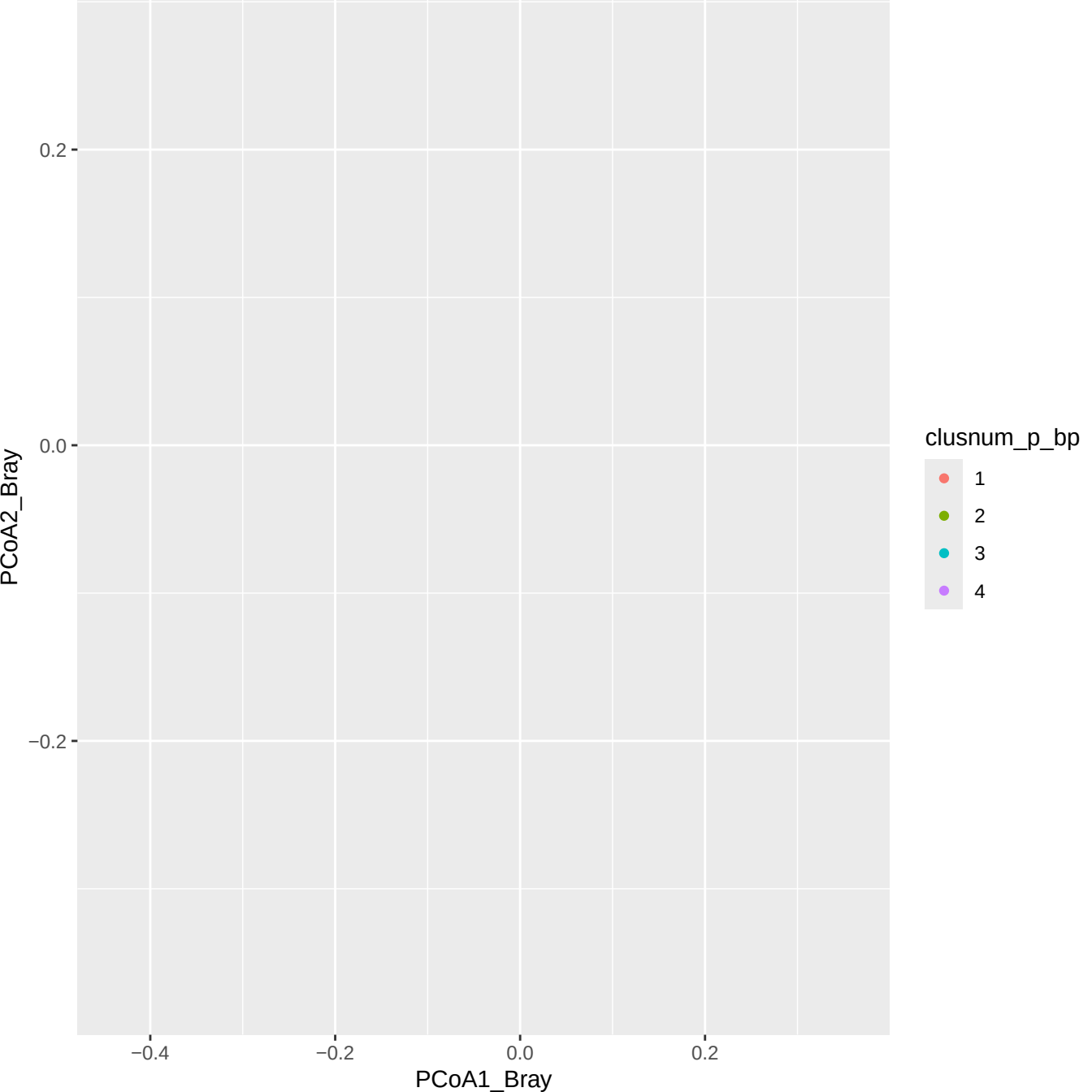

poc_pon_ratio_redfield
0.9
0.6
0.3
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4

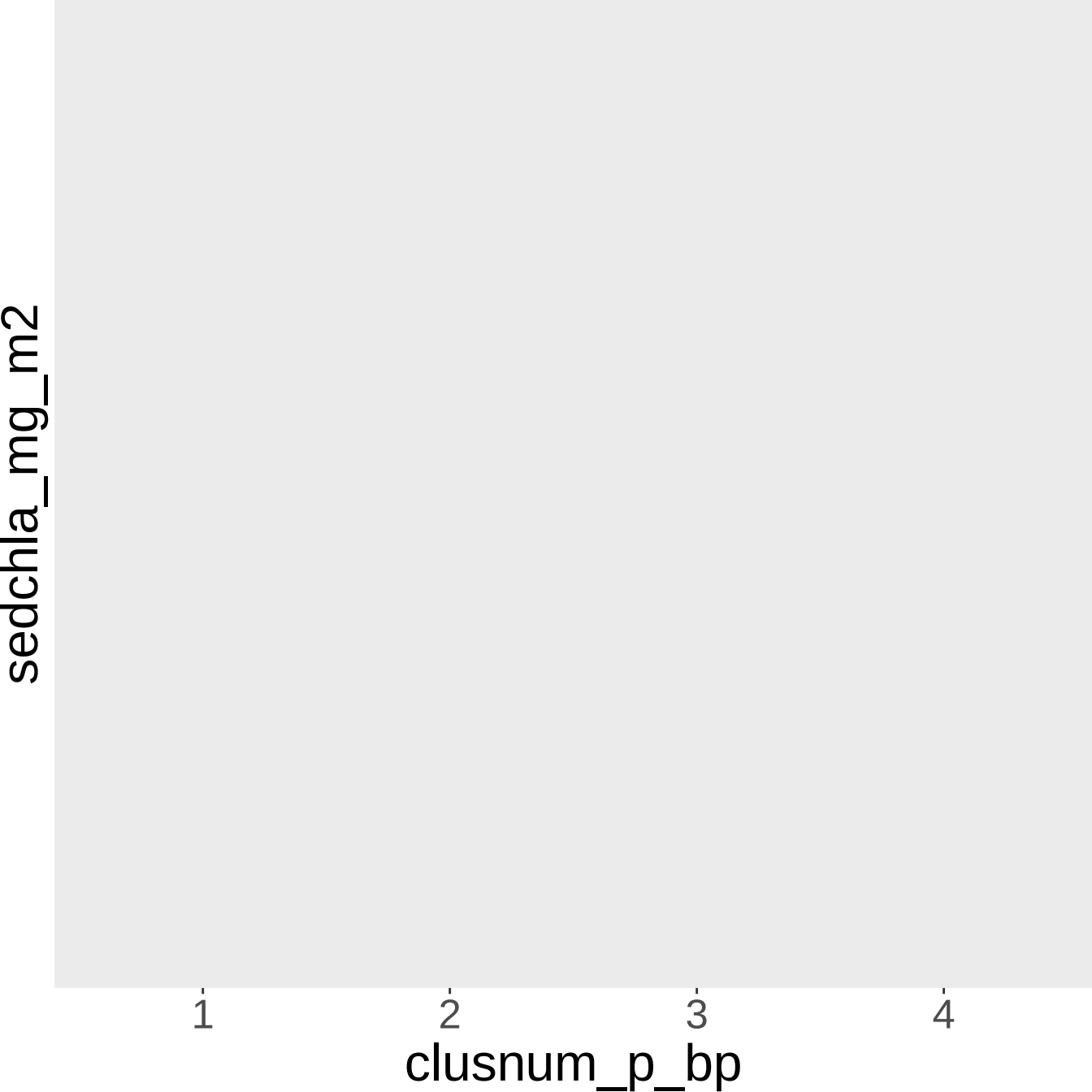

0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
poc_pon_ratio_redfield
0.2
0.4
0.6
0.8
1.0
clusnum_p_bp
1
2
3
4

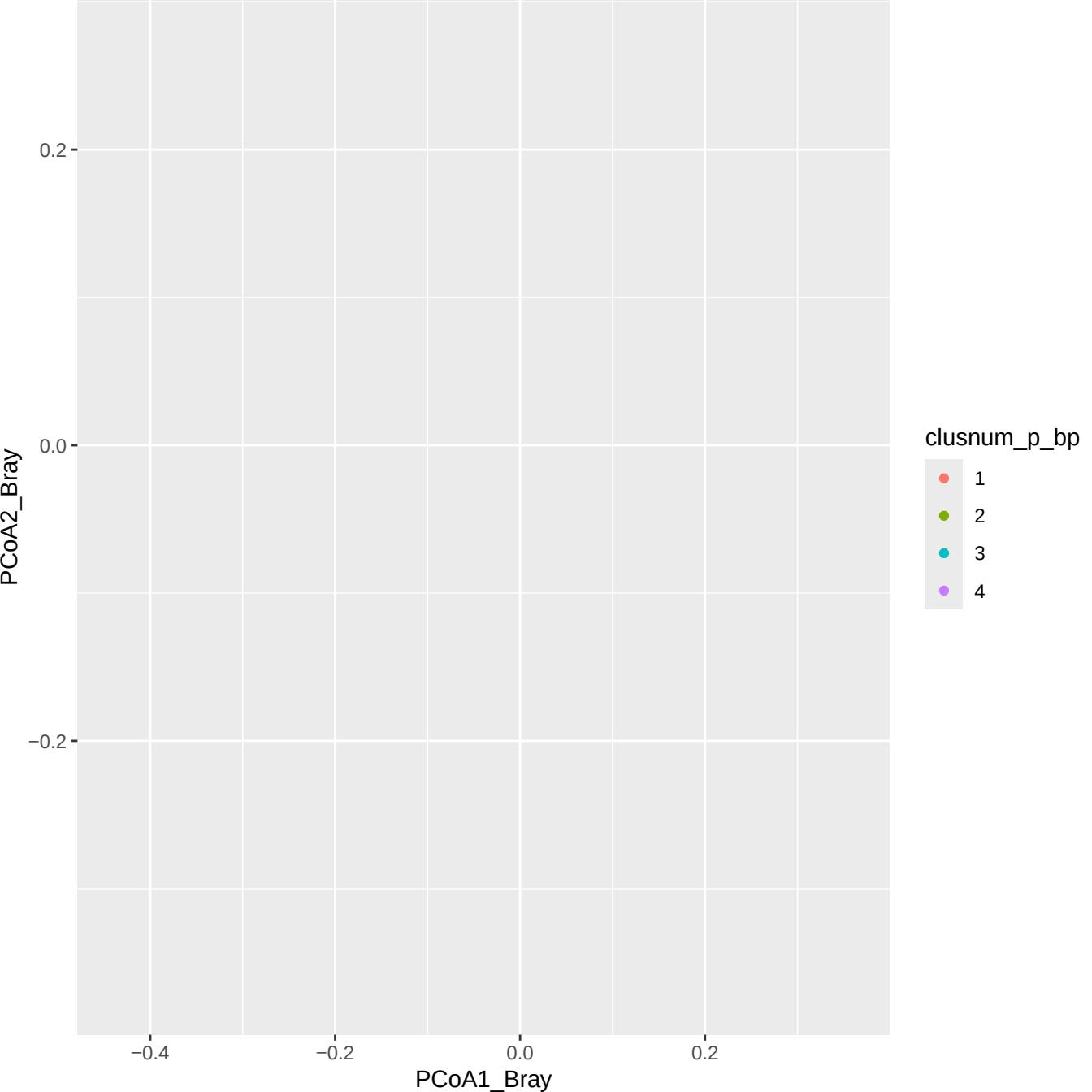

cast_depth
NA
1
2
3
4
clusnum_p_bp
clusnum_p_bp
1
2
3
4


0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
cast_depth
NA


delO18
1
2
3
4
clusnum_p_bp


0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4


sedchla_mg_m2
1
2
3
4
clusnum_p_bp


0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4

surf filter stack leaked about 2.5 L of water at

sea ice!!! No UVP :((LIS

e high flux of chlorophyll to bottom, took pycno/thermo instea

label says CNL3 with

filter was not centered at all in holder and was bent a

dropped in filter in sink while removing from

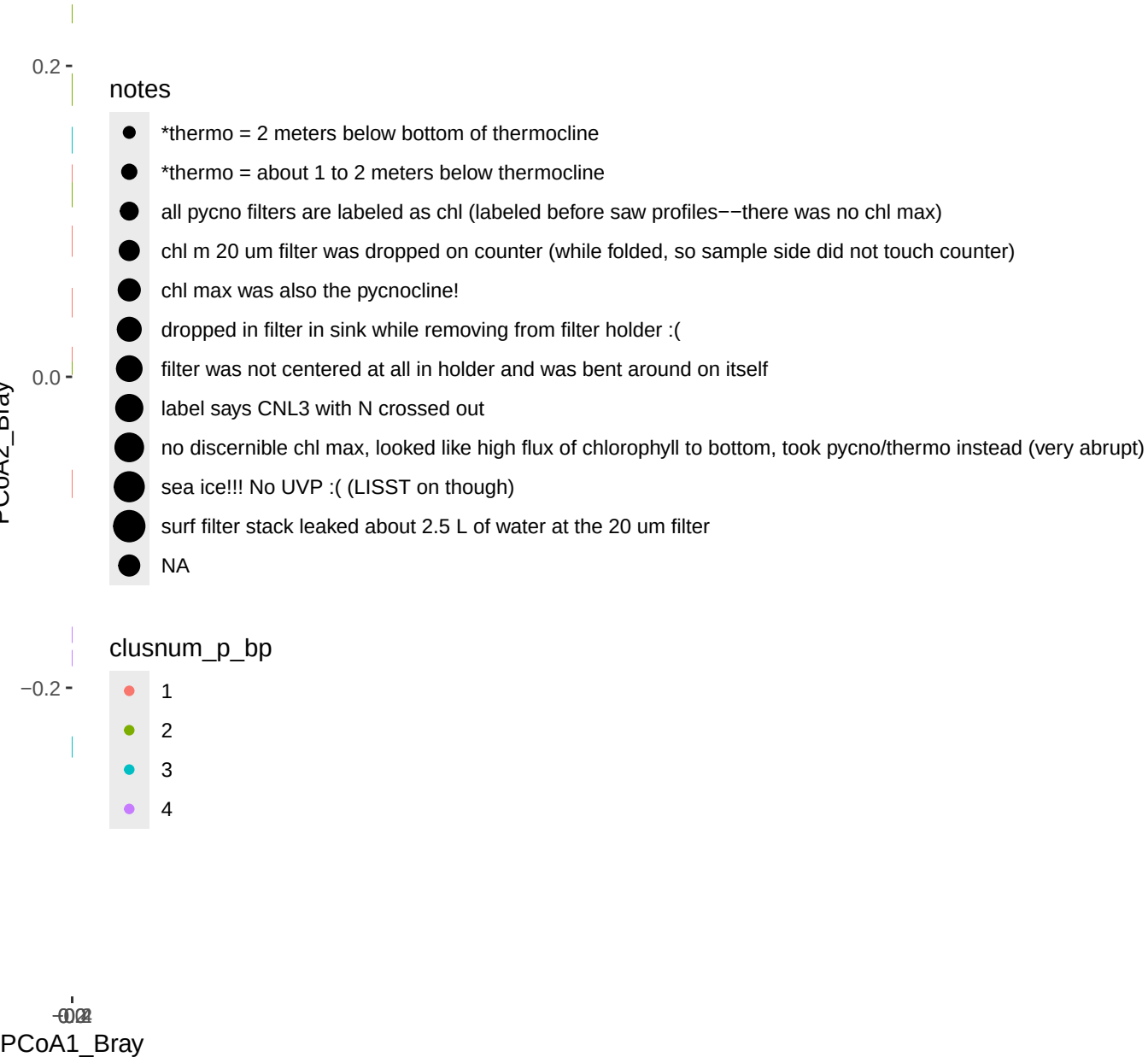
chl max was also

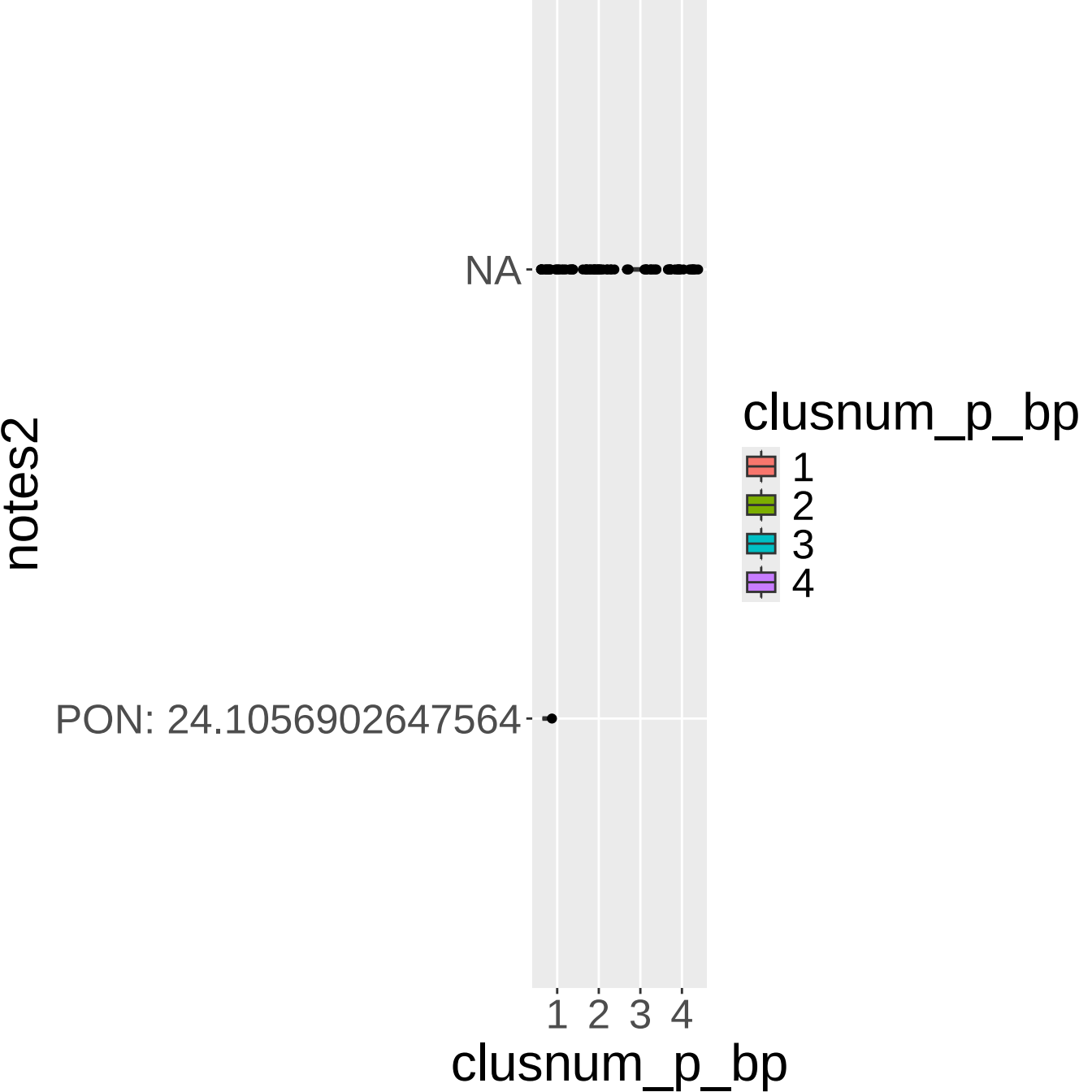
was dropped on counter (while folded, so sample side did not

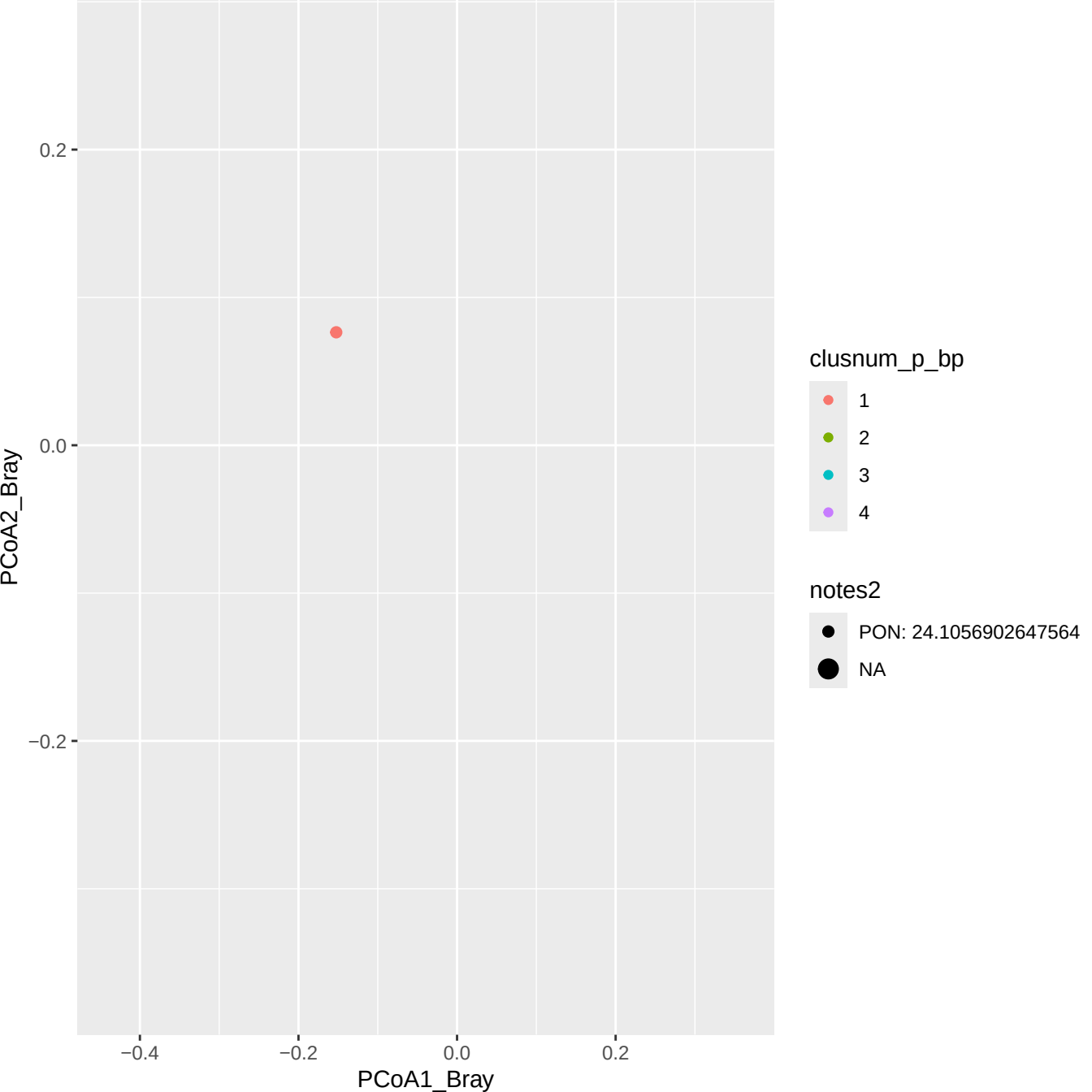
ters are labeled as chl (labeled before saw profiles−−there w

*thermo = about 1 to 2 meters belo

*thermo = 2 meters below bottom


0.2
0.0
−0.2
PCoA2_Bray
PCoA1_Bray
notes
*thermo = 2 meters below bottom of thermocline
*thermo = about 1 to 2 meters below thermocline
all pycno filters are labeled as chl (labeled before saw profiles−−there was no chl max)
chl m 20 um filter was dropped on counter (while folded, so sample side did not touch counter)
chl max was also the pycnocline!
dropped in filter in sink while removing from filter holder :(
filter was not centered at all in holder and was bent around on itself
label says CNL3 with N crossed out
no discernible chl max, looked like high flux of chlorophyll to bottom, took pycno/thermo instead (very abrupt)
sea ice!!! No UVP :((LISST on though)
surf filter stack leaked about 2.5 L of water at the 20 um filter
NA
clusnum_p_bp
1
2
3
4


NA
PON: 24.1056902647564
notes2
clusnum_p_bp
1
2
3
4
1 2 3 4
clusnum_p_bp


0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_p_bp
1
2
3
4
notes2
PON: 24.1056902647564
NA